



IEEE Standard for Ethernet

Amendment 8: Physical Layer Specifications and Management Parameters for 25 Gb/s Electrical Automotive Ethernet

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3cy™-2023

(Amendment to IEEE Std 802.3™-2022
as amended by IEEE Std 802.3dd™-2022,
IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022,
IEEE Std 802.3ck™-2022, IEEE Std 802.3de™-2022,
IEEE Std 802.3cx™-2023, and IEEE Std 802.3cz™-2023)

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IEEE Std 802.3cs™-2022,
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and IEEE Std 802.3cz™-2023)

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Approved 29 June 2023
IEEE SA Standards Board

Abstract: This amendment to IEEE Std 802.3-2022 adds Physical Layer specifications and management parameters for 25 Gb/s operation on a single balanced pair of conductors suitable for automotive applications.

Keywords: 25GBASE-T1, amendment, automotive Ethernet, IEEE 802.3™, IEEE 802.3cy™, Medium Dependent Interface, Physical Coding Sublayer, Physical Layer, Physical Medium Attachment

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Introduction

This introduction is not part of IEEE Std 802.3cy-2023, IEEE Standard for Ethernet—Amendment 8: Physical Layer Specifications and Management Parameters for 25 Gb/s Electrical Automotive Ethernet.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3cy-2023 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

IEEE Std 802.3ck™-2022

Amendment 4—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 161 through Clause 163, Annex 120F, Annex 120G, Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

IEEE Std 802.3de™-2022

Amendment 5—This amendment includes changes to IEEE Std 802.3-2022 to add 10 Mb/s Single-Pair Ethernet point-to-point PHYs to the PHYs supporting the MAC Merge function and the Time Synchronization Service Interface (TSSI).

IEEE Std 802.3cx™-2023

Amendment 6—This amendment to IEEE Std 802.3-2022 modifies Clause 30, Clause 45, and Clause 90, and adds Annex 90A to enhance support for time synchronization protocols by providing options for sub-nanosecond reporting of the transmit and receive path data delays, selection of the data delay measurement point, and dynamic reporting of path data delay variation.

IEEE Std 802.3cz™-2023

Amendment 7—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 166. This amendment adds 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, and 50 Gb/s Physical Layer specifications and management parameters for optical automotive Ethernet using graded-index glass optical fiber.

IEEE Std 802.3cy™-2023

Amendment 8—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 165 and Annex 165A. This amendment adds Physical Layer specifications and management parameters for operation at 25 Gb/s over a single balanced pair of conductors.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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IEEE Standard for Ethernet

Amendment 8: Physical Layer Specifications and Management Parameters for 25 Gb/s Electrical Automotive Ethernet

(This amendment is based on IEEE Std 802.3™-2022 as amended by IEEE Std 802.3dd™-2022, IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022, IEEE Std 802.3ck™-2022, IEEE Std 802.3de™-2022, IEEE Std 802.3cx™-2023, and IEEE Std 802.3cz™-2023.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~striketrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.⁶

⁶ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1. Introduction

1.4 Definitions

Insert 1.4.128a after 1.4.128 “25GBASE-T” as follows:

1.4.128a 25GBASE-T1: IEEE 802.3 Physical Layer specification for a 25 Gb/s Ethernet using a single balanced pair of conductors. (See IEEE Std 802.3, Clause 165.)

Change 1.4.407 as follows:

1.4.407 MultiGBASE-T1: PHYs that belong to the set of specific BASE-T1 PHYs at speeds in excess of 1000 Mb/s, including 2.5GBASE-T1, 5GBASE-T1, ~~and 10GBASE-T1,~~ and 25GBASE-T1. (See IEEE Std 802.3, Clause 149 and Clause 165.)

Change 1.4.472 as follows:

1.4.472 PoDL PD: A Powered Device that is intended to receive power from a link section consisting of a single ~~twisted conductor~~ conductor pair. (See IEEE Std 802.3, Clause 104.)

Change 1.4.473 as follows:

1.4.473 PoDL PSE: A device that provides power to a PoDL PD, connected via a link section consisting of a single ~~twisted conductor~~ conductor pair. ~~DTE powering is intended to provide a single 100BASE-T1 or 1000BASE-T1 device with a unified interface for both the reception and transmission of data as well as the power to operate.~~ (See IEEE Std 802.3, Clause 104.)

30. Management

30.3 Layer management for DTEs

30.3.2 PHY device managed object class

30.3.2.1 PHY device attributes

30.3.2.1.2 aPhyType

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “25GBASE-T” as follows:

25GBASE-T1	Clause 165 25 Gb/s PAM4
------------	-------------------------

30.3.2.1.3 aPhyTypeList

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “25GBASE-T” as follows:

25GBASE-T1	Clause 165 25 Gb/s PAM4
------------	-------------------------

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “25GBASE-T” as follows:

25GBASE-T1	Single balanced pair of conductors PHY as specified in Clause 165
------------	---

30.6 Management for link Auto-Negotiation

30.6.1 Auto-Negotiation managed object class

30.6.1.1 Auto-Negotiation attributes

30.6.1.1.5 aAutoNegLocalTechnologyAbility

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.6.1.1.5 after the entry for “25GBASE-T” as follows:

25GBASE-T1	25GBASE-T1 as specified in Clause 165
------------	---------------------------------------

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

45.2.1.7 PMA/PMD status 2 register (Register 1.8)

45.2.1.7.4 Transmit fault (1.8.11)

Insert a new row in Table 45–9 after the row for 25GBASE-T as follows (unchanged rows not shown):

Table 45–9—Transmit fault description location

PMA/PMD	Description location
...	
25GBASE-T1	165.4.2.2
...	

45.2.1.7.5 Receive fault (1.8.10)

Insert a new row in Table 45–10 after the row for 25GBASE-T as follows (unchanged rows not shown):

Table 45–10—Receive fault description location

PMA/PMD	Description location
...	
25GBASE-T1	165.4.2.3
...	

45.2.1.16 BASE-T1 PMA/PMD extended ability register (1.18)

Change the row for bits 1.18.15:7 in Table 45–19 and insert a new row below it as follows (unchanged rows not shown):

Table 45–19—BASE-T1 PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.18.15:8	Reserved	Value always 0	RO
1.18.7	25GBASE-T1 ability	1 = PMA/PMD is able to perform 25GBASE-T1 0 = PMA/PMD is not able to perform 25GBASE-T1	RO
...			

^aRO = Read only

Insert 45.2.1.16.a before 45.2.1.16.1 as follows:

45.2.1.16.a 25GBASE-T1 ability (1.18.7)

When read as a one, bit 1.18.7 indicates that the PMA/PMD is able to operate as a 25GBASE-T1 PMA type.

When read as a zero, bit 1.18.7 indicates that the PMA/PMD is not able to operate as a 25GBASE-T1 PMA type.

45.2.1.214 BASE-T1 PMA/PMD control register (Register 1.2100)

Change the row for bits 1.2100.3:0 in Table 45–178 as follows (unchanged rows not shown):

Table 45–178—BASE-T1 PMA/PMD control register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.2100.3:0	Type Selection	3 2 1 0 1 x x x = Reserved 0 1 1 1 = Reserved <u>0 1 1 1 = 25GBASE-T1</u> 0 1 1 0 = 10GBASE-T1 0 1 0 1 = 5GBASE-T1 0 1 0 0 = 2.5GBASE-T1 0 0 1 1 = 10BASE-T1S 0 0 1 0 = 10BASE-T1L 0 0 0 1 = 1000BASE-T1 0 0 0 0 = 100BASE-T1	R/W

^aRO = Read only, R/W = Read/Write

45.2.1.214.2 Type selection (1.2100.3:0)

Change 45.2.1.214.2 as follows:

Bits 1.2100.3:0 are used to set the mode of operation when Auto-Negotiation enable bit 7.512.12 is set to zero, or if Auto-Negotiation is not implemented. When these bits are set to 0000, the mode of operation is 100BASE-T1. When these bits are set to 0001, the mode of operation is 1000BASE-T1. When these bits are set to 0010, the mode of operation is 10BASE-T1L. When these bits are set to 0011, the mode of operation is 10BASE-T1S. When these bits are set to 0100, the mode of operation is 2.5GBASE-T1. When these bits are set to 0101, the mode of operation is 5GBASE-T1. When these bits are set to 0110, the mode of operation is 10GBASE-T1. When these bits are set to 0111, the mode of operation is 25GBASE-T1. These bits shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.

45.2.1.244 MultiGBASE-T1 training register (1.2311)

Change the row for bits 1.2311.12:11 in Table 45–206 as follows (unchanged rows not shown):

Table 45–206—MultiGBASE-T1 training register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.2311.12:11	Interleave request	12 11 0 0: L=1, no interleaving, default for 2.5GBASE-T1 0 1: L=2, RS-FEC interleaving factor two, default for 5GBASE-T1, undefined defined for 2.5GBASE-T1 1 0: L=4, RS-FEC interleaving factor four, default for 10GBASE-T1, undefined defined for 2.5GBASE-T1 and 5GBASE-T1 1 1: Reserved <u>L=8, RS-FEC interleaving factor eight, default for 25GBASE-T1, not defined for 2.5GBASE-T1, 5GBASE-T1, and 10GBASE-T1</u>	R/W
...			

^aRO = Read only, R/W = Read/Write

45.2.1.244.1 Interleave request (1.2311.12:11)

Change 45.2.1.244.1 as follows:

Bits 1.2311.12:11 control the Reed-Solomon interleave setting of this PHY. Reed-Solomon interleaving is described in [149.3.2.2.15](#) and [165.3.2.2.15](#). This is communicated to the link partner via Infobfields as specified in [149.4.2.4.5](#) and [165.4.2.4.5](#).

The values of L = 2, ~~and L = 4, and L = 8~~ are not defined for 2.5GBASE-T1 PHYs, ~~and the values of L = 4 and L = 8 are~~ is not defined for 5GBASE-T1 PHYs, and the value of L = 8 is not defined for 10GBASE-T1 PHYs. If bits 1.2311.12:11 are set to these ~~undefined~~ values, the PHY ~~will~~ communicates these values to the link partner, but the requested interleaver depth is out of scope of this standard and may not be supported by the link partner.

45.2.1.245 MultiGBASE-T1 link partner training register (1.2312)

Change the row for bits 1.2312.12:11 in Table 45–207 as follows (unchanged rows not shown):

Table 45–207—MultiGBASE-T1 link partner training register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.2312.12:11	Link partner interleave request	12 11 0 0: L=1, no interleaving, default for 2.5GBASE-T1 0 1: L=2, RS-FEC interleaving factor two, default for 5GBASE-T1, undefined defined for 2.5GBASE-T1 1 0: L=4, RS-FEC interleaving factor four, default for 10GBASE-T1, undefined defined for 2.5GBASE-T1 and 5GBASE-T1 1 1: Reserved L=8, RS-FEC interleaving factor eight, default for 25GBASE-T1, not defined for 2.5GBASE-T1, 5GBASE-T1, and 10GBASE-T1	RO
...			

^aRO = Read only

45.2.1.245.1 Link partner interleave request (1.2312.12:11)

Change 45.2.1.245.1 as follows:

Bits 1.2312.12:11 contain the Reed-Solomon interleave setting requested by the link partner. Reed-Solomon interleaving is described in [149.3.2.2.15](#) and [165.3.2.2.15](#). This is communicated by the link partner via Infocfields as specified in [149.4.2.4.5](#) and [165.4.2.4.5](#).

The values of L = 2, ~~and L = 4, and L = 8~~ are not defined for 2.5GBASE-T1 PHYs, ~~and the values of L = 4 and L = 8~~ are not defined for 5GBASE-T1 PHYs, and the value of L = 8 is not defined for 10GBASE-T1 PHYs. Bits 1.2312.12:11 ~~will~~ indicate whatever value is received from the link partner, but if the ~~undefined~~ not defined values are received, the requested interleaver depth is out of scope of this standard and may not be supported by the local PHY.

45.2.1.246 MultiGBASE-T1 test mode control register (1.2313)

45.2.1.246.1 Test mode control (1.2313.15:13)

Change 45.2.1.246.1 as follows:

Transmitter test mode operations defined by bits 1.2313.15:13, are described in [149.5.1](#) ~~and~~ [Table 149–17](#), [165.5.1](#), and [Table 165–12](#). The default value for bits 1.2313.15:13 is zero.

45.2.1.246.2 Local transmitter precoder override (1.2313.11)

Change 45.2.1.246.2 as follows:

When bit 1.2313.11 is set to one, the local transmitter's precoder shall be controlled by the value of bits 1.2313.10:9, and the precoder requested by the link partner in PrecodeSel shall be ignored. When bit

1.2313.11 is set to zero, the transmitter shall ignore bits 1.2313.10:9, and the precoder is set according to the value of PrecoderSel received from the link partner as specified in [149.3.2.2.20](#) and [165.3.2.2.20](#). The default value of 1.2313.11 is zero.

45.2.1.246.3 Local transmit precoder setting (1.2313.10:9)

Change 45.2.1.246.3 as follows:

When bit 1.2313.11 is set to one, bits 1.2313.10:9 control the precoder setting of the local transmitter, as defined in [149.3.2.2.20](#) and [165.3.2.2.20](#) in the variable precoder_type.

45.2.1.246.4 Jitter test control (1.2313.1:0)

When the transmitter is in test mode 2, bits 1.2313.1:0 control the pattern of the jitter test signal. When the transmitter is not in test mode 2, the setting of bits 1.2313.1:0 has no effect. A value of 00 transmits a square wave from the transmitter, a value of 01 transmits the JP03A pattern, and a value of 10 transmits the JP03B pattern. See [149.5.2.3.1](#) and [149.5.2.3.2](#), [165.5.3.3.1](#), and [165.5.3.3.2](#) for more information.

45.2.3 PCS Registers

45.2.3.87 MultiGBASE-T1 PCS status 2 register (Register 3.2324)

45.2.3.87.2 PCS high RFER (3.2324.9)

Change 45.2.3.87.2 as follows:

When read as a one, bit 3.2324.9 indicates that the MultiGBASE-T1 PCS receiver is detecting 16 or more RS-FEC errored blocks within 88 RS-FEC frames in 312 500 bit times (one rfer_timer interval). When read as a zero, bit 3.2324.9 indicates that the MultiGBASE-T1 PCS is detecting fewer than 16 RS-FEC errored blocks within 88 RS-FEC frames in 312 500 bit times. Bit 3.2324.9 is a reflection of the state of the hi_rfer variable defined in [149.3.7.2.2](#) ~~149.3.8.1~~.

78. Energy-Efficient Ethernet (EEE)

78.1 Overview

78.1.4 PHY types optionally supporting EEE

Insert a row for 25GBASE-T1 after the row for 25GBASE-CR in Table 78–1 as follows (unchanged rows not shown):

Table 78–1—Clauses associated with each PHY or interface type

PHY or interface type	Clause
...	
25GBASE-T1	165
...	

78.2 LPI mode timing parameters description

Insert a row for 25GBASE-T1 above the row for 25GBASE-T in Table 78–2 as follows:

Table 78–2—Summary of the key EEE parameters for supported PHYs or interfaces

PHY or interface type	T_s (μ s)		T_q (μ s)		T_r (μ s)	
	Min	Max	Min	Max	Min	Max
...						
25GBASE-T1	5.3248	5.3248	31.616	31.616	0.3328	0.3328
...						

78.3 Capabilities Negotiation

Insert a new sentence at the end of the first paragraph of 78.3 as follows:

The EEE capability for 25GBASE-T1 is advertised during link training according to 165.4.2.4.

Change the last sentence of the second paragraph of 78.3 as follows:

The same applies to 2.5GBASE-T, 2.5GBASE-T1, 5GBASE-T, 5GBASE-T1, 10GBASE-T1, 25GBASE-T, 25GBASE-T1, and 40GBASE-T except that the EEE capabilities are exchanged and resolved during link training instead of during Auto-Negotiation.

78.5 Communication link access latency

Insert a row in Table 78–4 for 25GBASE-T1 above the row for 25GBASE-T as follows (unchanged rows not shown):

Table 78–4—Summary of the LPI timing parameters for supported PHYs or interfaces

PHY or interface type	Case	$T_{w_sys_tx}$ (min) (μs)	T_{w_phy} (min) (μs)	$T_{phy_shrink_tx}$ (max) (μs)	$T_{phy_shrink_rx}$ (max) (μs)	$T_{w_sys_rx}$ (min) (μs)
...						
25GBASE-T1	Case-1	15.9744	15.9744	10.6496	0	5.325
	Case-2	9.318	9.318	3.994	0	5.325
	Case-3	45.2608	45.2608	39.936	0	5.325
	Case-4	38.6	38.6	33.28	0	5.325
...						

98. Auto-Negotiation for single differential-pair media

98.5 Detailed functions and state diagrams

98.5.1 State diagram variables

Insert a new item at the end of the dashed list after the first paragraph of 98.5.1 as follows:

- 25GigT1; represents that the 25GBASE-T1 PMA is the signal source.

104. Power over Data Lines (PoDL) of Single-Pair Ethernet

104.1 Overview

104.1.3 PoDL system types

Change the last sentence of the second paragraph in 104.1.3 as follows:

A Type F PSE and Type F PD are compatible with 2.5GBASE-T1, 5GBASE-T1, ~~and~~ 10GBASE-T1, and
25GBASE-T1 PHYs.

105. Introduction to 25 Gb/s networks

105.1 Overview

105.1.2 Relationship of 25 Gigabit Ethernet to the ISO OSI reference model

Replace Figure 105–1 as amended by IEEE Std 802.3cz-2023 with the following figure, which adds 25GBASE-T1 and NOTE 2:

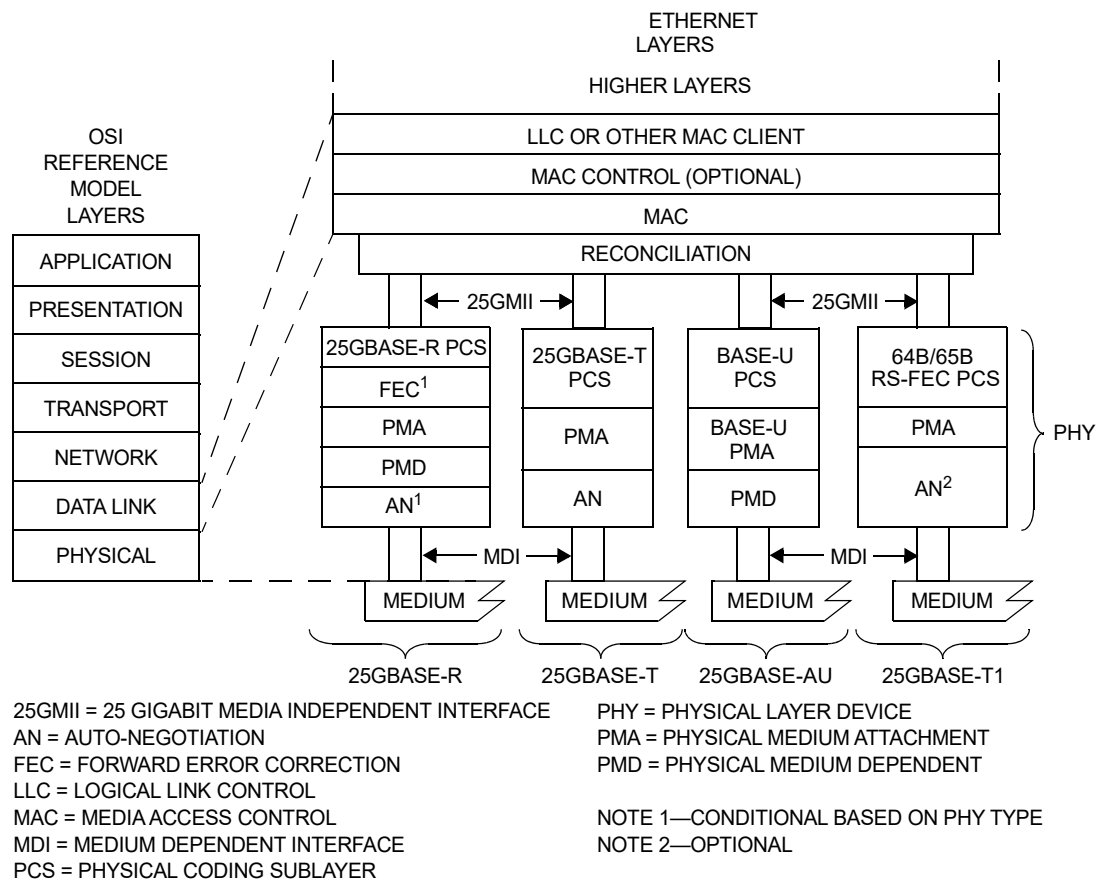


Figure 105–1—Architectural positioning of 25 Gigabit Ethernet

Insert a new item e) at the end of the lettered list in 105.1.2 as follows:

- e) The MDI as specified in Clause 165 for 25GBASE-T1 uses a single-lane data path.

105.1.3 Nomenclature

Insert a new paragraph after the fourth paragraph of 105.1.3 that begins “25GBASE-AU represents” (as inserted by IEEE Std 802.3cz-2023) as follows:

25GBASE-T1 represents Physical Layer devices using Clause 165 Physical Coding Sublayer (PCS) and Physical Medium Attachment (PMA) sublayer, for data communication at 25 Gb/s over a point-to-point single balanced pair of conductors. 25GBASE-T1 uses Reed-Solomon FEC and PAM4 modulation.

Insert a row for 25GBASE-T1 after the row for 25GBASE-CR in Table 105–1 (as amended by IEEE Std 802.3cz-2023) as follows (unchanged rows not shown):

Table 105–1—25 Gb/s PHYs

Name	Description
...	
25GBASE-T1	25 Gb/s PHY using Reed-Solomon encoding and PAM4 modulation over a single balanced pair of conductors (see Clause 165).
...	

105.2 Physical Layer signaling systems

Change Table 105–2 (as amended by IEEE Std 802.3cz-2023) as follows:

Table 105–2—Nomenclature and clause correlation, 25 Gb/s Ethernet PHYs

Nomenclature	Clause/Annex																	
	28	73	98	74	78	106	107	108	109	109A	109B	110	111	112	113	114	165	166
	Auto-Negotiation	Auto-Negotiation	Auto-Negotiation	BASE-R FEC	EEE	RS	25GMII	25GBASE-R PCS	25GBASE-R RS-FEC	PMA	25GAUI C2C	25GAUI C2M	25GBASE-CR PMD	25GBASE-CR-S PMD	25GBASE-KR PMD	25GBASE-KR-S PMD	25GBASE-SR PMD	25GBASE-T PCS/PMA
25GBASE-KR-S	M ^a			M	O ^a	M	O	M		M	O					M		
25GBASE-KR		M		M	O	M	O	M	M	O				M				
25GBASE-CR-S		M		M	O	M	O	M		M	O		M					
25GBASE-CR		M		M	O	M	O	M	M	O		M						
25GBASE-T1			O		O	M	O										M	
25GBASE-T	M				O	M	O									M		
25GBASE-AU					O	M	O											M
25GBASE-SR					O	M	O	M	M	M	O	O				M		
25GBASE-LR					O	M	O	M	M	M	O	O					M	
25GBASE-ER					O	M	O	M	M	M	O	O					M	

^a O = Optional, M = Mandatory

105.3 Summary of 25 Gigabit Ethernet sublayers

105.3.2 Physical Coding Sublayer (PCS)

Insert a new paragraph at the end of 105.3.2 as follows:

A 25GBASE-T1 PHY uses the PCS specified in Clause 165. The 25GBASE-T1 PCS performs encoding of data from the 25GMII to 64B/65B RS-FEC code blocks and PAM4 modulation. It then transfers the symbols to the PMA and performs error correction and decoding of PAM4 symbols from the PMA and transfers the decoded data to the 25GMII.

105.3.4 Physical Medium Attachment (PMA) sublayer

Insert a new paragraph at the end of 105.3.4 as follows:

A 25GBASE-T1 PHY uses the PMA specified in Clause 165. The PMA provides for full duplex communications over a single balanced pair of conductors.

105.3.6 Auto-Negotiation (AN)

Insert new paragraph at the end of 105.3.6 as follows:

Clause 98 Auto-Negotiation may be used by the 25GBASE-T1 PHY, but is not required.

105.5 Delay constraints

Insert new rows for 25GBASE-T1 after the row for 25GBASE-T in Table 105–3 (as amended by IEEE Std 802.3cz-2023) as follows (unchanged rows not shown):

Table 105–3—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
...				
25GBASE-T1 PHY	25 600	50	1024	No interleave, see 165.10.
	36 864	72	1474.56	2x interleave, see 165.10.
	58 880	115	2355.2	4x interleave, see 165.10.
	102 400	200	4096	8x interleave, see 165.10.
...				

^a 1 bit time (BT) is equal to 40 ps. (See 1.4.215 for the definition of bit time.)

^b 1 pause_quantum is equal to 20.48 ns. (See 31B.2 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

105.7 Protocol implementation conformance statement (PICS) proforma

Change the first paragraph of 105.7 (as amended by IEEE Std 802.3cz-2023) as follows:

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 45, [Clause 73](#), [Clause 74](#), [Clause 106](#) through [Clause 112](#), [Clause 114](#), [Clause 166](#), [Clause 165](#), and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

149. Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 2.5GBASE-T1, 5GBASE-T1, and 10GBASE-T1

149.3 Physical Coding Sublayer (PCS) functions

149.3.7 Detailed functions and state diagrams

149.3.7.2 State diagram parameters

149.3.7.2.2 Variables

Change the definition of “hi_rfer” in 149.3.7.2.2 as follows:

hi_rfer

Boolean variable that is asserted TRUE when the rfer_cnt reaches 16 errors in one RFRX_CNT_LIMIT ~~rfer_timer~~ interval.

Insert new Clause 165 after Clause 164 as follows:

165. Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 25GBASE-T1

165.1 Overview

This clause defines the type 25GBASE-T1 Physical Coding Sublayer (PCS) as well as the 25GBASE-T1 Physical Medium Attachment (PMA) sublayer. Together, the corresponding PCS, PMA, and optional Auto-Negotiation sublayers comprise a 25GBASE-T1 Physical Layer device (PHY). Provided in this clause are functional and electrical specifications for the type 25GBASE-T1 PCS and PMA.

A 25GBASE-T1 PHY is intended to be operated over a single balanced pair of conductors. The link segment specifications defined in 165.7 were derived from automotive requirements, but can also be used for non-automotive applications. The conductors supporting the operation of a 25GBASE-T1 PHY are defined in terms of performance requirements between the Medium Dependent Interfaces (MDIs) allowing implementers to provide their own conductors to operate a 25GBASE-T1 PHY as long as the normative requirements included in 165.7 are met.

This clause also specifies an optional Energy-Efficient Ethernet (EEE) capability. A 25GBASE-T1 PHY that supports this capability may enter a Low Power Idle (LPI) mode of operation during periods of low link utilization as described in Clause 78.

165.1.1 Nomenclature

The 25GBASE-T1 PHY defined in this clause is a member of the MultiGBASE-T1 PHY set, as defined in 1.4.407. PHYs in this set share control registers and characteristics. The term ‘MultiGBASE-T1’, when used in this clause, refers to registers and features common to PHYs in this set.

165.1.2 Relationship of 25GBASE-T1 to other standards

The relationship between a 25GBASE-T1 PHY, the ISO Open Systems Interconnection (OSI) reference model, and the IEEE 802.3 Ethernet model is shown in Figure 165–1. The PHY sublayers shown shaded in Figure 165–1 are specified in this clause, while the optional Auto-Negotiation sublayer for a 25GBASE-T1 PHY is defined in Clause 98. The 25GBASE-T1 PHY connects one [Clause 4](#) Media Access Control (MAC) layer to the medium.

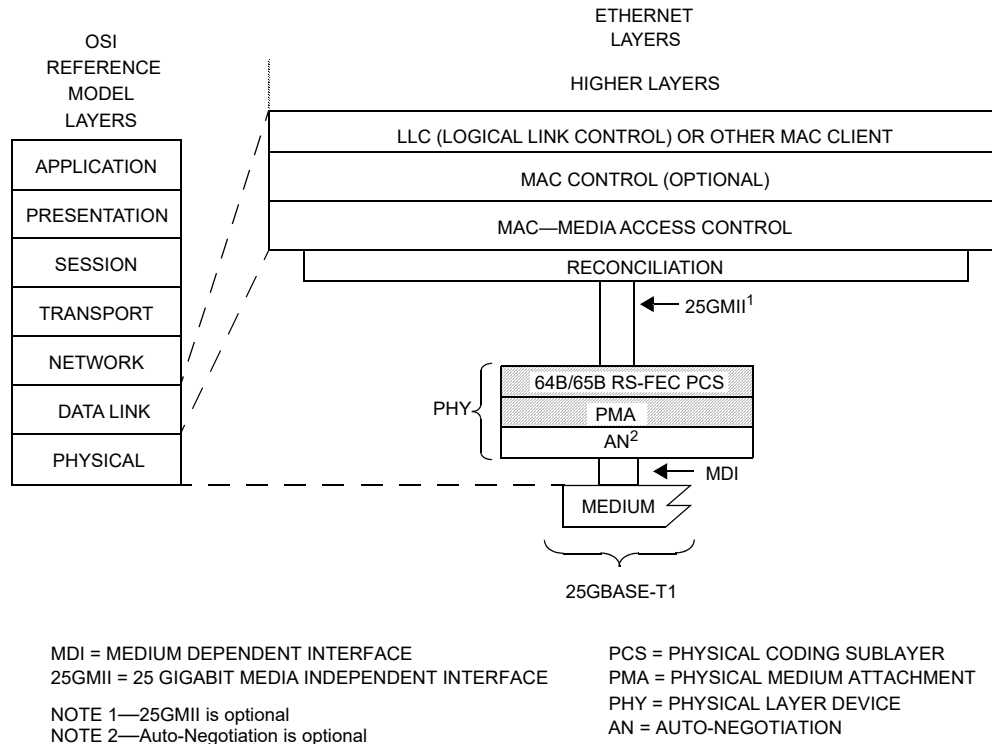


Figure 165-1—Relationship of the 25GBASE-T1 PHY to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

165.1.3 Operation of 25GBASE-T1

The 25GBASE-T1 PHY operates using full-duplex communications over a shielded balanced pair of conductors with an effective rate of 25 Gb/s in each direction simultaneously while meeting the requirements (EMC, temperature, etc.) of automotive environments. The PHY supports operation on an automotive link segment supporting up to two in-line connectors using a single shielded balanced pair of conductors for up to at least 11 m. The 25GBASE-T1 PHY utilizes 4-level pulse amplitude modulation (PAM4) transmitted at 14.0625 GBd. A 33-bit scrambler is used to improve the EMC performance. 25GMII TXD and TXC are encoded together using 64B/65B encoding. To maintain a frame loss ratio (FLR) equivalent to a bit error ratio (BER) of less than or equal to 10^{-12} , a 25GBASE-T1 PHY adds 900 bits of Reed-Solomon forward error correction (RS-FEC) parity to each group of 130 64B/65B blocks. The PAM4 mapping, scrambler, RS-FEC, interleaver, and PAM4 encoder/decoder are all contained in the PCS (see 165.3).

Auto-Negotiation (Clause 98) may optionally be used by 25GBASE-T1 devices to detect the abilities (modes of operation) supported by the device at the other end of a link segment, determine common abilities, and configure for normal operation. Auto-Negotiation is performed upon link startup through the use of half-duplex differential Manchester encoding. The implementation of the Auto-Negotiation function is optional. If Auto-Negotiation is implemented, it shall meet the requirements of Clause 98.

A 25GBASE-T1 PHY shall be capable of operating as MASTER or SLAVE, per runtime configuration. A MASTER PHY uses a local clock to determine the timing of transmitter operations. A SLAVE PHY recovers the clock from the received signal and uses it to determine the timing of transmitter operations. When Auto-Negotiation is used, the MASTER-SLAVE relationship between two devices sharing a link segment is established during Auto-Negotiation (see Clause 98). If Auto-Negotiation is not used, a MASTER-SLAVE relationship shall be established by management or hardware configuration of a PHY,

and the MASTER and SLAVE are synchronized by the PHY Link Synchronization function in the PHY (see 165.4.2.6).

NOTE—Annex K defines optional alternative terminology for “master” and “slave”.

A 25GBASE-T1 PHY may optionally support Energy-Efficient Ethernet (see Clause 78) and advertise the EEE capability as described in 165.4.2.4.5. The EEE capability is a mechanism by which a 25GBASE-T1 PHY adapts signaling during periods of low link utilization to provide opportunities for reduced power consumption independently for each direction of the link.

The 25GBASE-T1 PHY may optionally support the PCS-based operations, administration, and maintenance (OAM). The OAM is useful for monitoring link operation by exchanging PHY link health status and messages. The OAM information for 25GBASE-T1 is exchanged between two 25GBASE-T1 PHYs by interspersing it with the 25 Gb/s Ethernet data stream. The OAM for 25GBASE-T1 is specified in 165.3.9, and the 25GBASE-T1 PHY advertises the OAM for 25GBASE-T1 capability as described in 165.4.2.4.5.

The 25GBASE-T1 PMA couples messages from the PCS to the MDI and provides clock recovery, link management, and PHY Control functions. The PMA provides full duplex communications at 14.0625 GBd for 25GBASE-T1 over a link segment meeting the specifications of 165.7. PMA functionality is described in 165.4. The MDI is specified in 165.8. Figure 165–2 shows the functional block diagram.

165.1.3.1 Physical Coding Sublayer (PCS)

The PCS couples a 25 Gigabit Media Independent Interface (25GMII), as described in Clause 106, to the 25GBASE-T1 Physical Medium Attachment (PMA) sublayer.

In addition to the normal mode of operation, the PCS supports a training mode. Furthermore, the PCS contains a management interface.

In the transmit direction, in normal mode, the PCS receives eight 25GMII data octets provided by two consecutive transfers on the 25GMII service interface on TXD<31:0> and groups them into 64-bit blocks with the 64-bit block boundaries aligned with the boundary of the two 25GMII transfers. Each group of eight octets along with the data/control indications is transcoded into a 65-bit block. These 65-bit blocks are then aggregated into groups of 130 blocks. The contents of each group are contained in a vector tx_group130x65B. Next, a 10-bit OAM field is appended to form an 8460-bit RS-FEC input frame. A number, L (see 165.3.2.2.15), of these 8460-bit blocks are formed into an RS-FEC input superframe, then encoded by the RS-FEC(936,846) over $GF(2^{10})$ and the round-robin interleaving as described in 165.3.2.2.16. The RS-FEC output superframe consists of $L \times 9360$ bits. The duration of the superframe is $L \times 332.8$ ns. Finally these bits are exclusive OR'd with a degree 33 scrambler to create the payload. PCS transmit functions are described in 165.3.2.2.

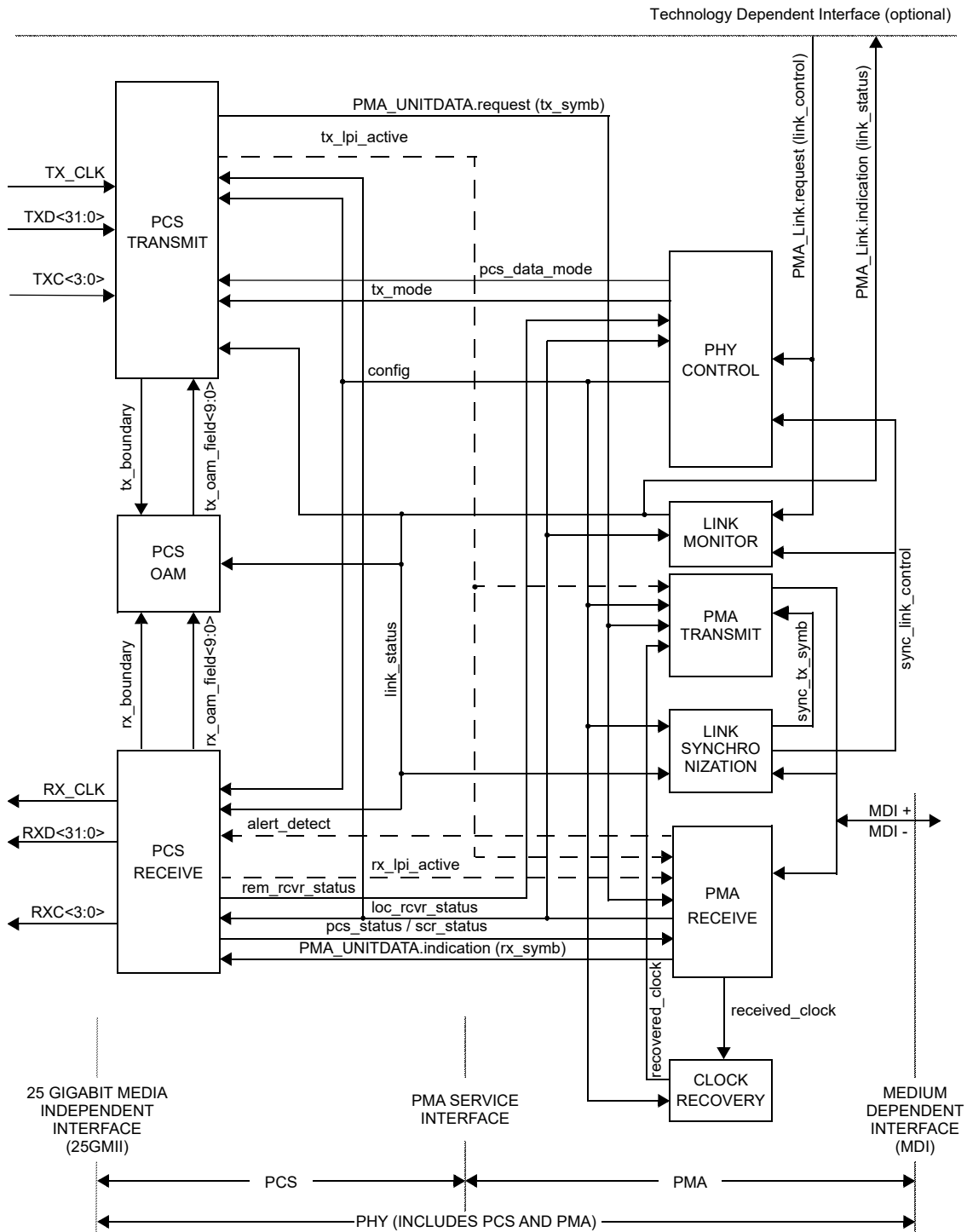
tx_group130x65B<8449:0> is defined as:

$$\text{tx_group130x65B}\langle 65 \times i + j \rangle = \text{tx_coded}_i\langle j \rangle$$

where $i = 0$ to 129, $j = 0$ to 64, and tx_coded _{i} <64:0> is the i th 64B/65B block where tx_coded₀<64:0> is the first block transmitted.

In the training mode (see 165.4.2.4), the PCS transmits and receives PAM2 training frames to synchronize to the PHY frame and exchanges EEE and OAM capabilities.

Details of the PCS functions and state diagrams are covered in 165.3. The interface to the PMA is an abstract message-passing interface specified in 165.4.



NOTE 1—The recovered_clock arc is shown to indicate delivery of the received clock signal back to PMA TRANSMIT for loop timing.

NOTE 2—Signals and functions shown with dashed lines are optional.

NOTE 3—The optional AN sublayer is not shown between the PMA sublayer and the MDI.

Figure 165–2—Functional block diagram

165.1.3.2 Physical Medium Attachment (PMA) sublayer

The PMA couples messages from the PCS service interface onto the single balanced pair of conductors via the Medium Dependent Interface (MDI) and provides the link management and PHY Control functions. The PMA provides full duplex communications at 14.0625 GBd.

The PMA PHY Control function generates signals that control the PCS and PMA sublayer operations. PHY Control is enabled following the completion of Auto-Negotiation or PHY Link Synchronization and provides the startup functions required for successful operation. It determines whether the PHY operates in a disabled state, a training state, or a data state where MAC frames can be exchanged between the link partners.

The Link Monitor determines the status of the underlying link channel and communicates this status to other functional blocks. A failure of the receive channel causes the data mode operation to stop and Auto-Negotiation or Link Synchronization to restart.

PMA functions and state diagrams are specified in 165.4. The electrical parameters of the PMA (i.e., test modes and electrical specifications for the transmitter and receiver) are specified in 165.5.

165.1.3.3 EEE Capability

A 25GBASE-T1 PHY may optionally support the EEE capability, as described in 78.3. The EEE capability is a mechanism by which a 25GBASE-T1 PHY adapts signaling during periods of low link utilization to provide opportunities for reduced power consumption. A PHY can enter the LPI mode of operation after completing training. Each direction of the full duplex link is able to enter and exit the LPI mode independently, supporting symmetric and asymmetric LPI operation. This allows power savings when only one side of the full duplex link is in a period of low utilization. The transition to or from LPI mode is not expected to cause any MAC frames to be lost or corrupted.

Support for the EEE capability is advertised in the Infobfield (Octet 10 bit 6) during link startup. Transitions to and from the LPI transmit mode are controlled via 25GMII signaling. Transitions to and from the LPI receive mode are controlled by the link partner using sleep and wake signaling.

When the PHY Health status received from the link partner indicates that LPI is insufficient to maintain PHY SNR, the PHY may temporarily exit LPI mode and send idles.

The PCS 64B/65B Transmit state diagram in [Figure 149–16](#) and [Figure 149–17](#) includes additional states for EEE. The PCS 64B/65B Receive state diagram in [Figure 149–18](#) and [Figure 149–19](#) includes additional states for EEE. The EEE transmit state diagram is contained in the PCS Transmit function and is specified in [Figure 165–14](#).

165.1.3.4 Link Synchronization

The Link Synchronization function is used when Auto-Negotiation is disabled or not implemented to detect the presence of the link partner, time and control link failure, and act as the data source for the PHY Control state diagram. Link Synchronization operates in a half-duplex fashion. Link Synchronization is defined in 165.4.2.6.

165.1.4 Signaling

25GBASE-T1 signaling is performed by the PCS generating continuous code-group sequences that the PMA transmits over a link segment meeting the specifications of 165.7. The signaling scheme achieves a number of objectives including:

- a) Forward error correction (FEC) coded symbol mapping for data.
- b) Algorithmic mapping from TXD<31:0> and TXC<3:0> to PAM4 symbols in the transmit path.
- c) Algorithmic mapping from the received signal on the MDI port to RXD<31:0> and RXC<3:0>.
- d) Uncorrelated symbols in the transmitted symbol stream.
- e) No correlation between symbol streams traveling both directions.
- f) Block framing and other control signals.
- g) Ability to signal the status of the local receiver to the remote PHY to indicate that the local receiver is not operating reliably and requires retraining.
- h) Ability to automatically detect and correct for incorrect polarity in the connection.
- i) Optionally, ability to support refresh, quiet, and alert signaling during LPI operation.

The PHY may operate in three basic modes: the normal data mode, the training mode, or an optional LPI mode.

In normal mode, the PCS generates a continuous stream of PAM4 symbols that are transmitted via the PMA at one of four voltage levels. In training mode, the PCS is directed to generate only PAM2 symbols for transmission by the PMA. (See Figure 165–22.)

A PHY may also support the EEE capability as described in 165.1.3.3. Transitions to the LPI mode are supported after reaching normal mode.

165.1.5 Interfaces

All 25GBASE-T1 PHY implementations are compatible at the MDI and at the 25GMII, if implemented. Implementation of the 25GMII is optional. Designers are free to implement circuitry within the PCS and PMA in an application-dependent manner provided that the MDI and 25GMII (if the 25GMII is implemented) specifications are met. System operation from the perspective of signals at the MDI and management objects are identical whether the 25GMII is implemented or not.

165.1.6 Conventions in this clause

The body of this clause contains state diagrams, including definitions of variables, constants, and functions. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

The notation used in the state diagrams follows the conventions of state diagrams as described in 21.5, along with the extensions described in 145.2.5.2. State diagram timers follow the conventions of 14.2.3.2. The notation ++ after a counter or integer variable indicates that its value is to be incremented. A plus symbol within a circle denotes an XOR operation.

Default initializations, unless specified, are left to the implementer.

165.2 25GBASE-T1 service primitives and interfaces

25GBASE-T1 transfers data and control information across the following four service interfaces:

- a) 25 Gigabit Media Independent Interface (25GMII)
- b) Technology Dependent Interface

- c) PMA service interface
- d) Medium Dependent Interface (MDI)

The 25GMII is specified in [Clause 106](#); the Technology Dependent Interface is specified in [98.4](#). The PMA service interface is defined in 165.2.2 and the MDI is defined in 165.8.

165.2.1 Technology Dependent Interface

25GBASE-T1 uses the following service primitives to exchange status indications and control signals across the Technology Dependent Interface, required in a PHY that implements Auto-Negotiation, as specified in [98.4](#):

```
PMA_LINK.request(link_control)
PMA_LINK.indication(link_status)
```

165.2.1.1 PMA_LINK.request

This primitive allows the Auto-Negotiation to enable and disable operation of the PMA, as specified in [98.4.2](#).

165.2.1.1.1 Semantics of the primitive

```
PMA_LINK.request(link_control)
```

The link_control parameter can take on one of two values: DISABLE or ENABLE.

DISABLE Used by the Auto-Negotiation function to disable the PHY.

ENABLE Used by the Auto-Negotiation function to enable the PHY.

165.2.1.1.2 When generated

Auto-Negotiation generates this primitive to indicate a change in link_control as described in [98.4](#).

165.2.1.1.3 Effect of receipt

This primitive affects the operation of the PMA Link Monitor function as defined in 165.4.2.5, the PMA PHY Control function as defined in 165.4.2.4, and the PMA Receive function defined in 165.4.2.3.

165.2.1.2 PMA_LINK.indication

This primitive is generated by the PMA to indicate the status of the underlying medium as specified in [98.4.1](#). This primitive informs the PCS, PMA PHY Control function, and the Auto-Negotiation functions about the status of the underlying link.

165.2.1.2.1 Semantics of the primitive

```
PMA_LINK.indication(link_status)
```

The link_status parameter can take on one of two values: FAIL or OK.

FAIL No valid link established.

OK The Link Monitor function indicates that a valid 25GBASE-T1 link is established. Reliable reception of signals transmitted from the remote PHY is possible.

165.2.1.2.2 When generated

The PMA generates this primitive to indicate a change in link_status in compliance with the state diagram given in Figure 165–23.

165.2.1.2.3 Effect of receipt

The effect of receipt of this primitive is specified in 98.4.1.

165.2.2 PMA service interface

25GBASE-T1 uses the following service primitives to exchange symbol vectors, status indications, and control signals across the service interfaces:

```
PMA_TXMODE.indication(tx_mode)
PMA_CONFIG.indication(config)
PMA_UNITDATA.request(tx_symb)
PMA_UNITDATA.indication(rx_symb)
PMA_SCRSTATUS.request(scr_status)
PMA_PCSSTATUS.request(pcs_status)
PMA_RXSTATUS.indication(loc_rcvr_status)
PMA_REMRXSTATUS.request(rem_rcvr_status)
PMA_PCSDATAMODE.indication(pcs_data_mode)
```

The use of these primitives is illustrated in Figure 165–3. Connections from the management interface (signals MDC and MDIO) to the sublayers are pervasive and are not shown in Figure 165–3.

An EEE-capable PHY additionally supports the following service primitives:

```
PMA_PCS_RX_LPI_STATUS.request(rx_lpi_active)
PMA_PCS_TX_LPI_STATUS.request(tx_lpi_active)
PMA_ALERTDETECT.indication(alert_detect)
```

165.2.2.1 PMA_TXMODE.indication

The transmitter in a 25GBASE-T1 link normally sends over the MDI symbols that represent a 25GMII data stream with framing, scrambling and encoding of data, control information, or idles.

165.2.2.1.1 Semantics of the primitive

```
PMA_TXMODE.indication(tx_mode)
```

PMA_TXMODE.indication specifies to PCS Transmit via the parameter tx_mode what sequence of symbols the PCS should be transmitting. The parameter tx_mode can take on one of the following values of the form:

- | | |
|--------|--|
| SEND_N | This value is continuously asserted during transmission of sequences of symbols representing a 25GMII data stream in the data mode. |
| SEND_T | This value is continuously asserted in the case that transmission of sequences of symbols representing the training mode is to take place. |
| SEND_Z | This value is continuously asserted in the case that transmission of zeros is required. |

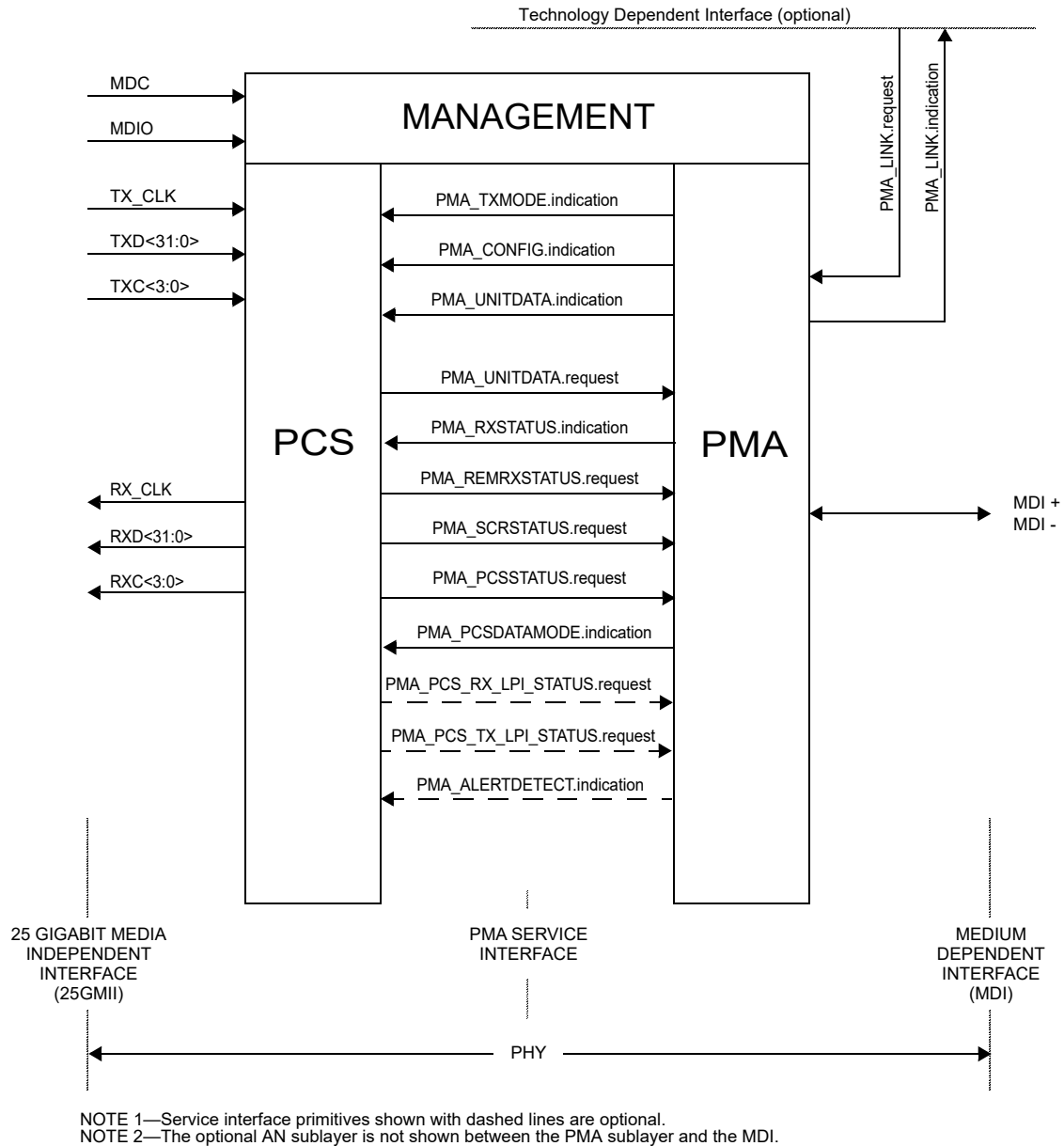


Figure 165-3—25GBASE-T1 service interfaces

165.2.2.1.2 When generated

The PMA PHY Control function generates PMA_TXMODE.indication messages to indicate a change in tx_mode.

165.2.2.1.3 Effect of receipt

Upon receipt of this primitive, the PCS performs its transmit function as described in 165.3.2.2.

165.2.2.2 PMA_CONFIG.indication

Each PHY in a 25GBASE-T1 link is capable of operating as a MASTER PHY and as a SLAVE PHY. If the Auto-Negotiation process is enabled, PMA_CONFIG MASTER-SLAVE configuration is determined during Auto-Negotiation (see Clause 98) and the result is provided to the PMA. If the Auto-Negotiation process is not implemented or not enabled, PMA_CONFIG MASTER-SLAVE configuration is predetermined to be MASTER or SLAVE via management control during initialization or via default hardware setup.

165.2.2.2.1 Semantics of the primitive

PMA_CONFIG.indication(config)

PMA_CONFIG.indication specifies to the PCS and PMA Transmit via the parameter config whether the PHY operates as a MASTER PHY or as a SLAVE PHY. The parameter config can take on one of the following two values of the form:

MASTER This value is continuously asserted when the PHY operates as a MASTER PHY.

SLAVE This value is continuously asserted when the PHY operates as a SLAVE PHY.

165.2.2.2.2 When generated

PMA generates PMA_CONFIG.indication messages to indicate a change in configuration.

165.2.2.2.3 Effect of receipt

PCS and PMA Clock Recovery perform their functions in MASTER or SLAVE configuration according to the value assumed by the parameter config.

165.2.2.3 PMA_UNITDATA.request

This primitive defines the transfer of symbols in the form of the tx_symb parameter from the PCS to the PMA. The symbols are obtained in the PCS Transmit function using the encoding rules defined in 165.3.2.2 to represent 25GMII data and control streams or other sequences.

165.2.2.3.1 Semantics of the primitive

PMA_UNITDATA.request(tx_symb)

During transmission, the PMA_UNITDATA.request simultaneously conveys to the PMA via the parameter tx_symb the value of the symbols to be sent over the MDI. The tx_symb may take on one of the following values:

$\{-1, -1/3, +1/3, +1\}$ in normal operation.

0 when zeros are to be transmitted in the following two cases:

- 1) when PMA_TXMODE.indication is SEND_Z during PMA training, and
- 2) after data mode is reached, the transmit function is in the LPI transmit mode, and lpi_tx_mode is QUIET.

165.2.2.3.2 When generated

The PCS generates PMA_UNITDATA.request(tx_symb) synchronously with every transmit clock cycle.

165.2.2.3.3 Effect of receipt

Upon receipt of this primitive, the PMA transmits on the MDI the signals corresponding to the indicated symbols processed to conform to 165.5.3.

165.2.2.4 PMA_UNITDATA.indication

This primitive defines the transfer of symbols in the form of the rx_symb parameter from the PMA to the PCS.

165.2.2.4.1 Semantics of the primitive

PMA_UNITDATA.indication(rx_symb)

During reception the PMA_UNITDATA.indication conveys to the PCS via the parameter rx_symb the value of symbols detected on the MDI during each cycle of the recovered clock.

165.2.2.4.2 When generated

The PMA generates PMA_UNITDATA.indication(rx_symb) messages synchronously for every symbol received at the MDI. The nominal rate of the PMA_UNITDATA.indication primitive is 14.0625 GHz for 25GBASE-T1; as governed by the recovered clock.

165.2.2.4.3 Effect of receipt

The effect of receipt of this primitive is unspecified.

165.2.2.5 PMA_SCRSTATUS.request

This primitive is generated by PCS Receive to communicate the status of the descrambler for the local PHY. The parameter scr_status conveys to the PMA Receive function the information that the training mode descrambler has achieved synchronization.

165.2.2.5.1 Semantics of the primitive

PMA_SCRSTATUS.request(scr_status)

The scr_status parameter can take on one of two values of the form:

OK The training mode descrambler has achieved synchronization.

NOT_OK The training mode descrambler is not synchronized.

165.2.2.5.2 When generated

PCS Receive generates PMA_SCRSTATUS.request messages to indicate a change in scr_status.

165.2.2.5.3 Effect of receipt

The effect of receipt of this primitive is specified in 165.4.2.3 and 165.4.2.4.

165.2.2.6 PMA_PCSSTATUS.request

This primitive is generated by PCS Receive to indicate the fully operational state of the PCS for the local PHY. The parameter `pcs_status` conveys to the PMA Receive function the information that the PCS is operating reliably in the data mode.

165.2.2.6.1 Semantics of the primitive

PMA_PCSSTATUS.request(`pcs_status`)

The `pcs_status` parameter can take on one of two values of the form:

- OK The PCS is operating reliably in the data mode.
- NOT_OK The PCS is not operating reliably in the data mode.

165.2.2.6.2 When generated

PCS Receive generates PMA_PCSSTATUS.request messages to indicate a change in `pcs_status`.

165.2.2.6.3 Effect of receipt

The effect of receipt of this primitive is specified in 165.4.4.1.

165.2.2.7 PMA_RXSTATUS.indication

This primitive is generated by PMA Receive to indicate the status of the receive link at the local PHY. The parameter `loc_rcvr_status` conveys to the PCS Transmit, PCS Receive, PMA PHY Control function, and Link Monitor the information on whether the status of the overall receive link is satisfactory or not. Note that `loc_rcvr_status` is used by the PCS Receive decoding functions. The criterion for setting the parameter `loc_rcvr_status` is left to the implementer. It can be based, for example, on observing the mean-square error at the decision point of the receiver and detecting errors during reception of the symbol stream.

165.2.2.7.1 Semantics of the primitive

PMA_RXSTATUS.indication(`loc_rcvr_status`)

The `loc_rcvr_status` parameter can take on one of two values of the form:

- OK This value is asserted and remains true during reliable operation of the receive link for the local PHY.
- NOT_OK This value is asserted whenever operation of the link for the local PHY is unreliable.

165.2.2.7.2 When generated

PMA Receive generates PMA_RXSTATUS.indication messages to indicate a change in `loc_rcvr_status` on the basis of signals received at the MDI.

165.2.2.7.3 Effect of receipt

The effect of receipt of this primitive is specified in Figure 165–23, 165.3.2.3, 165.4.2.4, and 165.4.5.

165.2.2.8 PMA_REMRXSTATUS.request

This primitive is generated by PCS Receive to indicate the status of the receive link at the remote PHY as communicated by the remote PHY via its encoding of its `loc_rcvr_status` parameter. The parameter `rem_rcvr_status` conveys to the PMA PHY Control function the information on whether reliable operation of the remote PHY is detected or not. The parameter `rem_rcvr_status` is set to the value received in the `loc_rcvr_status` bit in the Infofield from the remote PHY. The `rem_rcvr_status` is set to `NOT_OK` if the PCS has not decoded a valid Infofield from the remote PHY.

165.2.2.8.1 Semantics of the primitive

PMA_REMRXSTATUS.request(`rem_rcvr_status`)

The `rem_rcvr_status` parameter can take on one of two values of the form:

- OK The receive link for the remote PHY is operating reliably.
- NOT_OK Reliable operation of the receive link for the remote PHY is not detected.

165.2.2.8.2 When generated

The PCS generates PMA_REMRXSTATUS.request messages to indicate a change in `rem_rcvr_status` based on the PCS decoding the `loc_rcvr_status` bit in Infofield messages received from the remote PHY during training.

165.2.2.8.3 Effect of receipt

The effect of receipt of this primitive is specified in Figure 165–22.

165.2.2.9 PMA_PCSDATAMODE.indication

This primitive indicates whether or not the PCS state diagrams are able to transition from their initialization states. The `pcs_data_mode` variable is generated by the PMA PHY Control function. It is passed to the PCS Control function via the PMA_PCSDATAMODE.indication primitive.

165.2.2.9.1 Semantics of the primitive

PMA_PCSDATAMODE.indication(`pcs_data_mode`)

The `pcs_data_mode` parameter can take on one of the following two values of the form:

- TRUE PHY is in state PCS_Data (see Figure 165–22)
- FALSE PHY is not in state PCS_Data (see Figure 165–22)

165.2.2.9.2 When generated

The PMA PHY Control function generates PMA_PCSDATAMODE.indication messages continuously.

165.2.2.9.3 Effect of receipt

Upon receipt of this primitive, the PCS performs its transmit function as described in 165.3.2.2.

165.2.2.10 PMA_PCS_RX_LPI_STATUS.request

When the PHY supports the EEE capability this primitive is generated by the PCS Receive function to indicate the status of the receive link at the local PHY. The parameter PMA_PCS_RX_LPI_STATUS.request conveys to the PCS Transmit and PMA Receive functions information regarding whether the receive function is in the LPI receive mode.

165.2.2.10.1 Semantics of the primitive

PMA_PCS_RX_LPI_STATUS.request(rx_lpi_active)

The rx_lpi_active parameter can take on one of two values of the form:

- TRUE The receive function is in the LPI receive mode.
- FALSE The receive function is not in the LPI receive mode.

165.2.2.10.2 When generated

The PCS generates PMA_PCS_RX_LPI_STATUS.request messages to indicate a change in the rx_lpi_active variable as described in 165.3.2.3 and 165.3.7.2.2.

165.2.2.10.3 Effect of receipt

The effect of receipt of this primitive is specified in 165.3.7.3.

165.2.2.11 PMA_PCS_TX_LPI_STATUS.request

When the PHY supports the EEE capability this primitive is generated by the PCS Transmit function to indicate the status of the transmit link at the local PHY. The parameter PMA_PCS_TX_LPI_STATUS.request conveys to the PCS Transmit and PMA Receive functions information regarding whether the transmit function is in the LPI transmit mode.

165.2.2.11.1 Semantics of the primitive

PMA_PCS_TX_LPI_STATUS.request(tx_lpi_active)

The tx_lpi_active parameter can take on one of two values of the form:

- TRUE The transmit function is in the LPI transmit mode.
- FALSE The transmit function is not in the LPI transmit mode.

165.2.2.11.2 When generated

The PCS generates PMA_PCS_TX_LPI_STATUS.request messages to indicate a change in the tx_lpi_active variable as described in 165.3.6 and 165.3.7.2.2.

165.2.2.11.3 Effect of receipt

The effect of receipt of this primitive is specified in 165.3.7.3.

165.2.2.12 PMA_ALERTDETECT.indication

This primitive is generated by PMA Receive to indicate the status of the receive link at the local PHY when rx_lpi_active is TRUE. The parameter alert_detect conveys to the PCS Receive function information

regarding the detection of the LPI alert signal by the PMA Receive function. The criterion for setting the parameter `alert_detect` is left to the implementer.

165.2.2.12.1 Semantics of the primitive

`PMA_ALERTDETECT.indication(alert_detect)`

The `alert_detect` parameter can take on one of two values of the form:

- | | |
|-------|--|
| TRUE | The alert signal has been reliably detected at the local receiver. |
| FALSE | The alert signal at the local receiver has not been detected. |

165.2.2.12.2 When generated

The PMA generates `PMA_ALERTDETECT.indication` messages to indicate a change in the `alert_detect` status.

165.2.2.12.3 Effect of receipt

The effect of receipt of this primitive is specified in 165.3.2.3, [Figure 149–17](#).

165.3 Physical Coding Sublayer (PCS) functions

165.3.1 PCS service interface (25GMII)

The PCS service interface allows the 25GBASE-T1 PCS to transfer information to and from a PCS client. The PCS service interface is precisely defined as the 25 Gigabit Media Independent Interface (25GMII) in [Clause 106](#).

165.3.2 PCS functions

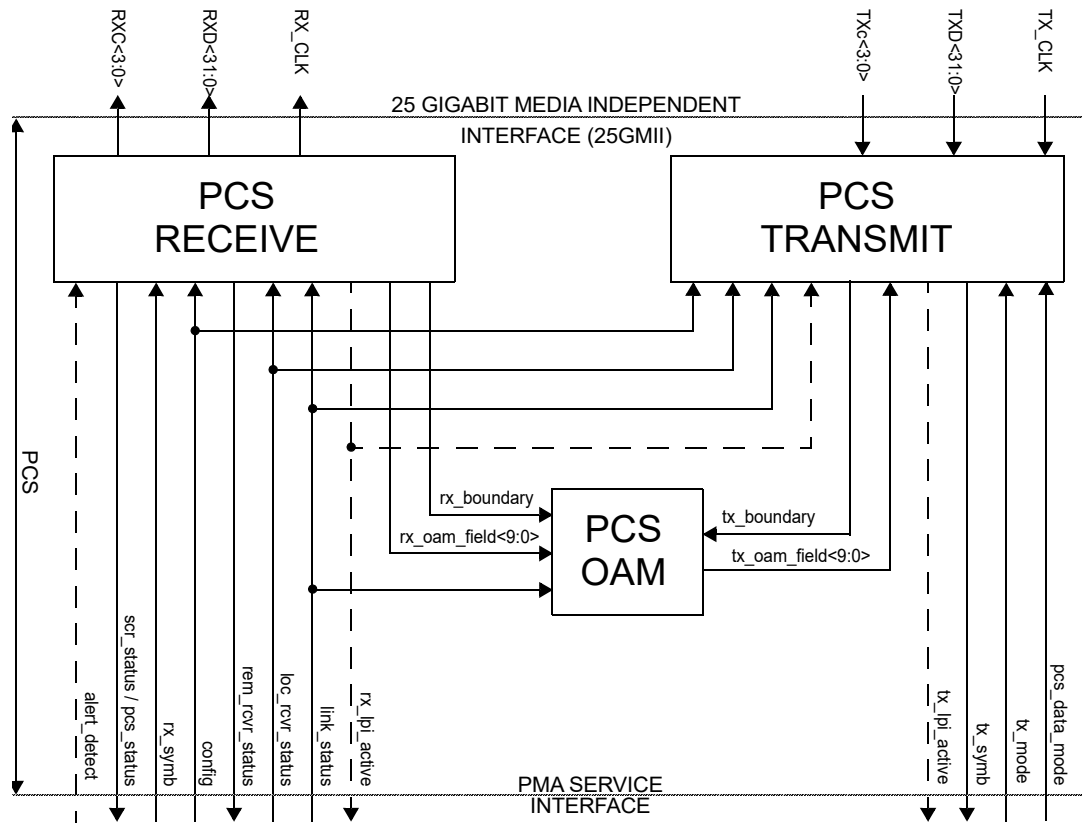
The PCS comprises one PCS Reset function and two simultaneous and asynchronous operating functions, and the PCS OAM function. The PCS operating functions are: PCS Transmit and PCS Receive. All operating functions start immediately after the successful completion of the PCS Reset function.

The information in Table 165–1 shows the period and relative offset of the start of various frames. The values are given in terms of partial frame count (PFC24, see 165.3.5), which are synchronized between master and slave.

The PCS reference diagram, Figure 165–4, shows how the two operating functions relate to the messages of the PCS-PMA interface. Connections from the management interface (signals MDC and MDIO) to other layers are pervasive and are not shown in Figure 165–4.

Table 165–1—Frame alignment parameters

Align start of	PFC24	Offset	Comment
Partial PHY frame	1	0	One PFC24 count every 1170 PMA_UNITDATA clocks
RS-FEC frame	4	0	There are 4 PFC24 in each frame
Superframe $L=1$	4	0	One RS-FEC frame
Superframe $L=2$	8	0	Two RS-FEC frames
Superframe $L=4$	16	0	Four RS-FEC frames
Superframe $L=8$	32	0	Eight RS-FEC frames
OAM field	4	8450/2340	OAM field comes after first 8450 bits in RS-FEC frame
Training frame	16	0	
Training alignment bit	1	1	Alignment bit at start of every partial frame
Training Infocfield	16	15	Infocfield is in last partial frame of training frame
QR cycle	384	0	Each QR cycle is 96 RS-Frames
Alert (fast wake)	32	16	4 th frame of every 8 frame period
Alert (slow wake)	384	368	The 92 nd frame of QR cycle
Refresh (slave)	384	168	The 42 nd frame of QR cycle
Refresh (master)	384	368	The 90 th frame of QR cycle
NOTE—Sleep and wake signals have to be aligned on superframe boundary.			



NOTE—alert_detect, rx_lpi_active, and tx_lpi_active are only required for the EEE capability.

Figure 165-4—PCS reference diagram

165.3.2.1 PCS Reset function

PCS Reset initializes all PCS functions. The PCS Reset function shall be executed whenever one of the following conditions occur:

- Power on (see 165.3.7.2.2).
- The receipt of a request for reset from the management entity.

PCS Reset sets pcs_reset = TRUE while any of the above reset conditions hold true. All state diagrams take the open-ended pcs_reset branch upon execution of PCS Reset. The reference diagrams do not explicitly show the PCS Reset function.

The control and management interface shall be restored to operation within 10 ms from the setting of bit 3.2322.15.

165.3.2.2 PCS Transmit function

The PCS Transmit function shall conform to the PCS 64B/65B Transmit state diagram in Figure 149-16 and Figure 149-17, and to the PCS Transmit bit ordering in Figure 165-6.

Dashed rectangles in Figure 149–16 and Figure 149–17 are used to indicate states and state transitions in the transmit process state diagram that are supported by a PHY with the EEE capability. A PHY without the EEE capability does not support these states or transitions.

When communicating with the 25GMII, the 25GBASE-T1 PCS uses a four octet-wide, synchronous data path, with packet delimiting being provided by transmit control signals and receive control signals. Alignment of pairs of 25GMII transfers to 64B/65B blocks is performed in the PCS. The PMA sublayer operates independently of PCS block, RS-FEC frames, and higher-layer packet boundaries. The PCS provides the functions necessary to map packets between the 25GMII format and the PMA service interface format.

After mapping the 25GMII transfers to 64B/65B blocks, the subsequent functions of the PCS Transmit process take L groups of 130 64B/65B blocks and append a 10-bit OAM field to each group. This forms the input to an L -interleaved RS-FEC, which adds $L \times 900$ parity bits. The resulting $L \times 9360$ bits are then scrambled. These bits are then mapped, two at a time, into a PAM4 symbol. Transmit data-units are sent to the PMA service interface via the PMA_UNITDATA.request primitive.

In each symbol period, when communicating with the PMA, the PCS Transmit generates a PAM4 symbol that is transferred to the PMA via the PMA_UNITDATA.request primitive. The symbol period, T , is one period of the transmitter clock frequency, specified in 165.5.3.6.

The operation of the PCS Transmit function is controlled by the PMA_TXMODE.indication message received from the PMA PHY Control function.

If a PMA_TXMODE.indication message has the value SEND_Z, PCS Transmit shall pass a vector of zeros at each symbol period to the PMA via the PMA_UNITDATA.request primitive.

If a PMA_TXMODE.indication message has the value SEND_T, PCS Transmit shall generate a sequence (T_n) defined in 165.3.5.1 to the PMA via the PMA_UNITDATA.request primitive. This sequence is used for training and contains only the symbols $\{-1, +1\}$.

During training mode an Infocfield is transmitted at regular intervals containing messages for startup operation. By this mechanism, a PHY indicates the status of its own receiver to the link partner and makes requests for remote transmitter settings. (See 165.4.2.4.)

If a PMA_TXMODE.indication message has the value SEND_N, the PCS is in the normal mode of operation and the PCS Transmit function shall use the 65B coding technique specified in 149.3.2.2.4 to represent data or control. During transmission, the 130 blocks of 65B encoded bits are appended with a 10-bit OAM field to form the RS-FEC input frame. During data encoding, PCS Transmit utilizes L -interleaved ($L = 1, 2, 4, \text{ or } 8$) Reed-Solomon encoders to generate and append 900 parity check bits to form 9360-bit RS-FEC(936,846) frames that are interleaved into an L -interleaved RS-FEC superframe.

Each RS-FEC input superframe consists of $8460 \times L$ bits, or $846 \times L$ Reed-Solomon message symbols. The interleaving function is integrated with the RS-FEC encoding, applying a round-robin interleaving scheme and distributing the 10-bit Reed-Solomon message symbols into L RS-FEC encoders. After encoding, the RS-FEC frames from each encoder are recombined into one single interleaved RS-FEC superframe, which consists of $936 \times L$ symbols, or $9360 \times L$ bits. The bits of the RS-FEC superframe are then scrambled by the PCS using an additive scrambler, encoded in PAM4 symbols, and transferred to the PMA.

L is called the interleaving depth, and the possible choices of L are 1, 2, 4, and 8. The interleaver settings requested in each direction of transmission may be different, and the value of L used by the transmitter is determined by the link partner and signaled during the PAM2 training mode Infocfield exchange.

After reaching the normal mode of operation, an EEE-capable PHY may enter the LPI transmit mode under the control of the Reconciliation Sublayer via the 25GMII. The EEE Transmit state diagram is contained within the PCS Transmit function. The EEE capability is described in 165.3.2.2.22.

A block diagram of the PCS Transmit function is shown in Figure 165–5.

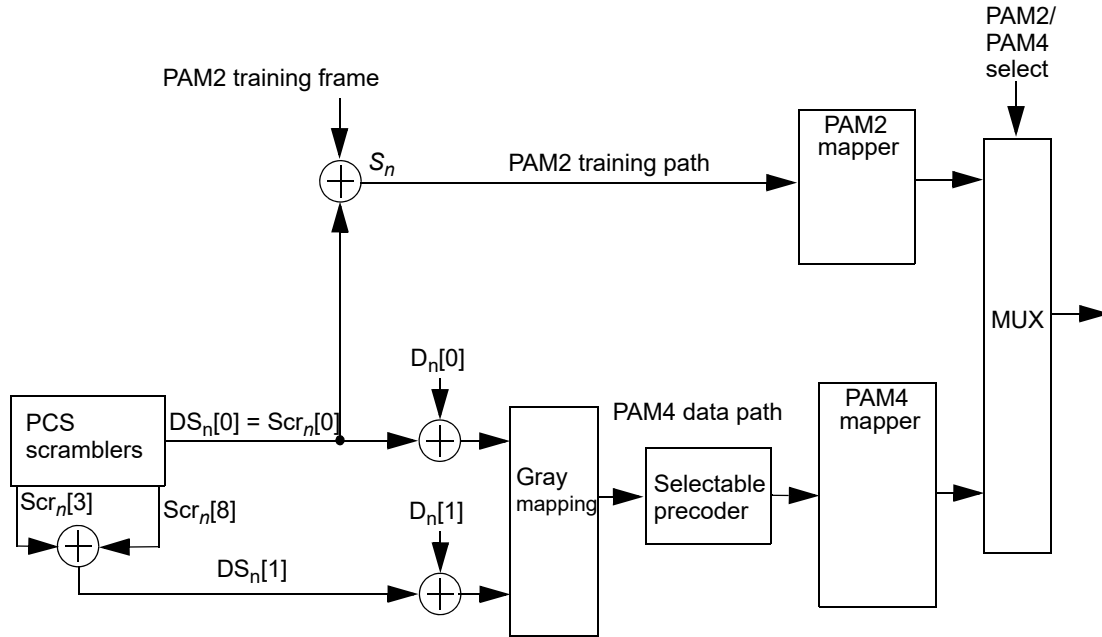


Figure 165–5—PCS Transmit function block diagram

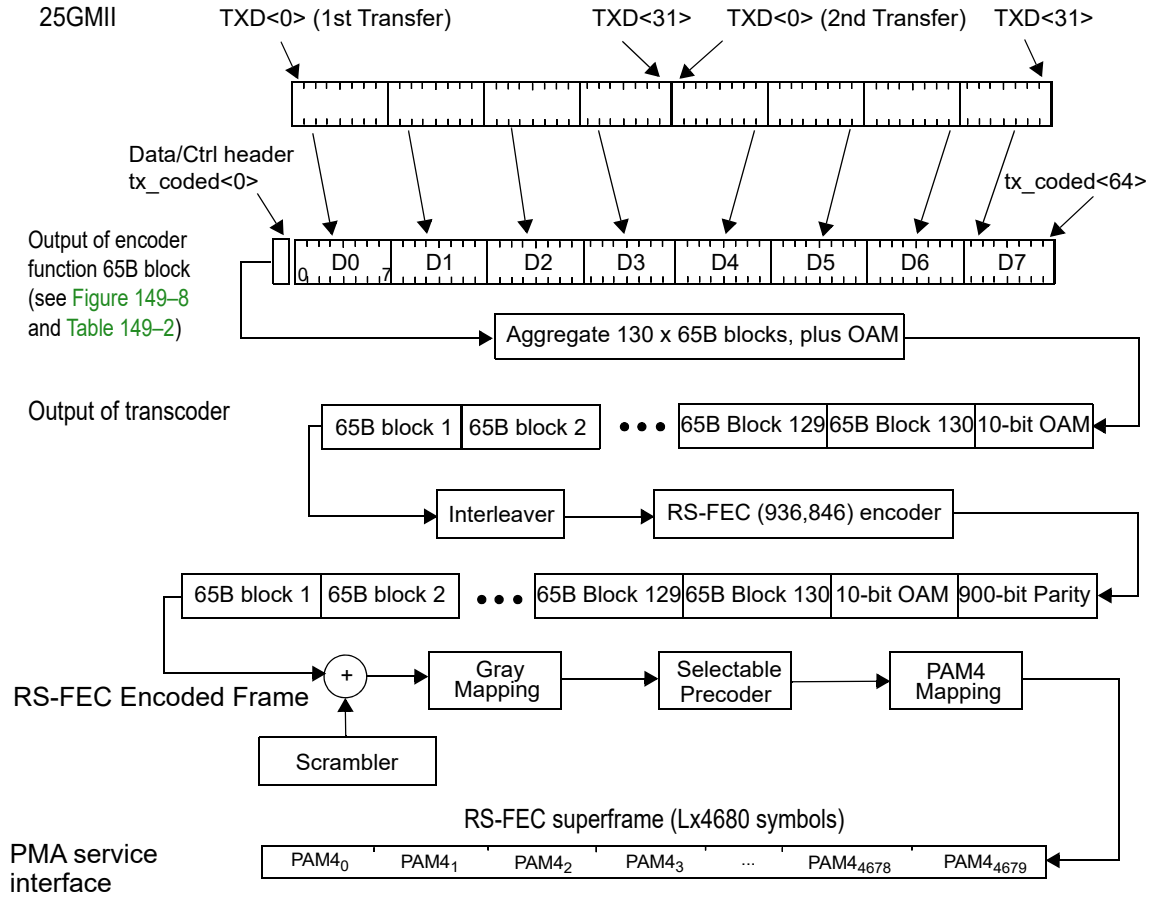
165.3.2.2.1 Use of blocks

The PCS maps 25GMII signals into 65-bit blocks inserted into an RS-FEC frame, and vice versa, using a 65B RS-FEC coding scheme. The PAM2 PMA training frame synchronization allows establishment of RS-FEC frame and 65B boundaries by the PCS Synchronization process. Blocks and frames are unobservable and have no meaning outside the PCS. During the LPI mode, RS-FEC frame boundaries delimit sleep, wake, refresh, quiet, and alert cycles. The PCS functions ENCODE and DECODE generate, manipulate, and interpret blocks and frames as provided by the rules in 165.3.2.2.2.

165.3.2.2.2 65B RS-FEC transmission code

The PCS uses a transmission code to improve the transmission characteristics of information to be transferred across the link and to support transmission of control and data characters.

The relationship of block bit positions to 25GMII, PMA, and other PCS constructs is illustrated in Figure 165–6 for transmit and Figure 165–7 for receive. These figures illustrate the processing of a multiplicity of blocks containing 8 data octets. See 165.3.2.2.4 for information on how blocks containing control characters are mapped.



NOTE 1—This figure shows the mapping from the 25GMII to a 64B/65B block for a block containing eight data characters.

NOTE 2—Figure shown for $L = 1$.

Figure 165–6—PCS Transmit bit ordering

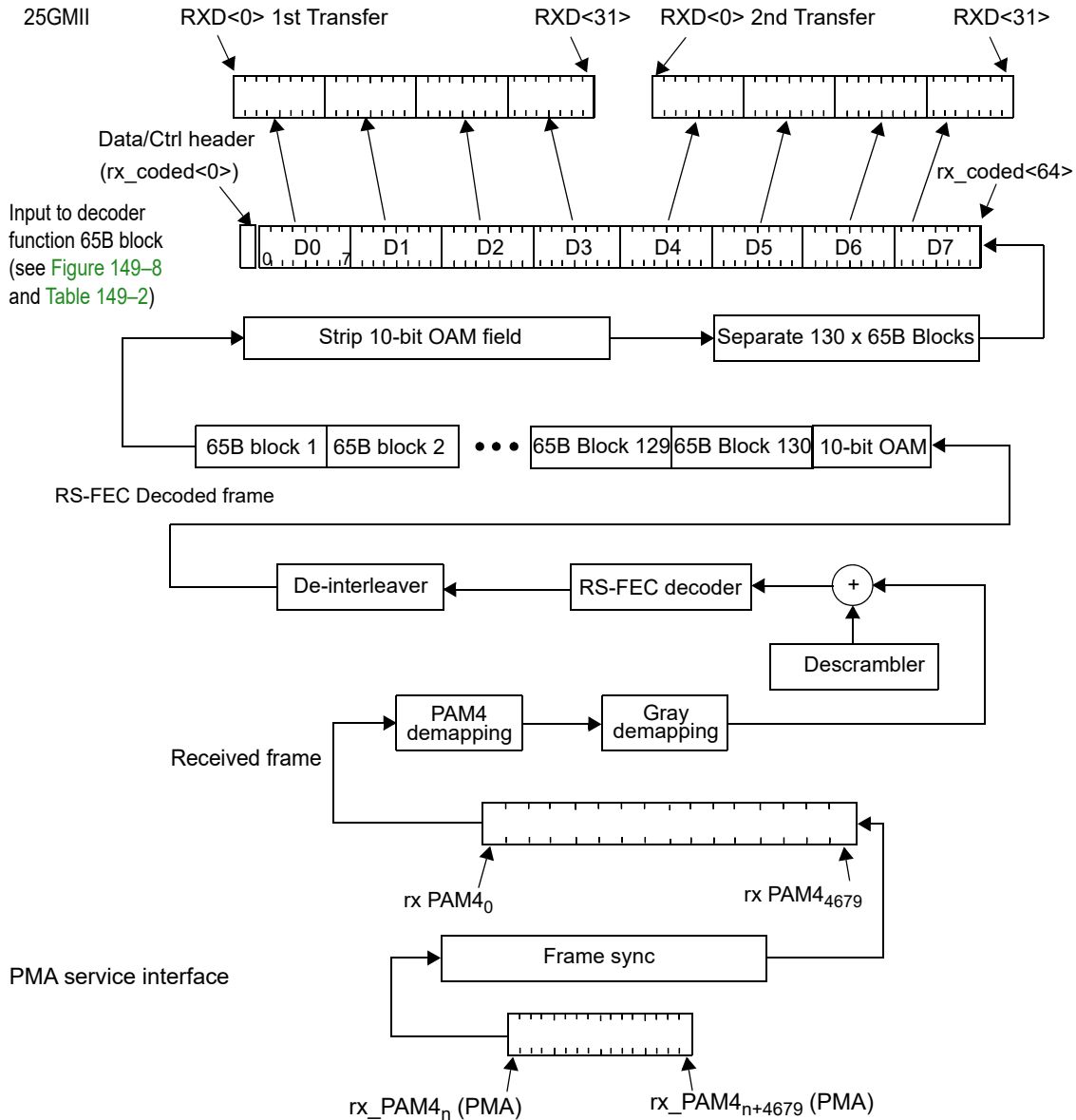
165.3.2.2.3 Notation conventions

For values shown as binary, the leftmost bit is the first transmitted bit.

64B/65B encodes 8 data octets or control characters into a block. Blocks containing control characters also contain a block type field. Data octets are labeled D_0 to D_7 . Control characters other than /O/, /S/, and /T/ are labeled C_0 to C_7 . The control character for ordered set is labeled as O_0 or O_4 since it is only valid on the first octet of the 25GMII. The control character for start is labeled as S_0 or S_4 for the same reason. The control character for terminate is labeled as T_0 to T_7 .

For 25GBASE-T1, two 25GMII transfers provide eight characters that are encoded into one 65-bit transmission block. The subscript in the above labels indicates the position of the character in the eight characters from the 25GMII transfer(s).

Contents of block type fields, data octets, and control characters are shown as hexadecimal values. The LSB of the hexadecimal value represents the first transmitted bit. For instance, the block type field 0x1E is sent from left to right as 01111000. The bits of a transmitted or received block are labeled tx_coded<31:0> and rx_coded<31:0> where tx_coded<0> and rx_coded<0> represent the first transmitted bit.



NOTE 1—This figure shows the mapping from a 64B/65B block a block containing eight data characters to the 25GMII.

NOTE 2—Figure shown for $L = 1$.

Figure 165-7—PCS Receive bit ordering

165.3.2.2.4 Block structure

The 65-bit block structure specified in 149.3.2.2.4 is used by 25GBASE-T1, with the block format shown in Figure 149-8.

165.3.2.2.5 Control codes

The mapping of control characters used to map the 25GMII and 25GBASE-T1 PCS is as specified for MultiGBASE-T1 PHYs in 149.3.2.2.5 and shown in Table 149-2. All 25GMII control code values that do not appear in the table shall not be transmitted and shall be treated as an error if received.

165.3.2.2.6 Ordered sets

The use of ordered sets is as specified for MultiGBASE-T1 PHYs in [149.3.2.2.6](#).

165.3.2.2.7 Idle (/I/)

Idle control characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.7](#).

165.3.2.2.8 LPI (/LI/)

Low Power Idle control characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.8](#).

165.3.2.2.9 Start (/S/)

Start control characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.9](#).

165.3.2.2.10 Terminate (/T/)

Terminate control characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.10](#).

165.3.2.2.11 Ordered set (/O/)

Ordered set control characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.11](#).

165.3.2.2.12 Error (/E/)

Error characters are as specified for MultiGBASE-T1 PHYs in [149.3.2.2.12](#). See R_BLOCK_TYPE and T_BLOCK_TYPE function definitions in 165.3.7.2.4 for further information.

165.3.2.2.13 Transmit process

The transmit process generates blocks based upon the TXD and TXC signals received from the 25GMII. 260 25GMII data transfers are encoded into an RS-FEC frame. It takes 4680 PMA_UNITDATA transfers to send an RS-FEC frame of data. Therefore, for 25GBASE-T1, if the PCS is connected to a 25GMII and PMA sublayer where the ratio of their transfer rates is exactly 1:18, then the transmit process does not need to perform rate adaptation. Where the 25GMII and PMA sublayer data rates are not synchronized to that ratio, the transmit process needs to insert idles, delete idles, or delete sequence ordered sets to adapt between the rates.

The transmit process generates blocks as specified in the PCS 64B/65B Transmit state diagram (see [Figure 149–16](#) and [Figure 149–17](#)). The contents of each block are contained in a vector tx_coded<64:0>, which is passed to the transcoder and scrambler. tx_coded<0> contains the data/ctrl header and the remainder of the bits contain the block payload.

165.3.2.2.14 RS-FEC framing and RS-FEC encoder

The resulting RS-FEC frame of 130 64B/65B blocks, followed by the 10-bit OAM field and 900 parity bits is 9360 bits. See Figure 165–6 and 165.3.2.2.17 for details on PCS bit ordering and RS-FEC encoding.

The RS-FEC encoding shall generate the 90 10-bit parity symbols (900 bits total) using the 8460-bit vector, consisting of tx_group130x65B and the 10-bit OAM_field as the message symbols.

165.3.2.2.15 RS-FEC superframe and round-robin interleaving

The interleaver depth L of the transmitter shall be set to the InterleaverDepth requested by the link partner during Infofield exchange, as specified in 165.4.2.4.5.

When the selected interleaving depth $L = 1$, there is no interleaving, and the RS-FEC superframe is the same as the RS-FEC frame.

When the selected interleaving depth $L > 1$, the round-robin interleaving scheme as shown in Figure 165–8 shall be applied.

25GBASE-T1 supports $L = 1, 2, 4$, and 8 .

The PCS Transmit shall aggregate L RS-FEC input frames into an interleaved RS-FEC input superframe. There are $8460 \times L$ bits, or $846 \times L$ Reed-Solomon message symbols in total in the input superframe. The corresponding message symbols are $m_{846 \times L - 1}, m_{846 \times L - 2}, \dots, m_1, m_0$. These message symbols are distributed to L RS-FEC encoders. When $L > 1$, each RS-FEC encoder receives one out of every L message symbols from the superframe; otherwise, the RS-FEC encoder operates exactly the same as specified in 165.3.2.2.17.

165.3.2.2.16 RS-FEC recombine

The L encoded RS-FEC frames are recombined into an interleaved RS-FEC superframe. The output symbols are as follows:

$m_{846 \times L - 1}, m_{846 \times L - 2}, \dots, m_1, m_0, p_{1,89}, \dots, p_{L,89}, \dots, p_{1,0}, \dots, p_{L,0}$, where $p_{i,r}$ is the r th parity symbol of the i th encoder.

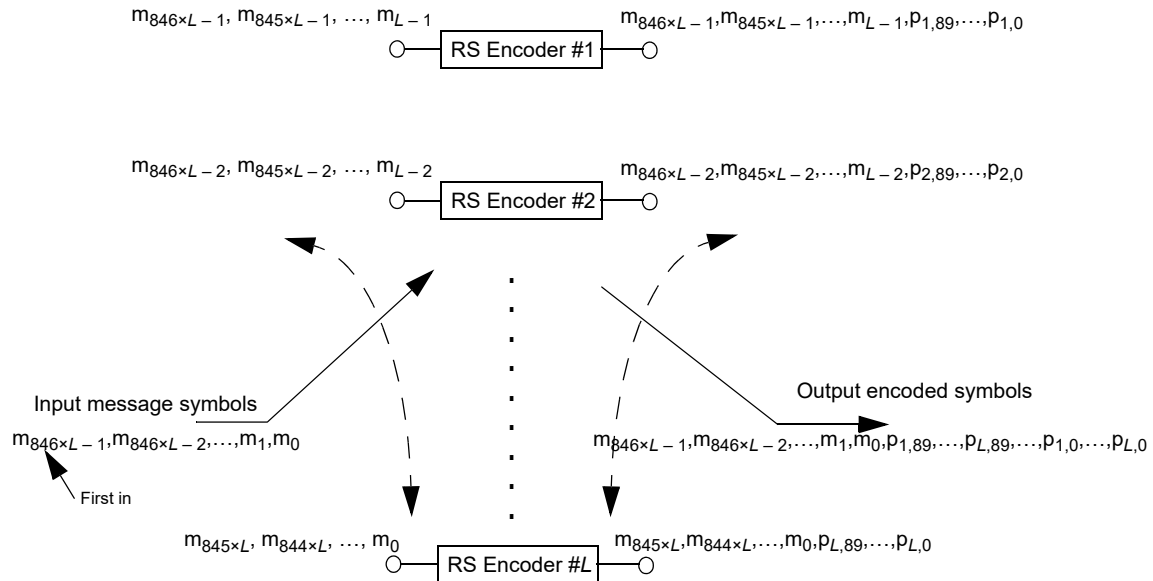


Figure 165–8—Interleaving block diagram with interleaving depth L

165.3.2.2.17 Reed-Solomon encoder

The group of 8460 bits are encoded using a Reed-Solomon encoder operating over the Galois Field $GF(2^{10})$ where the symbol size is 10 bits. The encoder processes 846 RS-FEC message symbols to generate 90 RS-FEC parity symbols, which are then appended to the message to produce a codeword of 936 RS-FEC symbols. For the purposes of this clause, the particular Reed-Solomon code is denoted as RS-FEC(936,846).

The code is based on the generating polynomial given by Equation (165–1).

$$g(x) = \prod_{j=0}^{89} (x - \alpha^j) = g_{90}x^{90} + g_{89}x^{89} + \dots + g_4x^4 + g_3x^3 + g_2x^2 + g_1x + g_0 \quad (165-1)$$

In Equation (165–1), α , is a primitive element of the finite field defined by the primitive polynomial $x^{10} + x^3 + 1$.

The contents of the RS-FEC message are represented by a polynomial $m(x)$ whose coefficients are the message symbols m_{845} to m_0 as shown in Equation (165–2).

$$m(x) = m_{845}x^{935} + m_{844}x^{934} + \dots + m_1x^{91} + m_0x^{90} \quad (165-2)$$

Each message symbol m_i is the bit vector $(m_{i,9}, m_{i,8}, \dots, m_{i,1}, m_{i,0})$, which is identified with the element of the finite field. $m_{i,0}$ is the first bit transmitted. The message symbols are composed of the bits in `tx_RSmessage<8459:0>` where

$m_{i,j} = \text{tx_RSmessage} \langle (845 - i) \times 10 + j \rangle$, for $i = 0$ to 845, and $j = 0$ to 9.

`tx_RSmessage<8459:0>` prior to the RS-FEC (936,846) encoder is formed as follows:

`tx_RSmessage<8449:0> = tx_group130x65B<8449:0>.`
`tx_RSmessage<8459:8450> = OAM_field<9:0>.`

The first symbol input to the encoder is m_{845} .

The parity polynomial $p(x)$ is calculated as the remainder of polynomial division of $m(x)$ by $g(x)$. Its coefficients, p_{89} to p_0 , as shown in Equation (165–3), are the parity symbols.

$$p(x) = p_{89}x^{89} + p_{88}x^{88} + \dots + p_2x^2 + p_1x + p_0 \quad (165-3)$$

Each parity symbol p_i is the bit vector $(p_{i,9}, p_{i,8}, \dots, p_{i,1}, p_{i,0})$, which is identified with the element of the finite field. $p_{i,0}$ is the first bit transmitted.

The calculation of the coefficients of $p(x)$ is illustrated in Figure 165–9. The outputs of the delay elements are initialized to zero prior to the computation of the parity for a given message. After the last message symbol, m_0 , is processed by the encoder, the outputs of the delay elements are the parity symbols for that message.

The codeword polynomial $c(x)$ is then the sum of $m(x)$ and $p(x)$ where the coefficient of the highest power of x , $c_{935} = m_{845}$, is transmitted first and the coefficient of the lowest power of x , $c_0 = p_0$, is transmitted last. The first bit transmitted from each symbol is bit 0.

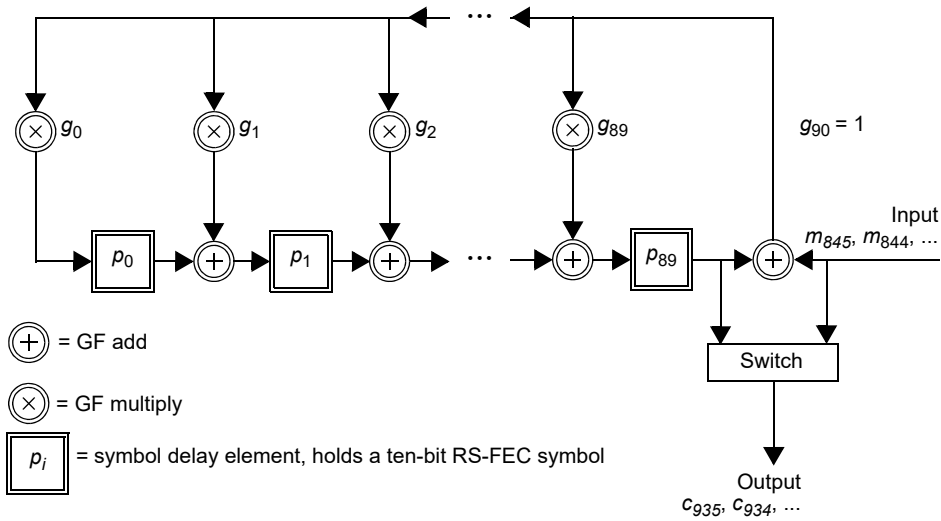


Figure 165–9—Reed-Solomon encoder functional model

The coefficients of the generator polynomial for the code are presented in Table 165–2.

Table 165–2—Coefficients of the generator polynomial g_i (decimal)

RS-FEC(936, 846)						
$i=0$ to 12	$i=13$ to 25	$i=26$ to 38	$i=39$ to 51	$i=52$ to 64	$i=65$ to 77	$i=78$ to 90
572	751	336	866	251	979	233
291	545	62	500	698	662	471
466	86	679	61	798	332	750
539	114	515	966	249	500	796
360	922	268	848	529	929	558
644	974	399	740	354	29	407
508	471	545	216	239	69	1003
534	880	575	790	679	185	997
873	423	976	172	835	182	149
519	1000	300	331	600	905	183
722	1013	654	479	301	989	814
698	1022	458	154	1011	484	944
785	822	246	334	886	797	1

165.3.2.2.18 PCS scrambler

The PCS scrambler operates as specified in 149.3.2.2.18.

165.3.2.2.19 Gray mapping for PAM4 encoding

The PCS transmit process maps pairs of bits to Gray-coded PAM4 symbols as specified in 149.3.2.2.19.

165.3.2.2.20 Selectable precoder

The PCS transmit process precodes the Gray-coded PAM4 symbols as specified in 149.3.2.2.20.

165.3.2.2.21 PAM4 encoding

The PCS transmit process encodes each precoder output symbol as specified in 149.3.2.2.21.

165.3.2.2.22 EEE capability

The optional 25GBASE-T1 EEE capability allows a compliant PHY to transition to an LPI mode of operation when link utilization is low in either direction of transmission.

A PHY that supports EEE shall conform to the EEE transmit state diagram, shown in Figure 165–14, within the PCS.

In the transmit direction, the transition to the LPI transmit mode begins when the PCS Transmit function detects an LPI control character in the last 64B/65B block of an RS-FEC frame. Following this event, the PMA transmits the sleep signal to indicate to the link partner that it is transitioning to the LPI transmit mode. The sleep signal is composed of 16 RS-FEC frames that contain only LP_IDLE 64B/65B blocks. Once initiated, the complete sleep signal consisting of 16 RS-FEC frames of LP_IDLE shall be transmitted.

Following the transmission of the sleep signal, quiet-refresh signaling begins, as described in 165.3.6.

After the sleep signal is transmitted, LPI control characters shall be input to the PCS scrambler continuously until the PCS Transmit function exits the LPI transmit mode.

While the PMA asserts SEND_N, the lpi_tx_mode variable shall control the transmit signal through the PMA_UNITDATA.request primitive described as follows:

- When the PHY is not in the PCS_DATA state, the lpi_tx_mode variable is ignored.
- When the lpi_tx_mode variable takes the value NORMAL and the PMA asserts SEND_N, the PCS passes coded data to the PMA via the PMA_UNITDATA.request primitive as described in 165.3.2.2.
- When the lpi_tx_mode variable takes the value QUIET and the PMA asserts SEND_N, the PCS passes zeros to the PMA through the PMA_UNITDATA.request primitive.
- When the lpi_tx_mode variable takes the value REFRESH and the PMA asserts SEND_N, the PCS passes the PMA training signal to the PMA, to allow both the local and remote PHY to refresh adaptive filters and timing loops.
- When the lpi_tx_mode variable takes the value ALERT, the PMA transmits the link synchronization sequence onto the MDI as provided by the link synchronization block via sync_tx_symb.

The quiet-refresh cycle is repeated until codewords other than LP_IDLE are detected at the 25GMII. These codewords indicate that the local system is requesting a transition back to the normal operational mode. Following this event, the PMA_UNITDATA.request parameter tx_symb_vector is set to the value ALERT.

After transmitting the alert signal, the PCS completes the transition from LPI mode to normal mode by sending a wake signal containing lpi_wake_time RS-FEC frames composed of IDLE 64B/65B blocks.

`lpi_wake_time` is a fixed parameter that is defined in Table 165–3.

Table 165–3—LPI wake time

<code>lpi_wake_time</code>	<code>lpi_tx_wake_time</code> when wake starts before sleep signal is complete		<code>lpi_tx_wake_time</code> when wake starts after sleep signal is complete	
(frames)	(frames)	(μ s)	(frames)	(μ s)
16	48	15.9744	28	9.3184

165.3.2.3 PCS Receive function

The PCS Receive function shall conform to the PCS 64B/65B Receive state diagram in Figure 149–18 and Figure 149–19, and the PCS Receive bit ordering in Figure 165–7 including compliance with the associated state variables as specified in 165.3.7.2.2.

The PCS Receive function accepts received symbols provided by the PMA Receive function via the parameter `rx_symb`. The PCS receiver uses knowledge of the encoding rules and PMA training alignment to correctly align the 65B RS-FEC frames. The received PAM4 symbols are demapped and descrambling is performed.

Following descrambling, the *L*-interleaved RS-FEC superframe is de-interleaved and the RS-FEC frames are decoded with Reed-Solomon error correction. Frames that cannot be corrected are marked with error symbols by the decoder. The RS-FEC decoded frame is then separated into a 10-bit OAM field and 130 64B/65B blocks. This process generates the 64B/65B block vector `rx_coded<64:0>`, which is then decoded to form the 25GMII signals `RXD<31:0>` and `RXC<3:0>` as specified in the PCS 64B/65B Receive state diagram (see Figure 149–18 and Figure 149–19). Two 25GMII data transfers are decoded from each block. Where the 25GMII and PMA sublayer data rates are not synchronized, the receive process inserts idles, deletes idles, or deletes sequence ordered sets to adapt between rates.

During PMA training mode, PCS Receive checks the received PAM2 framing and signals the reliable acquisition of the descrambler state by setting the `scr_status` parameter of the `PMA_SCRSTATUS.request` primitive to OK.

When the PCS Synchronization process has obtained synchronization, the RS-FEC frame error ratio (RFER) monitor process monitors the signal quality and asserts `hi_rfer` to indicate excessive RS-FEC frame errors. If 40 consecutive RS-FEC frame errors are detected, the `block_lock` flag is de-asserted. The `block_lock` flag is re-asserted upon detection of a valid RS-FEC frame. When `block_lock` is asserted and `hi_rfer` is de-asserted, the PCS Receive process continuously accepts blocks. The PCS Receive process monitors these blocks and generates `RXD <31:0>` and `RXC <3:0>` on the 25GMII.

When the receive channel is in training mode, the PCS Synchronization process continuously monitors `PMA_RXSTATUS.indication(loc_rcvr_status)`. When `loc_rcvr_status` indicates OK, then the PCS Synchronization process accepts data-units via the `PMA_UNITDATA.indication` primitive. It attains frame and block synchronization based on the PMA training frames and conveys received blocks to the PCS Receive process. The PCS Synchronization process sets the `block_lock` flag to indicate whether the PCS has obtained synchronization. The PMA training frame includes an alignment bit every 1170 PAM2 symbols, which is aligned with the PCS partial frame boundary, as well as an Infofield, which is inserted in the 16th PCS partial frame. When the PCS Synchronization process is synchronized to this pattern, `block_lock` is asserted.

A PHY with the EEE capability supports transition to the LPI mode when the PHY has successfully completed training and `pcs_data_mode` is TRUE and is subject to the timing requirements of 106.1.7. Transitions to and from the LPI mode are allowed to occur independently in the transmit and receive functions. The PCS Receive function is responsible for detecting transitions to and from the LPI receive mode and indicating these transitions using signals defined in 165.2.2.

The link partner signals a transition to the LPI mode of operation by transmitting 16 RS-FEC frames composed entirely of 64B/65B blocks of /LI/. When blocks of /LI/ are detected at the output of the 64B/65B decoder, `rx_lpi_active` is asserted by the PCS Receive function and the /LI/ character is continuously asserted at the receive 25GMII. These frames may be preceded by a frame composed partially of /LI/ characters. After these frames, the link partner begins transmitting zeros, and it is recommended that the receiver power down receive circuits to reduce power consumption. The receive function uses RS-FEC frame counters to maintain synchronization with the remote PHY and receives periodic refresh signals that are used to update coefficients, so that the integrity of adaptive filters and timing loops in the PMA is maintained. LPI signaling is defined in 165.3.6. The quiet-refresh cycle continues until the PMA Receive function asserts `alert_detect` to indicate that the alert (link synchronization) sequence has been reliably detected. After the alert sequence, the link partner transmits repeated /I/ characters, representing a wake signal. The PHY receive function sends /I/ to the 25GMII for 16 RS-Frame periods (wake duration) and then resumes normal operation.

165.3.2.3.1 Frame and block synchronization

When operating in the data mode, the receiving PCS shall form a PAM4 stream from the `PMA_UNITDATA.indication` primitive by concatenating requests in order from `rx_PAM4_0` to `rx_PAM4_4679` (see Figure 165–7). It obtains block lock to the PHY frames during PAM2 training using synchronization bits provided in the training frames.

165.3.2.3.2 PCS descrambler

The descrambler processes the payload to reverse the effect of the scrambler using the same polynomial. The PCS descrambles the data stream and returns the proper sequence of symbols to the decoding process for generation of `RXD<31:0>` to the 25GMII. For side-stream descrambling, the MASTER PHY shall employ the receiver descrambler generator polynomial per Equation (149–6) and the SLAVE PHY shall employ the receiver descrambler generator polynomial per Equation (149–5).

165.3.2.3.3 Invalid blocks

A block is invalid if any of the following conditions exist:

- a) The block type field contains a reserved value.
- b) Any control character contains a value not in Table 149–2.
- c) Any O code contains a value not in Table 149–2.
- d) The block contains information from the payload of an invalid RS-FEC frame.

The PCS Receive function shall check the integrity of the RS-FEC parity bits defined in 165.3.2.2.14. If the check fails the RS-FEC frame is invalid.

The `R_BLOCK_TYPE` of an invalid block is set to E.

165.3.3 Test-pattern generators

The test-pattern generator mode shall operate as specified in 149.3.3.

165.3.4 Side-stream scrambler polynomials

The PCS Transmit function shall employ side-stream scrambling as specified in 149.3.4.

165.3.5 PMA training frame

During PMA training, the training frames are embedded with indicators to establish alignment to the RS-FEC superframe. The last partial frame, in each training frame, is embedded with an information field used to exchange messages between link partners. The timing relationship among training frame, partial frame, RS-FEC frame, superframe, and partial frame count (PFC24) is shown in Figure 165–10.

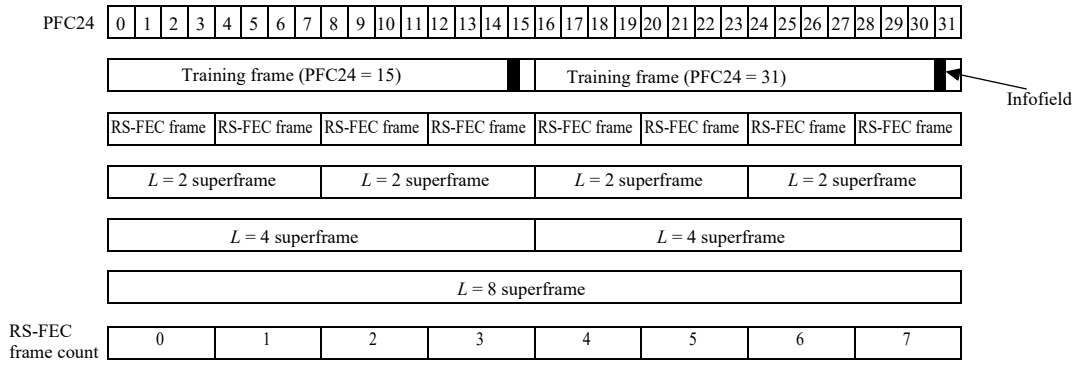


Figure 165–10—Timing relationship to PFC24

PMA training frame encoding is based on the generation, at time n , of the bit S_n . The first bit is inverted in the first 15 partial frames of each training frame. The first 96 bits of the 16th partial frame are XORed with the contents of the Infofield. Each partial frame is 1170 PMA_UNITDATA clocks. See Equation (165–4).

$$S_n = \begin{cases} Scr_n[0] \oplus InfoField_{(n \bmod 1170)} & 17550 \leq (n \bmod 18720) < 17646 \\ Scr_n[0] \oplus 1 & \text{else if } (n \bmod 1170) = 0 \\ Scr_n[0] & \text{otherwise} \end{cases} \quad (165-4)$$

165.3.5.1 Generation of symbol T_n

The bit S_n is mapped to the transmit symbol T_n as follows: if $S_n = 0$ then $T_n = +1$, if $S_n = 1$ then $T_n = -1$.

165.3.5.2 PMA training mode descrambler polynomials

The PHY shall acquire descrambler state synchronization to the PAM2 training sequence and report success through scr_status. For side-stream descrambling, the MASTER PHY employs the receiver descrambler generator polynomial per Equation (149–6) and the SLAVE PHY employs the receiver descrambler generator polynomial per Equation (149–5).

165.3.6 LPI signaling

A PHY with EEE capability has transmit and receive functions that can enter and leave the LPI mode independently. The PHY can transition to the LPI mode when the PHY has successfully completed training and pcs_data_mode is TRUE. The transmit function of the PHY initiates a transition to the LPI transmit

mode by generating the sleep signal composed of 16 RS-FEC frames, each composed entirely of LPI control characters, as described in 165.3.2.2.22. When the transmitter begins to send the sleep signal, it asserts `tx_lpi_active` and the transmit function enters the LPI transmit mode.

Within the LPI mode, a PHY uses a repeating quiet-refresh cycle (see Figure 165–11 and Figure 165–12). The LPI timing parameters are shown in Table 165–4. One part of this cycle is known as the quiet period and lasts for a time `lpi_quiet_time`. The quiet period is defined in 165.3.6.2. The other part of this cycle is known as the refresh period and lasts for a time `lpi_refresh_time`. The refresh period is defined in 165.3.6.3. A cycle composed of one quiet period and one refresh period is known as an LPI cycle and lasts for a time `lpi_qr_time`.

The parameters `lpi_master_refresh_start`, `lpi_slave_refresh_start`, `lpi_quiet_time`, `lpi_refresh_time`, and `lpi_qr_time` are timing parameters that are integer multiples of the RS-FEC frame period.

The end of LPI mode occurs at the transmission of the alert signal indicating the end of quiet-refresh cycle. The alert signal is restricted to starting at predetermined RS-FEC frame count values, where the allowed values depend on whether the `SlowWakeRequest` PHY capability bit is set to 1.

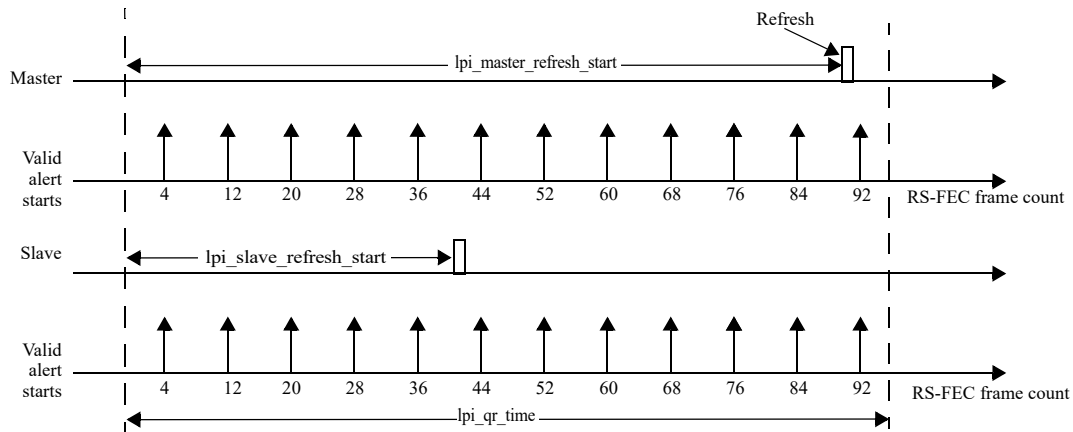


Figure 165–11—Timing periods for LPI signals when `SlowWakeRequest` is set to 0

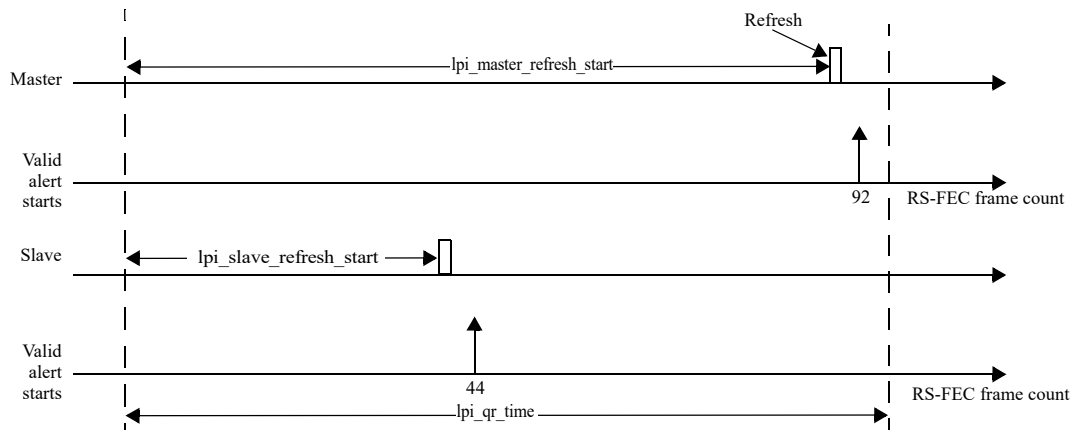


Figure 165–12—Timing periods for LPI signals when `SlowWakeRequest` is set to 1

Table 165–4—LPI timing parameters

Parameter	Number of RS-FEC frame periods
alert_length	4
alert_period	8
lpi_slave_refresh_start	42
lpi_master_refresh_start	90
lpi_qr_time	96
lpi_quiet_time	95
lpi_refresh_time	1

165.3.6.1 LPI synchronization

An EEE-capable PHY shall synchronize refresh intervals during the LPI mode. A PHY in SLAVE mode is responsible for synchronizing its partial frame count (PFC24) to the master’s PFC24 during PAM2 training. For the requirements on the SLAVE and the MASTER frame alignment, see 165.4.2.4.10.

Refresh signaling is derived by tracking the RS-FEC frame count as shown in Figure 165–11, where:

$$\text{RS-FEC frame count} = \text{floor}(\text{PFC24} / 4) \bmod 96.$$

Following the transition to PAM4, the PCS continues with the RS-FEC frame count and uses the count to generate refresh, alert, and wake control signals for the transmit functions.

Alert is a four RS-FEC frame long sequence (alert_length). When SlowWakeRequest is set to 0, alert can be transmitted such that it ends at any eighth RS-FEC frame boundary, counting from the start of the QR cycle (see Figure 165–11). When SlowWakeRequest is set to 1, alert can only be transmitted in a single QR cycle location, starting at RS-FEC frame 92 for master and at RS-FEC frame 44 for slave (see Figure 165–12). The synchronization for the alert signaling is described in Table 165–5 and Table 165–6.

Table 165–5—Synchronization logic derived from slave signal RS-FEC frame count

Slave-side variable	Condition (where u = RS-FEC frame count)	SlowWakeRequest
tx_refresh_active = TRUE	$\text{lpi_slave_refresh_start} \leq \text{mod}(u, \text{lpi_qr_time}) < \text{lpi_slave_refresh_start} + \text{lpi_refresh_time}$	0 or 1
tx_alert_start_next = TRUE	$\text{mod}(u, \text{alert_period}) = \text{alert_period} - \text{alert_length}$	0
tx_alert_start_next = TRUE	$\text{mod}(u, \text{lpi_qr_time}) = \text{lpi_slave_refresh_start} + \text{lpi_refresh_time} + 1$	1

Table 165–6—Synchronization logic derived from master signal RS-FEC frame count

Master-side variable	Condition (where v = RS-FEC frame count)	SlowWakeRequest
tx_refresh_active = TRUE	$\text{lpi_master_refresh_start} \leq \text{mod}(v, \text{lpi_qr_time}) < \text{lpi_master_refresh_start} + \text{lpi_refresh_time}$	0 or 1
tx_alert_start_next = TRUE	$\text{mod}(v, \text{alert_period}) = \text{alert_period} - \text{alert_length}$	0
tx_alert_start_next = TRUE	$\text{mod}(v, \text{lpi_qr_time}) = \text{lpi_master_refresh_start} + \text{lpi_refresh_time} + 1$	1

165.3.6.2 Quiet period signaling

During the quiet period the PCS transmitter shall pass zeros to the PMA via the PMA_UNITDATA.request primitive. The receiver should ignore transmissions during the first 150 ns following the transition to the quiet period and the last 150 ns before the start of the refresh signal or the allowed alert time slot.

165.3.6.3 Refresh period signaling

During the LPI mode the 25GBASE-T1 PHY uses staggered, out-of-phase refresh signaling. Two-level PAM refresh symbols are generated from the T_n mapping defined in 165.3.5.1 of S_n defined in 165.3.5, with the exception that the Infobits consists of zeros. The 10-bit OAM symbol to be transmitted is XORed with the last 10 bits of the PAM2 refresh transmission.

165.3.7 Detailed functions and state diagrams

165.3.7.1 State diagram conventions

The body of this subclause is composed of state diagrams, including the associated definitions of variables, constants, and functions. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

165.3.7.2 State diagram parameters

165.3.7.2.1 Constants

The PCS state diagram constants are as defined in 149.3.7.2.1 with the exception that 25GMII replaces XGMII.

165.3.7.2.2 Variables

The PCS state diagram variables are as defined in 149.3.7.2.2 with the exceptions that 25GMII replaces XGMII and the following modification:

- Reference to 149.4.4.1 is replaced by 165.4.4.1.

165.3.7.2.3 Timers

The PCS timers are as defined in 149.3.7.2.3 with the exceptions that 25GMII replaces XGMII and the definition of the `lpi_tx_sleep_timer` (required for PHYs that support the EEE capability) is:

`lpi_tx_sleep_timer`

This timer defines the time the local transmitter sends the sleep signal to the link partner.

Values: The condition `lpi_tx_sleep_timer_done` becomes TRUE upon timer expiration.

Duration: This timer shall have a period equal to 16 RS-FEC frame periods.

165.3.7.2.4 Functions

The PCS functions are as defined in 149.3.7.2.4 with the exceptions that 25GMII replaces XGMII and the following modification:

- Reference to 149.3.2.2.2 is replaced by 165.3.2.2.2.

165.3.7.2.5 Counters

The PCS counters are as defined in 149.3.7.2.5.

165.3.7.2.6 Messages

`RX_FRAME`

A signal sent to PCS Receive indicating that a full RS-FEC frame has been decoded and the variable `rf_valid` is updated. If the optional EEE is supported, `RX_FRAME` shall be FALSE during the first eight WAKE frames.

165.3.7.3 State diagrams

The RFER monitor state diagram shown in Figure 165–13 monitors the received signal for high RS-FEC frame error ratio.

The EEE transmit state diagram shown in Figure 165–14 controls transitions between normal operation and low power idle.

The PCS 64B/65B Transmit state diagram shown in Figure 149–16 and Figure 149–17 controls the encoding of 65B transmitted blocks. It makes exactly one transition for each 65B transmit block processed. Though the state diagram sends Local Fault ordered sets when reset is asserted, the scrambler and 65B RS-FEC are not guaranteed to be operational during reset. Thus, the Local Fault ordered sets are not guaranteed to appear on the PMA service interface.

The PCS 64B/65B Receive state diagram is shown in Figure 149–18 and Figure 149–19 and controls the decoding of 65B received blocks. It makes exactly one transition for each receive block processed except for the transition from `RX_WE` to `RX_E`, which occurs immediately after the `RX_WE` processes are complete.

The PCS shall perform the functions of RFER monitor, Transmit, and Receive as specified in these state diagrams. The PCS shall not perform the RFER monitor function during LPI receive operation from the time that the PCS 64B/65B Receiver enters the state `RX_L`, until the state `RX_W` is exited.

Transitions surrounded by dashed rectangles indicate requirements for 25GBASE-T1 EEE-capable implementations.

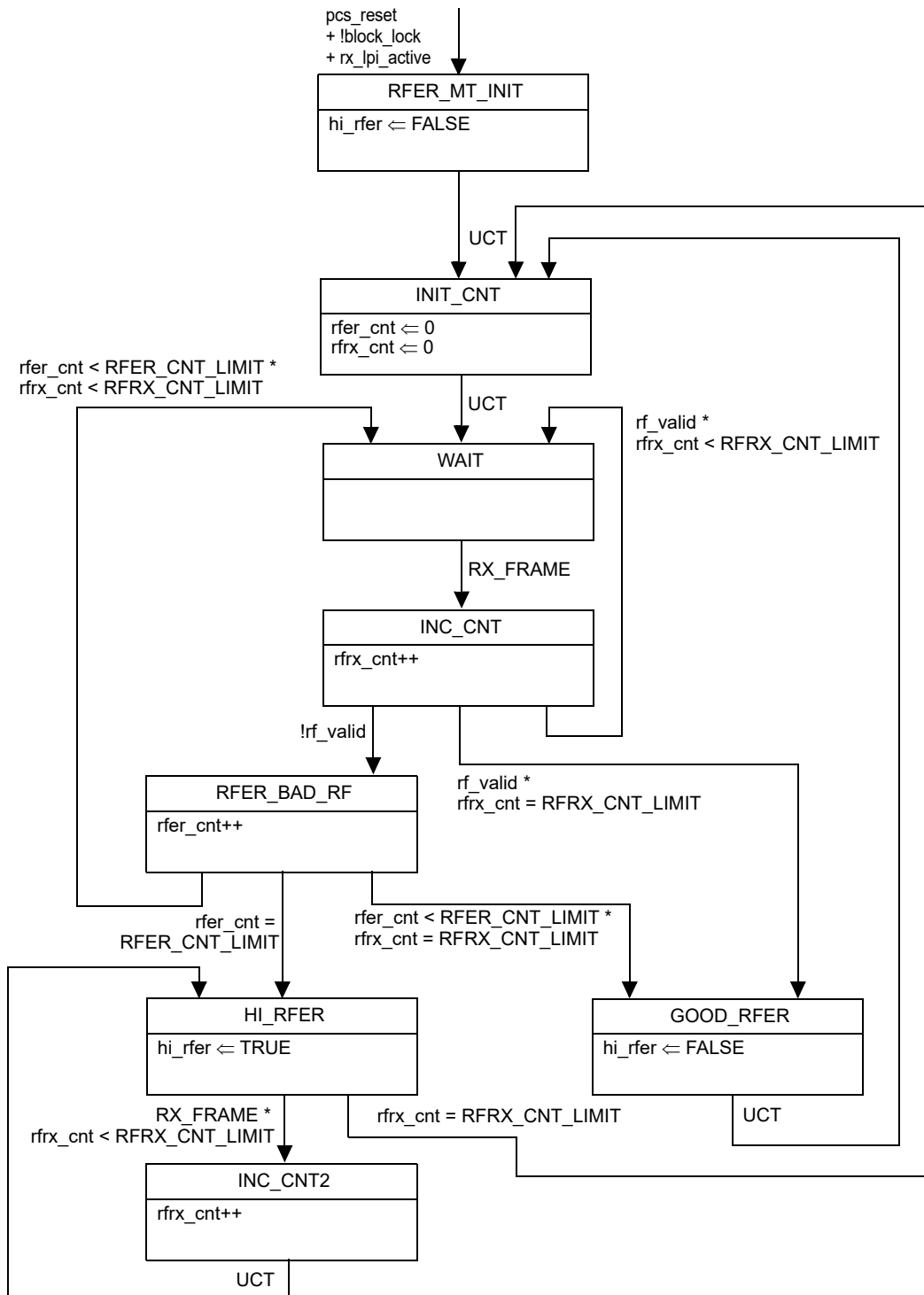
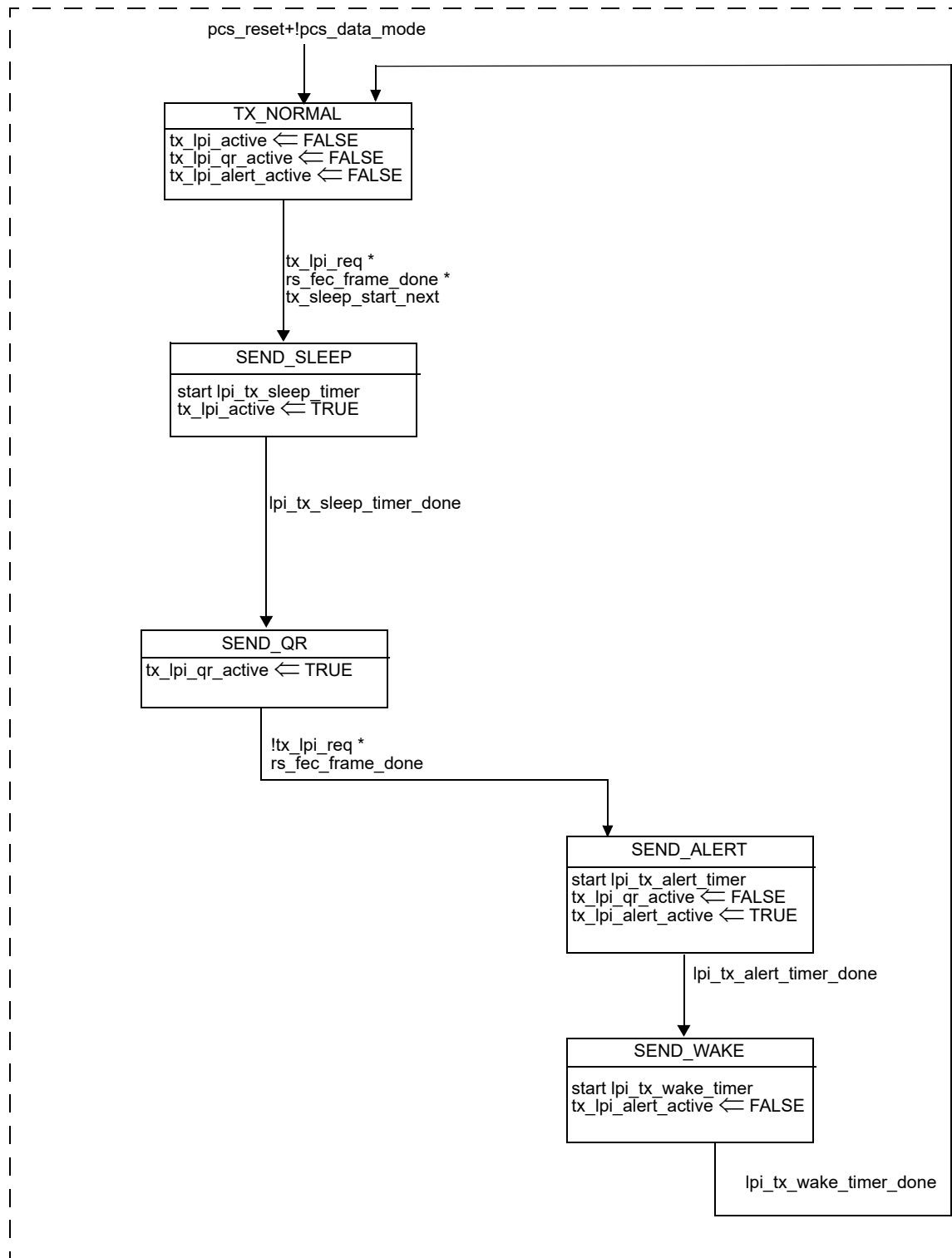


Figure 165–13—RFER monitor state diagram



NOTE—The functionality in this figure is mandatory for a PHY with the EEE capability.

Figure 165–14—EEE transmit state diagram

165.3.8 PCS management

PCS management is as defined in 149.3.8 with the exception that 25GMII replaces XGMII.

165.3.9 25GBASE-T1 operations, administration, and maintenance (OAM)

The 25GBASE-T1 PCS level operations, administration, and maintenance (OAM) provides an optional mechanism useful for monitoring link operation such as exchanging PHY link health status and message exchange. When OAM is implemented, behavior is defined in 149.3.9, including the state diagrams in Figure 149–24 and Figure 149–25.

The OAM frame data is carried in the OAM 10-bit field described in 165.3.2.2.14 for the normal power data mode and 165.3.6.3 for low power mode.

165.3.9.1 Definitions

The definitions for OAM are as defined in 149.3.9.1 for OAM frame, OAM symbol, OAM message, and OAM status.

OAM field: A 10-bit field in each RS-FEC frame reserved for the OAM symbol as described in 165.3.2.2.14 or in each refresh cycle as described in 165.3.6.3.

165.3.9.2 Functional specifications

25GBASE-T1 OAM functions are the same as those defined for a MultiGBASE-T1 PHY in 149.3.9.2 with the exception that the 25GBASE-T1 OAM frame structure is defined in 165.3.9.2.1.

165.3.9.2.1 25GBASE-T1 OAM frame structure

The 25GBASE-T1 OAM frame structure is defined in 149.3.9.2.1 with the addition of the behavior with 8x interleaving.

When the PCS frame is operating in interleaved mode of 2x, 4x, or 8x the first symbol (OAM<0>) shall be inserted in the first RS frame in the superframe so that the full OAM frame can be packed into eight superframes in the 2x interleaved mode, into four superframes in the 4x interleaved mode, and into two superframes in the 8x interleaved mode.

When transitioning from Low Power Idle to normal operation, it is possible that part of the OAM frame is sent over the LPI refreshes and the remainder of the OAM frame is packed in RS frames. In this case the first eight frames of Wake shall contain dummy OAM symbols. After that, in some instances, it is possible that the remainder of the OAM frame cannot be aligned where the next OAM frame will have its first symbol (OAM<0>) inserted in the first RS frame of the superframe. In such cases, the partially transmitted OAM frame shall be interrupted according to the following rules:

- 1x interleaving — no interruption is required.
- 2x interleaving — insert 0 or 1 dummy OAM symbol into the superframe for alignment before continuing.
- 4x interleaving — insert 0 to 3 dummy OAM symbols into the superframe for alignment before continuing.
- 8x interleaving — insert 0 to 7 dummy OAM symbols into the superframe for alignment before continuing.

The receiver can anticipate when the dummy OAM symbols will be inserted based on the sequence of refresh and wake events.

The dummy OAM symbol is all 0's and its value is ignored at the receiver.

An example, in Figure 165–15, shows the fixed eight dummy OAM symbols followed by the two dummy OAM symbols inserted to realign OAM<0> to the 4x interleaved superframe boundary.

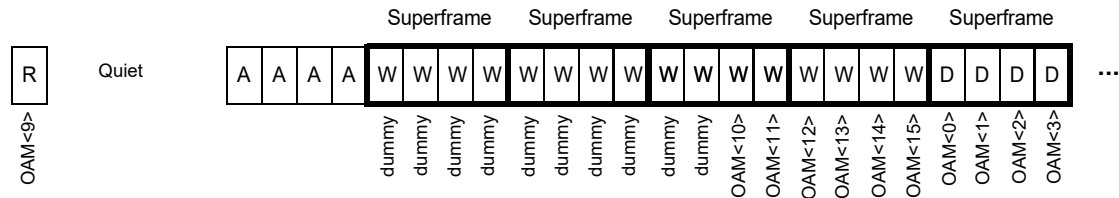


Figure 165–15—Example of 4x interleave dummy data insertion

165.3.9.3 State diagram variable to OAM register mapping

See 149.3.9.3.

165.3.9.4 Detailed functions and state diagrams

165.3.9.4.1 State diagram conventions

The body of this subclause specifies directly (and by reference) state diagrams, including the associated definitions of variables, counters, and functions. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

165.3.9.4.2 State diagram parameters

165.3.9.4.3 Variables

See 149.3.9.4.3 except for the variables rx_boundary and tx_boundary, which are defined as:

rx_boundary

This variable is set to TRUE whenever the receive data stream reaches the end of an RS-FEC frame and a dummy OAM symbol is not expected per 165.3.9.2.1 during normal power operation in the data mode, or at the end of a received refresh cycle during Low Power Idle operation. This variable is set to FALSE at other times.

Values:

FALSE: Receive stream is not at a boundary end.

TRUE: Receive stream is at a boundary end.

tx_boundary

This variable is set to TRUE whenever the transmit data stream reaches the start of an RS-FEC frame and a dummy OAM symbol is not transmitted per 165.3.9.2.1 during normal power operation in the data mode, or at the start of a transmit refresh cycle during Low Power Idle operation. This variable is set to FALSE at other times.

Values:

FALSE: Transmit stream is not at a boundary end.

TRUE: Transmit stream is at a boundary end.

165.3.9.4.4 Counters

See 149.3.9.4.4.

165.3.9.4.5 Functions

See 149.3.9.4.5.

165.3.9.4.6 State diagrams

See 149.3.9.4.6.

165.4 Physical Medium Attachment (PMA) sublayer

165.4.1 PMA functional specifications

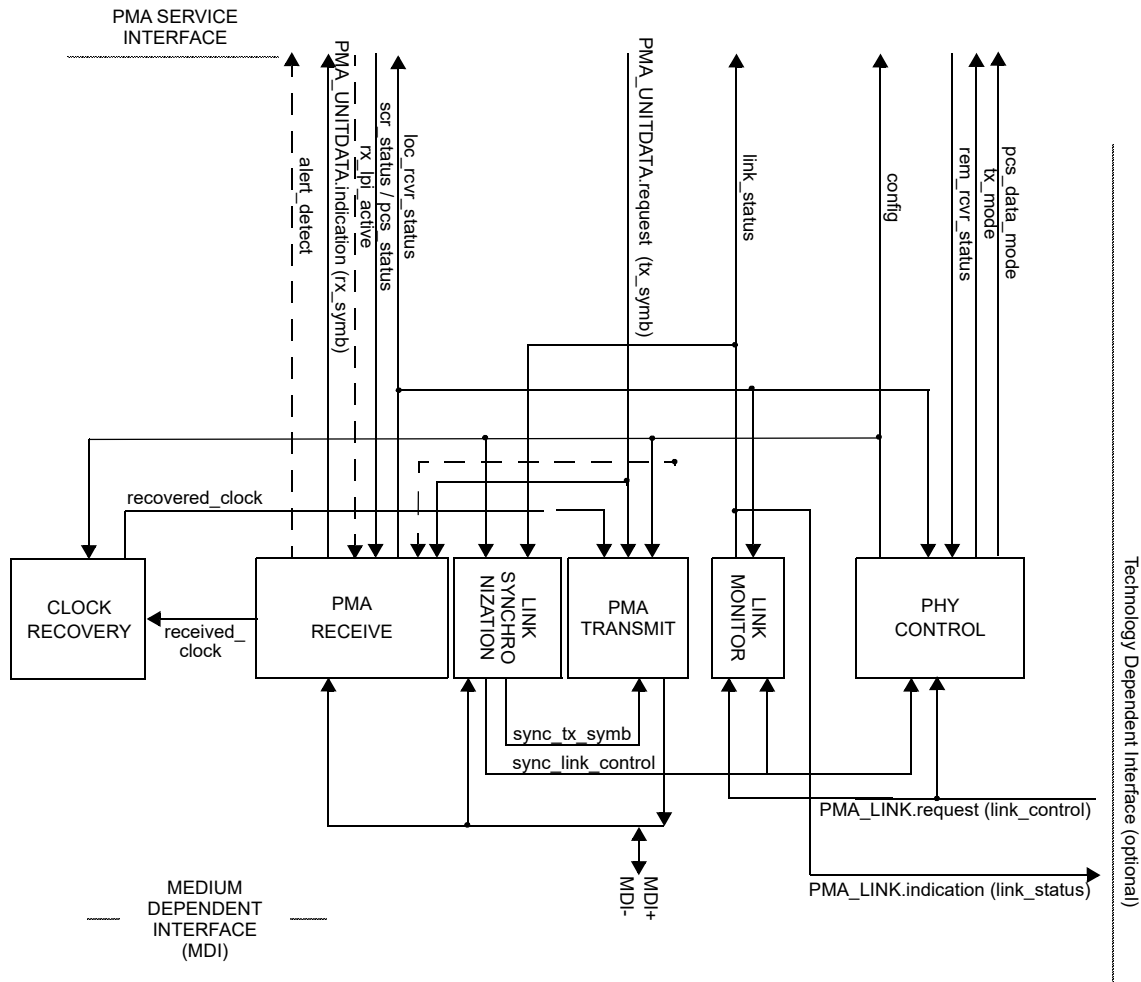
The PMA couples messages from the PMA service interface specified in 165.2.2 to the 25GBASE-T1 baseband medium, specified in 165.7.

The interface between the PMA and the baseband medium is the Medium Dependent Interface (MDI), which is specified in 165.8.

165.4.2 PMA functions

The PMA sublayer comprises one PMA Reset function and five simultaneous and asynchronous operating functions. The PMA operating functions are PHY Control, PMA Transmit, PMA Receive, Link Monitor, and Clock Recovery. All operating functions are started immediately after the successful completion of the PMA Reset function.

The PMA reference diagram, Figure 165–16, shows how the operating functions relate to the messages of the PMA service interface and the signals of the MDI. Connections from the management interface, comprising the signals MDC and MDIO, to other layers are pervasive and are not shown in Figure 165–16.



NOTE—The recovered_clock arc is shown to indicate delivery of the recovered clock signal back to PMA TRANSMIT for loop timing.

Figure 165–16—PMA reference diagram

165.4.2.1 PMA Reset function

The PMA Reset function shall be executed whenever one of the two following conditions occur:

- Power for the device containing the PMA has not reached the operating state.
- The receipt of a request for reset from the management entity.

PMA Reset sets pma_reset = ON while any of the above reset conditions hold TRUE. All state diagrams take the open-ended pma_reset branch upon execution of PMA Reset. The reference diagrams do not explicitly show the PMA Reset function.

The 25GBASE-T1 PMA takes no longer than 100 ms to enter the PCS_DATA state after exiting from reset or low power mode (see Figure 165–22).

165.4.2.2 PMA Transmit function

The PMA Transmit function comprises a transmitter to generate a four-level modulated signal on the single balanced pair of conductors. The PMA Transmit function also generates a zero output symbol when required for training, test, or EEE operation. When the PHY Control state diagram (Figure 165–22) is not in the DISABLE_TRANSMITTER state, PMA Transmit shall continuously transmit pulses modulated by the symbols given by tx_symb onto the MDI. During Link Synchronization, when sync_link_control = DISABLE and Auto-Negotiation is either not enabled or is not implemented, the sync_tx_symb output by the PHY Link Synchronization function shall be used in place of tx_symb as the data source for PMA Transmit. When lpi_tx_mode = ALERT, the PN sequence defined in 165.4.2.6 shall be used in place of tx_symb as the data source for PMA Transmit. The signals generated by PMA Transmit shall comply with the electrical specifications given in 165.5.3.

When the PMA_CONFIG.indication parameter config is MASTER, the PMA Transmit function shall source TX_TCLK from a local clock source while meeting the transmit jitter requirements of 165.5.3.3. The MASTER-SLAVE relationship shall include loop timing. If the PMA_CONFIG.indication parameter config is SLAVE, the PMA Transmit function shall source TX_TCLK from the recovered clock of 165.4.2.8 while meeting the jitter requirements of 165.5.3.3.

The PMA Transmit fault function is optional. The faults detected by this function are implementation specific. If the MDIO Interface is implemented, then this function shall be mapped to the transmit fault bit as specified in 45.2.1.7.4.

165.4.2.2.1 Global PMA transmit disable

When the PMA_transmit_disable variable is set to TRUE, this function shall turn off the transmitter so that the average launch power of the transmitter is less than –53 dBm.

165.4.2.3 PMA Receive function

The PMA Receive function comprises a receiver for PAM4 signals on the balanced pair. PMA Receive contains the circuits necessary to both detect symbol sequences from the signals received at the MDI over the receive pair and to present these sequences to the PCS Receive function. The PMA translates the signals received on the pair into the PMA_UNITDATA.indication parameter rx_symb. The quality of these symbols shall allow an RFER of less than 2×10^{-10} after RS-FEC decoding, over a channel meeting the requirements of 165.7.

To achieve the indicated performance, it is highly recommended that PMA Receive include the functions of signal equalization and echo cancellation. The sequence of symbols assigned to tx_symb is needed to perform echo cancellation.

The PMA Receive function uses the parameters pcs_status and scr_status, and the state of the equalization, cancellation, and estimation functions to determine the quality of the receiver performance, and generates the loc_rcvr_status variable accordingly. The loc_rcvr_status variable is expected to become NOT_OK when the link partner's tx_mode changes to SEND_Z from any other value (see the PHY Control state diagram in Figure 165–22). Note that during quiet-refresh signal periods, QUIET is represented by periods of zeros, and this should not, in itself, trigger the loc_rcvr_status variable to indicate NOT_OK. The precise algorithm for generation of loc_rcvr_status is implementation dependent.

The receiver uses the sequence of symbols during the training sequence to detect and correct for pair polarity swaps.

The PMA Receive fault function is optional. The PMA Receive fault function is the logical OR of the link_status = FAIL and any implementation specific fault. If the MDIO Interface is implemented, then this function shall contribute to the receive fault bit specified in 45.2.1.7.5 and 45.2.1.243.7.

165.4.2.4 PHY Control function

PHY Control generates the control actions that are needed to bring the PHY into a mode of operation during which frames can be exchanged with the link partner. PHY Control shall comply with the state diagram in Figure 165–22.

During PMA training (TRAINING and COUNTDOWN states in Figure 165–22), PHY Control information is exchanged between link partners with a 12-octet Infocfield, which is XORed with the first 96 bits of the 16th partial frame (bits 17550 to 17645) of each training frame. The Infocfield is also denoted IF. The link partner is not required to decode every IF transmitted but is required to decode IFs at a rate that enables the correct actions prior to the PAM2 to PAM4 transition.

The 12-octet Infocfield shall include the fields in 165.4.2.4.2 through 165.4.2.4.8, also shown in Figure 165–17 and Figure 165–18. The Infocfield shall be transmitted at least 256 times with each change to octets 7 to 10.

PMA_state = 00

octet 1	octet 2	octet 3	octets 4/5/6	octet 7	octets 8/9/10	octets 11/12
0xBB	0xA7	0x00	PFC24	Message	PHY capability bits	CRC16

Figure 165–17—Infocfield TRAINING format

PMA_state = 01

octet 1	octet 2	octet 3	octets 4/5/6	octet 7	octets 8/9/10	octets 11/12
0xBB	0xA7	0x00	PFC24	Message	DataSwPFC24	CRC16

Figure 165–18—Infocfield COUNTDOWN format

165.4.2.4.1 Infocfield notation

For all the Infocfield notations in the following subclauses, Reserved<bit location> represents any unused values and shall be set to zero on transmit and ignored when received by the link partner. The Infocfield is transmitted following the notation where the LSB of each octet is sent first and the octets are sent in increasing number order (that is, the LSB of Octet 1 is sent first).

165.4.2.4.2 Start of frame delimiter

The start of frame delimiter consists of three octets [Octet 1<7:0>, Octet 2<7:0>, Octet 3<7:0>] and shall use the hexadecimal value 0xBBA700. 0xBB corresponds to Octet 1<7:0> and so forth.

165.4.2.4.3 Partial frame count (PFC24)

The partial frame count consists of 3 octets [Octet 4<7:0>, Octet 5<7:0>, Octet 6<7:0>] and indicates the running count of partial frames sent LSB first. The partial frame count is increased by one for every 1170 PMA_UNITDATA clocks. There are 16 partial frames per training frame and the Infobfield is embedded within the 16th partial frame. The first partial frame is zero, thus the first partial frame count field after a reset is 15.

PFC24 continues to run uninterrupted for the duration of the link. The resolution of PFC24 is large enough that it does not rollover during the allotted training time. However, it will rollover if allowed to run indefinitely. PFC24 is defined to rollover to 0 after it reaches 16 776 959 to align with the IEEE QR cycle.

165.4.2.4.4 Message Field

The Message Field is one octet. For the MASTER, this field is represented by Octet 7 {PMA_state<7:6>, loc_rcvr_status<5>, en_slave_tx<4>, reserved<3:0>}. For the SLAVE, this field is represented by Octet 7 {PMA_state<7:6>, loc_rcvr_status<5>, timing_lock_OK<4>, reserved<3:0>}.

The two state-indicator bits PMA_state<7:6> shall communicate the state of the transmitting transceiver to the link partner. PMA_state<7:6> = 00 indicates TRAINING, and PMA_state<7:6> = 01 indicates COUNTDOWN.

All possible Message Field settings are listed in Table 165–7 for the MASTER and Table 165–8 for the SLAVE. Any other value shall not be transmitted and shall be ignored at the receiver. The Message Field setting for the first transmitted PMA frame shall be the first row of Table 165–7 for the MASTER and the first or second row of Table 165–8 for the SLAVE. Moreover, for a given Message Field setting, the next Message Field setting shall be the same Message Field setting or the Message Field setting corresponding to a row below the current setting. When loc_rcvr_status = OK the Infobfield variable is set to loc_rcvr_status<5> = 1 and set to 0 otherwise.

Table 165–7—Infobfield message field valid MASTER settings

PMA_state<7:6>	loc_rcvr_status	en_slave_tx	reserved	reserved	reserved	reserved
00	0	0	0	0	0	0
00	0	1	0	0	0	0
00	1	1	0	0	0	0
01	1	1	0	0	0	0

Table 165–8—Infobfield message field valid SLAVE settings

PMA_state<7:6>	loc_rcvr_status	timing_lock_OK	reserved	reserved	reserved	reserved
00	0	0	0	0	0	0
00	0	1	0	0	0	0
00	1	1	0	0	0	0
01	1	1	0	0	0	0

165.4.2.4.5 PHY capability bits

When PMA_state<7:6> = 00, [Octet 8<7:0>, Octet 9<7:0>, Octet 10<7:0>] contains the PHY capability bits. Each octet is sent LSB first. See Table 165–9 for the details.

Table 165–9—PHY capability bits

Octet 8							Octet 9							Octet 10									
0	1	2	3	4	5	6	7	0	1	2	3	4	5	6	7	0	1	2	3	4	5	6	7
VendorSpecificData															Reserved	InterleaverDepth	PrecodeSel	SlowWakeRequest	EEEn	OAMen			

The format of PHY capability bits is Octet 10<2:1> = InterleaverDepth[1:0], Octet 10<4:3> = PrecodeSel[1:0], Octet 10<5> = SlowWakeRequest, Octet 10<6> = EEEen, Octet 10<7> = OAMen, Octet 8<7:0> = VendorSpecificData[7:0], and Octet 9<7:0> = VendorSpecificData[15:8].

EEEen and OAMen indicate EEE and 25GBASE-T1 OAM capability enable, respectively. The PHY shall indicate the support of these two optional capabilities by setting the corresponding capability bits.

The optional EEE capability shall be enabled only if both PHYs set the capability bit EEEen = 1. The optional BASE-T1 OAM capability shall be enabled only if both PHYs set the capability bit OAMen = 1. InterleaverDepth indicates the requested data mode interleaving depth. PrecodeSel indicates the requested precoder. SlowWakeRequest = 1 indicates to the link partner that the PHY will transmit alert once per QR cycle (see 165.3.6.1).

The capability bit values shall be considered as valid only when the loc_rcvr_status bit is 1.

Reserved bits shall be transmitted as 0 and ignored upon receipt.

165.4.2.4.6 Data switch partial frame count

When PMA_state<7:6> = 01, [Octet 8<7:0>, Octet 9<7:0>, Octet 10<7:0>] contains the data switch partial frame count (DataSwPFC24) sent LSB first. DataSwPFC24 indicates the partial frame count when the transmitter switches from PAM2 to PAM4, which occurs at the start of an RS-FEC superframe. The last value of PFC24 prior to the transition is DataSwPFC24 – 1. DataSwPFC24 shall be set to an integer multiple of 32. DataSwPFC24 shall be a minimum of 4081 and a maximum of 4785 from the current PFC24 value.

165.4.2.4.7 Reserved fields

When PMA_state<7:6> is greater than 01, [Octet 8<7:0>, Octet 9<7:0>, Octet 10<7:0>] contains a reserved field. All reserved fields are transmitted as 0 and ignored upon receipt.

165.4.2.4.8 CRC16

The CRC16 (2 octets) shall implement the CRC16 polynomial $(x + 1)(x^{15} + x + 1)$ of the previous 7 octets, Octet 4<7:0>, Octet 5<7:0>, Octet 6<7:0>, Octet 7<7:0>, Octet 8<7:0>, Octet 9<7:0>, and Octet 10<7:0>. The CRC16 shall produce the same result as the implementation shown in Figure 165–19. In Figure 165–19 the 16 delay elements $S_0, \dots, S_{15}$, shall be initialized to zero. After initialization, the switch is set to CRCgen, as shown in Figure 165–19, and Octet 4 through Octet 10 are used to compute the CRC16 output. After all 7 octets have been processed, the switch is disconnected (setting CRCout) and the 16 values stored in the delay elements are transmitted in the order illustrated, first S_{15} , followed by S_{14} , and so on, until the final value S_0 .

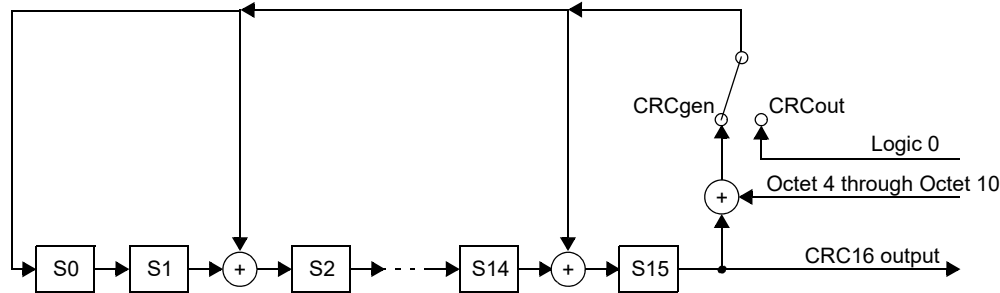


Figure 165–19—CRC16

165.4.2.4.9 PMA MDIO function mapping

See 149.4.2.4.9.

165.4.2.4.10 Startup sequence

The startup sequence shall comply with the state diagram description given in Figure 165–22. If the Auto-Negotiation function is not implemented, or disabled (`mr_autoneg_en = FALSE`), PMA_CONFIG is predetermined to be MASTER or SLAVE via management control during initialization or via default hardware setup. The Auto-Negotiation function is optional for a 25GBASE-T1 PHY. If the Auto-Negotiation function is implemented and enabled, Auto-Negotiation is the source of control (via `link_control`) and MASTER/SLAVE configuration; however, if Auto-Negotiation is either not enabled or is not implemented, the Link Synchronization function is the source of control (via `sync_link_control`) and MASTER/SLAVE configuration.

During startup, prior to entering the TRAINING state, the SLAVE shall align its transmit 65B RS-FEC frame to within +0/–4 partial PHY frames of the MASTER as seen at the SLAVE MDI. The SLAVE Infocfield partial frame count shall match the MASTER Infocfield partial frame count for the aligned frame.

In the TRAINING state, PAM 2 transmission is used and PHY capabilities are exchanged with Infocfields as specified in 165.4.2.4.5.

In the TRAINING state, whenever the SLAVE operating in loop timing locks the MASTER timing reference, it sets `timing_lock_OK=1`.

At any time following the TRAINING state, if the local receiver status (indicated by `loc_rcvr_status`) transitions to NOT_OK, PHY Control returns to the SILENT state and attempts a retrain.

The startup timing shall comply with Table 165–10 for MASTER and Table 165–11 for SLAVE.

Table 165–10—Startup timing maxima for MASTER

Timing interval	Maximum time (ms)
From entry to SILENT state until en_slave_tx = 1 is transmitted	29.616
From entry of SILENT state until entry to COUNTDOWN state	95.591
Entry to COUNTDOWN until entry of TX_SWITCH	0.384
Entry to exit of PCS TEST	1.025
Total (Entry to SILENT to exit of PCS TEST)	97

Table 165–11—Startup timing maxima for SLAVE

Timing interval	Maximum time (ms)
Entry to exit of SILENT state	30
Entry of SILENT state to exit of TRAINING state	95.591
Entry to COUNTDOWN until entry of TX_SWITCH	0.384
Entry to exit of PCS TEST	1.025
Total (Entry to SILENT to exit of PCS TEST)	97

165.4.2.5 Link Monitor function

Link Monitor determines the status of the underlying receive channel and communicates it via the variable link_status. Failure of the underlying receive channel causes the PMA to set link_status to FAIL, which in turn causes the PMA’s clients to stop exchanging frames and restart the Auto-Negotiation (if enabled) or Link Synchronization (if Auto-Negotiation is not enabled) process.

The Link Monitor function shall comply with the state diagram of Figure 165–23.

Upon power on, reset, or release from power down, the Auto-Negotiation function sets link_control = DISABLE, or PHY Link Synchronization algorithms set sync_link_control = DISABLE. During this period, link_status = FAIL is asserted. When the Auto-Negotiation function establishes the presence of a remote 25GBASE-T1 PHY, link_control is set to ENABLE, or when the PHY Link Synchronization finishes the synchronization function, sync_link_control is set to ENABLE, and the Link Monitor state diagram begins monitoring the PCS and receiver lock status. As soon as reliable transmission is achieved, the variable link_status = OK is asserted, upon which further PHY operations can take place.

165.4.2.6 PHY Link Synchronization

If the optional Clause 98 Auto-Negotiation function is disabled or not implemented, then the Link Synchronization function shall establish the start of PHY PMA training as defined in 165.4.2.4.

When operating, the Link Synchronization function is the data source for the PMA Transmit function (see 165.4.2.2), and uses a signal, SEND_S. This signal is used by the MASTER and SLAVE to discover the link partner and synchronize the start of PMA training.

Link Synchronization employs the SEND_S signal to achieve synchronization prior to link training. If the PHY is configured as MASTER, Link Synchronization shall employ Equation (165–5) as the PN sequence generator.

$$p_M(x) = x^8 + x^4 + x^3 + x^2 + 1 \quad (165-5)$$

If the PHY is configured as SLAVE, Link Synchronization shall employ Equation (165–6) as PN sequence generator.

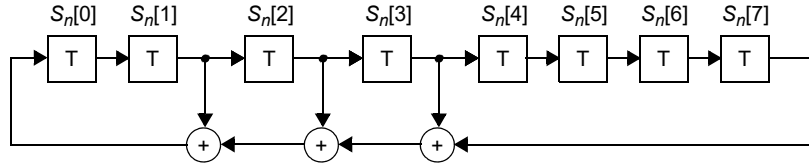
$$p_S(x) = x^8 + x^6 + x^5 + x^4 + 1 \quad (165-6)$$

The period of both PN sequences is 255.

An implementation of MASTER and SLAVE PHY SEND_S PN sequence generators by linear-feedback shift registers is shown in Figure 165–20. The bits stored in the shift register delay line at time n are denoted by $S_n[7:0]$. The symbols of the shift register sequence $S_n[]$ shall be generated at a rate of 703.125 MHz. At each symbol period, the shift register is advanced by one bit, and one new bit represented by $S_n[0]$ is generated. The PN sequence generator shift registers shall be reset to a value of $S_n[7:0] = 0000\ 0001$ upon entering into the TRANSMIT_DISABLE state (see Figure 165–20) or on the transmission of the first symbol of the alert sequence.

For 25GBASE-T1, the bit $S_n[0]$ shall be mapped to the transmit symbol T_n as follows: if $S_n[0] = 0$ then $T_n = +1\ +1\ \dots\ +1$ (repeated 20 times), if $S_n[0] = 1$ then $T_n = -1\ -1\ \dots\ -1$ (repeated 20 times).

MASTER PHY SEND_S PN sequence generator



SLAVE PHY SEND_S PN sequence generator

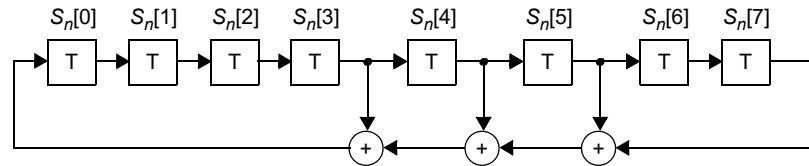


Figure 165–20—SEND_S PN sequence generator by linear feedback shift registers S_n

The PHY Link Synchronization state diagram in Figure 165–21 shall be used to synchronize 25GBASE-T1 PHYs prior to the 25GBASE-T1 link training. If the Clause 98 Auto-Negotiation function is enabled, then the Auto-Negotiation function shall be used as the mechanism for PHY synchronization and the PHY Link Synchronization state diagram in Figure 165–21 remains in the SYNC_DISABLE state.

165.4.2.6.1 State diagram variables

force_config

This variable indicates whether the PHY operates as a MASTER or as a SLAVE. The variable takes on one of the following values:

- MASTER: This value is continuously asserted when the PHY operates as a MASTER.
- SLAVE: This value is continuously asserted when the PHY operates as a SLAVE.

force_phy_type

This variable indicates what speed the PHY is to operate when Auto-Negotiation is disabled or not implemented. The variable takes on one of the following values:

- 25G-T1: If Auto-Negotiation is disabled or not implemented and 25GBASE-T1 is selected.

Other values are implementation-dependent and beyond the scope of this clause (for examples of other values see [Clause 97](#) and [Clause 149](#)).

link_status

The link_status parameter is set by PMA Link Monitor and passed to the PCS via the PMA_LINK.indication primitive. This variable takes the values of OK or FAIL.

min_send_s_cnt

The variable takes on the following value: 16, fixed value, not programmable.

mr_autoneg_enable:

see 98.5.1.

mr_main_reset

see 98.5.1.

power_on

see 98.5.1.

send_s_counter

An integer value that counts the number of frames of SEND_S signal being sent at SLAVE.

send_s_sigdet

This variable indicates whether the SEND_S signal was detected. This variable shall be set FALSE no later than 1 μ s after the signal goes quiet on the MDI.

Values:

- TRUE: SEND_S signal detected.
- FALSE: SEND_S signal not detected.

sync_link_control

This variable indicates the data source for the PMA Transmit function.

Values:

- DISABLE: The data source is the PHY Link Synchronization function (sync_tx_symb).
- ENABLE: The data source is PMA_UNITDATA.request (tx_symb).

165.4.2.6.2 State diagram timers

break_link_timer

see [98.5.2](#).

link_fail_inhibit_timer
see 98.5.2.

send_s_timer
This timer is used to control the duration SEND_S is transmitted. The timer shall expire $1.25\text{ }\mu\text{s} \pm 0.025\text{ }\mu\text{s}$ after being started.

sigdet_wait_timer
This timer is used to control the wait time after transmitting or detecting the end of SEND_S. The timer shall expire $5\text{ }\mu\text{s} \pm 0.025\text{ }\mu\text{s}$ after being started.

silent_wait_timer
This timer is used to avoid overlap of MASTER and SLAVE SEND_S signals. The timer shall expire $1.25\text{ }\mu\text{s} \pm 0.025\text{ }\mu\text{s}$ after being started.

165.4.2.6.3 Messages

sync_tx_symb
The value of sync_tx_symb is set by the Link Synchronization state diagram and indicates the symbols sent from the PHY Link Synchronization block to PMA Transmit. The Link Synchronization block generates sync_tx_symb synchronously with every transmit clock cycle.
Values:
SEND_S: Transmit the SEND_S signal defined in 165.4.2.6.
SEND_Z: Transmit a zero value.

165.4.2.6.4 State diagrams

The PHY Link Synchronization state diagram is shown in Figure 165–21.

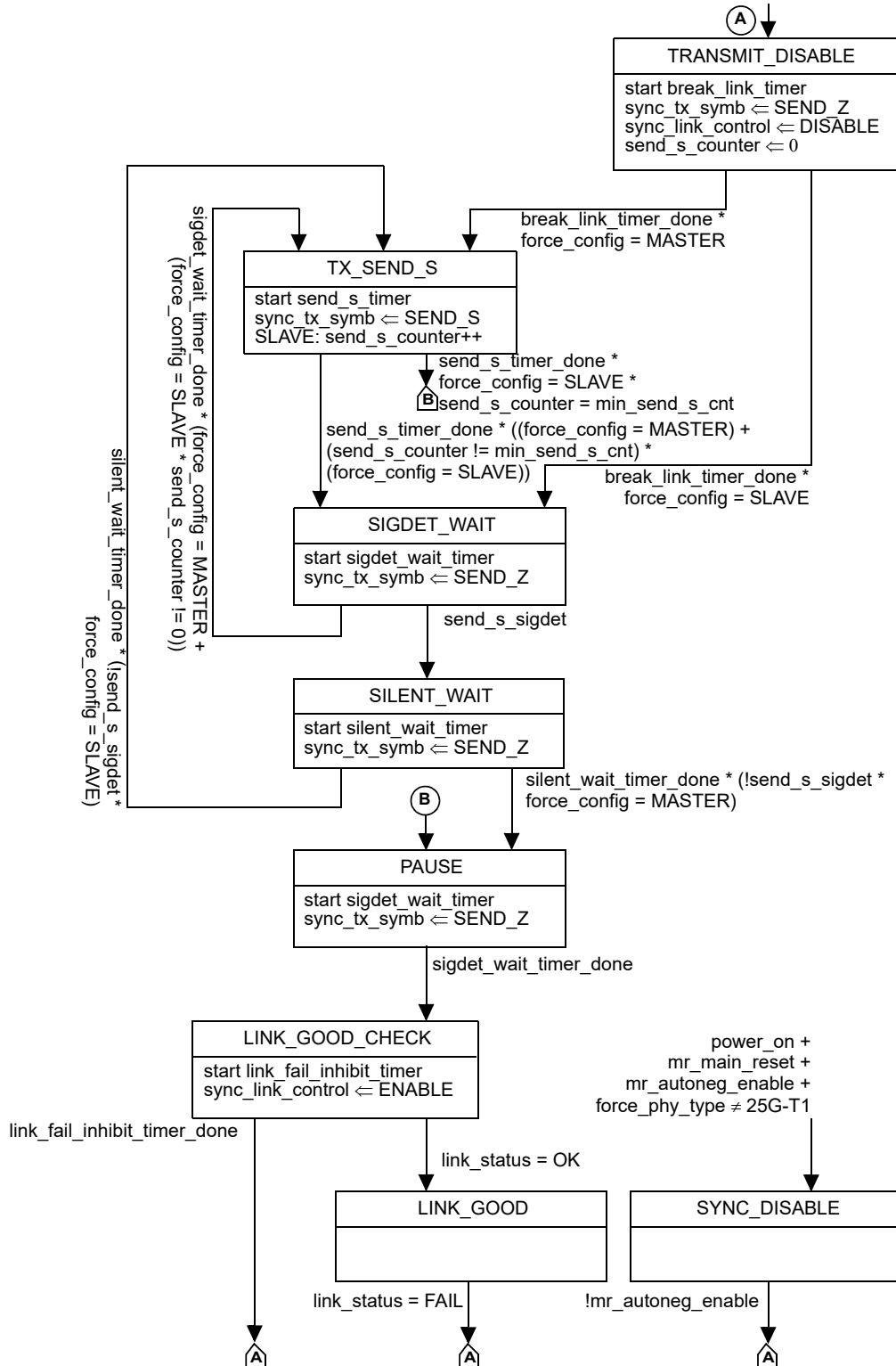


Figure 165–21—PHY Link Synchronization state diagram

165.4.2.7 Refresh monitor function

The Refresh monitor is required for a PHY that supports the EEE capability. The Refresh monitor operates when the PHY is in the LPI receive mode. The Refresh monitor shall comply with the state diagram of Figure 165–24. The Refresh monitor sets the PMA_refresh_status variable, which forces a link retrain if a refresh signal is not reliably detected within a moving time window equivalent to 50 complete quiet-refresh cycles (nominally equal to 1.59744 ms), when the PHY is in the lower power receive mode.

165.4.2.8 Clock Recovery function

The Clock Recovery function shall provide a clock suitable for signal sampling so that the RFER indicated in 165.4.2.3 is achieved. The received clock signal is expected to be stable and ready for use when training has been completed. The received clock signal is supplied to the PMA Transmit function by received_clock.

165.4.3 MDI

Communication through the MDI is summarized in 165.4.3.1 and 165.4.3.2.

165.4.3.1 MDI signals transmitted by the PHY

The symbols to be transmitted by the PMA are denoted by tx_symb. PMA Transmit generates a pulse-amplitude modulated signal in the following form:

$$s(t) = \sum_{n=0}^{\infty} a_n h_T(t - nT) \quad (165-7)$$

In Equation (165–7), a_n is the PAM4 modulation to be transmitted at time nT , and $h_T(t)$ denotes the system symbol response at the MDI. This symbol response shall comply with the electrical specifications given in 165.5.3.

When the tx_mode is SEND_N, tx_symb represents the PCS-encoded 25GMII data stream, a_n is the PAM4 modulation symbol from the set $\{-1, -1/3, +1/3, +1\}$.

During training or quiet-refresh signaling, tx_mode is SEND_Z or SEND_T. During training or quiet refresh signaling, the value of a_n is a PAM modulation symbol from the set $\{+1, 0, -1\}$.

165.4.3.2 Signals received at the MDI

Signals received at the MDI can be expressed as pulse-amplitude modulated signals that are corrupted by noise as follows:

$$r(t) = \sum_{n=0}^{\infty} a_n h_R(t - nT) + w(t) \quad (165-8)$$

In Equation (165–8), $h_R(t)$ denotes the symbol response of the overall channel impulse response between the transmit symbol source and the receive MDI and $w(t)$ represents the contribution of various noise sources including uncanceled echo. The receive signal is processed within the PMA Receive function to yield the received symbols rx_symb.

165.4.4 State variables

165.4.4.1 State diagram variables

auto_neg_imp

This variable indicates if an optional Auto-Negotiation sublayer is associated with the PMA.

Values:

- TRUE: An optional Auto-Negotiation sublayer is associated with the PMA.
- FALSE: An optional Auto-Negotiation sublayer is not associated with the PMA.

config

The PMA generates this variable continuously and passes it to the PCS via the PMA_CONFIG.indication primitive.

Values: MASTER or SLAVE.

en_slave_tx

The en_slave_tx variable in the Infocfield received by the SLAVE.

Values:

- 0: MASTER is not ready for the SLAVE to transmit.
- 1: MASTER is ready for the SLAVE to transmit.

infocfield_complete

This variable indicates that a complete set of Infocfield messages has been sent (see 165.4.2.4).

Values:

- FALSE: A complete set of Infocfield messages has not been sent.
- TRUE: A complete set of Infocfield messages has been sent.

link_control

This variable is defined in 165.2.1.1.1.

link_status

The link_status parameter set by PMA Link Monitor state diagram and communicated through the PMA_LINK.indication primitive.

Values: OK or FAIL.

loc_countdown_done

This variable is set to FALSE when the PHY Control state diagram is in the DISABLE_TRANSMITTER state and is set to TRUE immediately after transmitting the last bit of the DataSwPFC24 – 1 partial PHY frame.

loc_rcvr_status

Variable set by the PMA Receive function to indicate correct or incorrect operation of the receive link for the local PHY.

Values:

- OK: The receive link for the local PHY is operating reliably.
- NOT_OK: Operation of the receive link for the local PHY is unreliable.

loc_SNR_margin

This variable reports whether the local device has sufficient SNR margin to continue to the next state. The criterion for setting the parameter loc_SNR_margin is left to the implementer.

Values:

- OK: The local device has sufficient SNR margin.
- NOT_OK: The local device does not have sufficient SNR margin.

pcs_data_mode

Generated by the PMA PHY Control function and indicates whether or not the local PHY can transition its PCS state diagrams out of their initialization states. The current value of the pcs_data_mode is passed to the PCS via the PMA_PCSDATAMODE.indication primitive.

PMA_refresh_status

This variable indicates the status of the Refresh monitor as described in 165.4.2.7.

Values: OK or FAIL.

pma_reset

Allows reset of the PHY Control and Link Monitor state diagrams.

Values: ON or OFF.

PMA_state

Variable for the value transmitted in the PMA_state<7:6> of the Infobfield by the local PHY.

Values:

00: TRAINING state.

01: COUNTDOWN state.

rem_countdown_done

This variable is set to FALSE when the PHY Control state diagram is in the DISABLE_TRANSMITTER state or SILENT state and is set to TRUE once the receiver has transitioned from PAM2 to PAM4.

rem_rcvr_status

Variable set by the PCS Receive function to indicate whether correct operation of the receive link for the remote PHY is detected or not.

Values:

OK: The receive link for the remote PHY is operating reliably.

NOT_OK: Reliable operation of the receive link for the remote PHY is not detected.

sync_link_control

This variable is defined in 165.4.2.6.1.

tx_mode

The PMA generates this variable continuously and passes it to the PCS via the PMA_TXMODE.indication primitive.

Values:

SEND_N: This value is continuously asserted when transmission of sequences of symbols representing a 25GMII data stream takes place.

SEND_T: This value is continuously asserted when transmission of sequences of symbols representing the training sequences of symbols is to take place.

SEND_Z: This value is asserted when transmission of zero symbols is to take place.

165.4.4.2 Timers

All timers operate in the manner described in 14.2.3.2.

minwait_timer

A timer used to determine the minimum amount of time the PHY Control stays in the SILENT, TRAINING, PCS_TEST, and PCS_DATA states. The timer shall expire $975\ \mu\text{s} \pm 50\ \mu\text{s}$ after being started.

The following timer is required only for a PHY that support the EEE capability:

`lpi_refresh_rx_timer`

This timer is used to monitor link quality during the LPI receive mode. If the PHY does not detect reliable refresh signaling before this timer expires then a full retrain is performed.

Values: The condition `lpi_refresh_rx_timer_done` becomes TRUE upon timer expiration.

Duration: This timer shall have a period equal to 50 complete quiet-refresh signal periods, equivalent to 1.59744 ms.

165.4.5 State diagrams

The PHY Control state diagram is shown in Figure 165–22, the Link Monitor state diagram is shown in Figure 165–23, and the EEE Refresh monitor state diagram is shown in Figure 165–24.

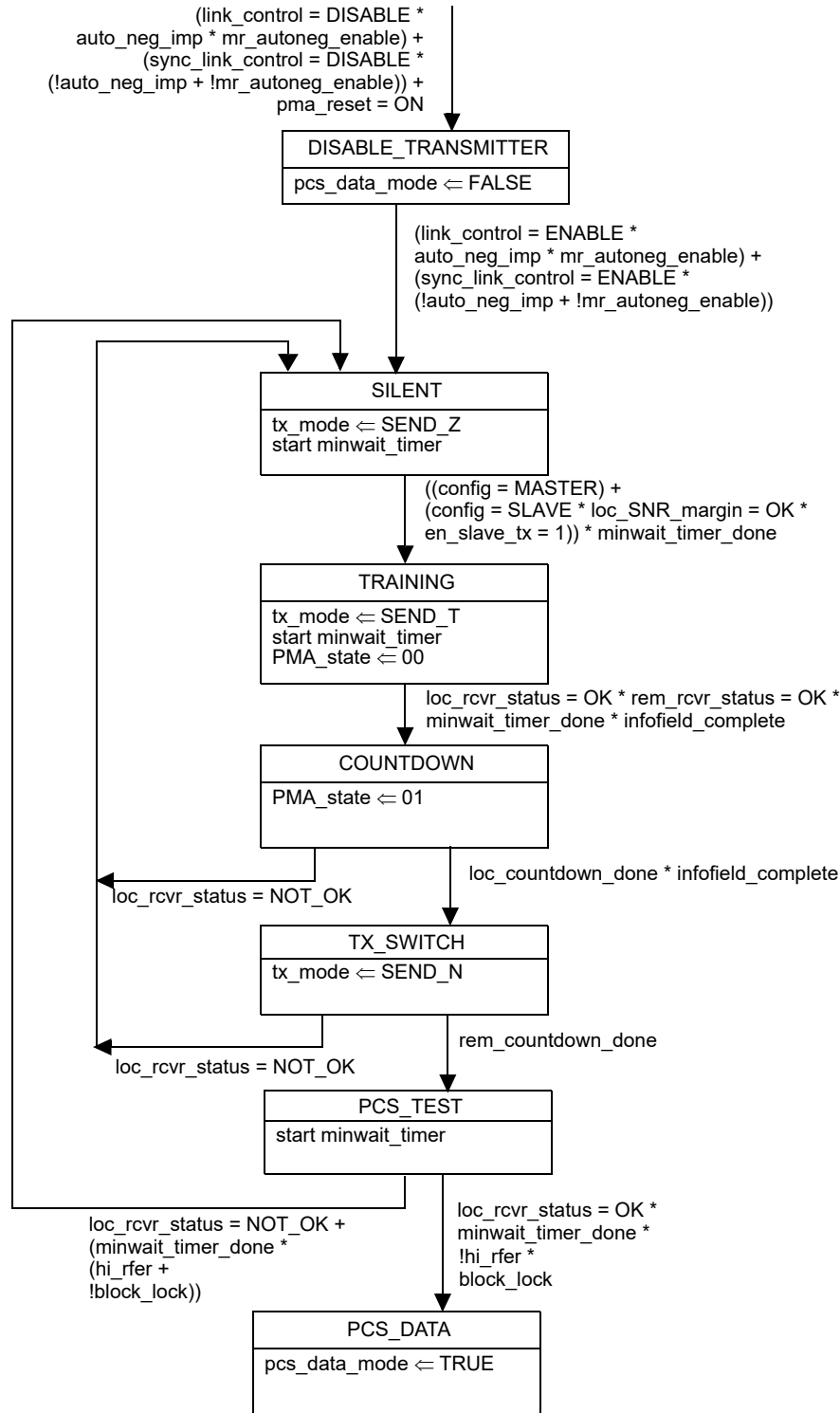
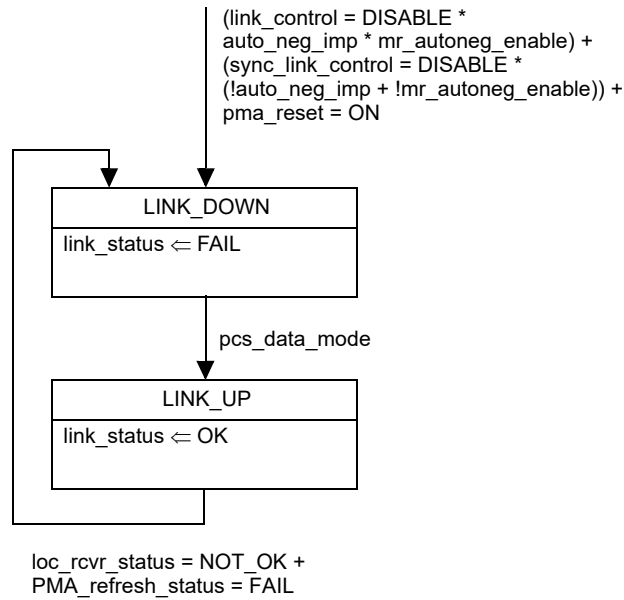
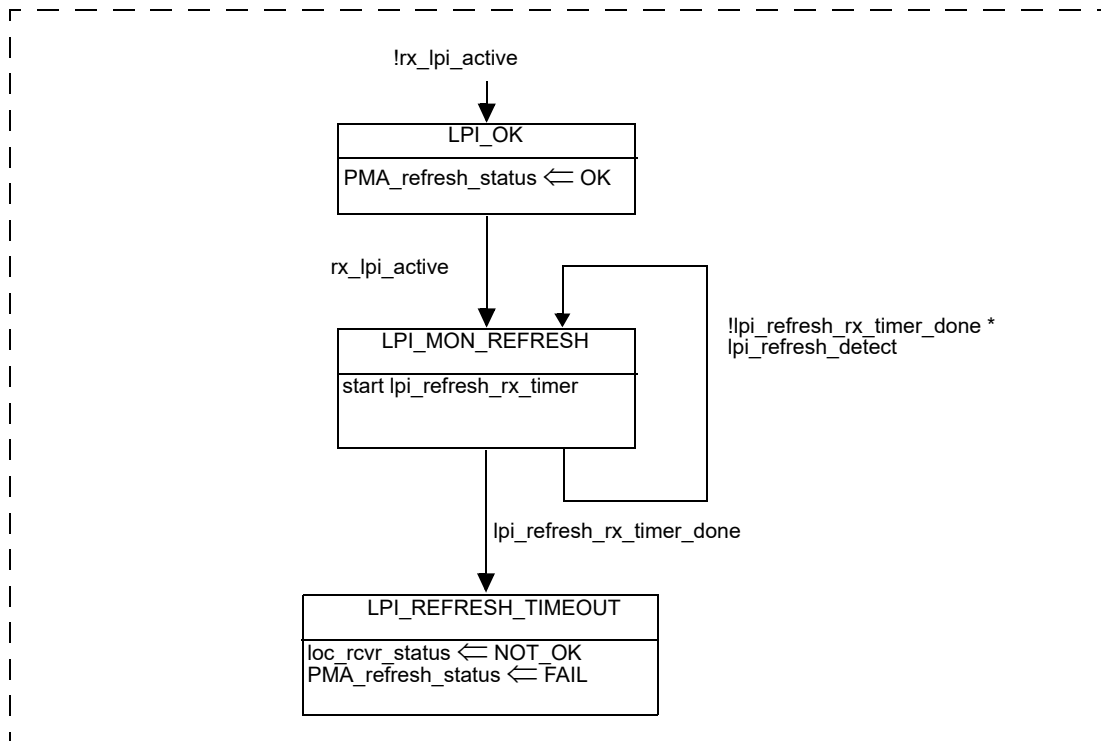


Figure 165–22—PHY Control state diagram



NOTE—The variables `link_control` and `link_status` are designated as `link_control_25GigT1` and `link_status_25GigT1`, respectively, by the Auto-Negotiation Arbitration state diagram (Figure 98–7) if the optional Auto-Negotiation function is implemented.

Figure 165–23—Link Monitor state diagram



NOTE—The functionality of this state diagram is only required when the PHY supports EEE.

Figure 165–24—EEE Refresh monitor state diagram

165.5 PMA electrical specifications

This subclause defines the electrical characteristics of the PMA and specifies PMA-to-MDI interface tests.

165.5.1 Test modes

The test modes described as follows shall be provided to allow for testing of the transmitter jitter, transmitter distortion, transmitter PSD, transmitter droop, and BER.

If MDIO is implemented, these test modes shall be enabled by setting a control register, 1.2313.15:13, as shown in Table 165–12. If MDIO is not implemented then equivalent functionality shall be provided. The test modes shall only change the data symbols provided to the transmitter circuitry and do not alter the electrical and jitter characteristics of the transmitter and receiver from those of normal (non-test mode) operation.

Table 165–12—MDIO management registers settings for test modes

Register value	Register description
000	Normal (non-test mode) operation.
001	Test mode 1—Setting MASTER and SLAVE PHYs for transmit clock jitter test in linked mode.
010	Test mode 2—Transmit MDI jitter test in MASTER mode.
011	Test mode 3—Precoder test mode.
100	Test mode 4—Transmitter linearity test.
101	Test mode 5—Normal operation in Idle mode. This is for the PSD Mask test.
110	Test mode 6—Transmitter droop test mode.
111	Test mode 7—Normal operation with zero data pattern. This is for BER monitoring.

Test mode 1 enables testing of timing jitter on MASTER and SLAVE transmitters. MASTER and SLAVE PHYs are connected over a link segment defined in 165.7. When in this mode, the PHY shall provide access to a frequency-divided version of its transmit symbol clock, with divisor 16, referred to as TX_TCLK_879. The nominal frequency of TX_TCLK_879 is 878.90625 MHz.

Test mode 2 is for transmitter jitter testing on the MDI when the transmitter is in MASTER timing mode. When test mode 2 is enabled, the PHY shall transmit a continuous pattern of JP03A (as specified in 94.2.9.1) or JP03B (as specified in 94.2.9.2) with the transmitted symbols timed from its local clock source.

Test mode 3 is for testing the precoder operation. When test mode 3 is enabled, the PCS shall generate a continuous pattern of {0, 3} symbols to be input to the transmit precoder specified in 165.3.2.2.20, to be precoded according to the transmit precoder settings as determined by the value set in register 1.2313:10:9, or equivalent functionality if MDIO is not implemented, and transmitted by the PMA timed from its local clock source.

Test mode 4 is for transmitter linearity testing. When test mode 4 is enabled, the PHY shall transmit a continuous pattern of PRBS13Q (as specified in 120.5.11.2.1).

Test mode 5 is for checking whether the transmitter is compliant with the transmit PSD mask and the transmit power level. When test mode 5 is enabled, the PHY shall transmit as in non-test operation and in the MASTER data mode with data set to normal interframe idle signals.

When test mode 6 is enabled, the PHY shall transmit a continuous pattern of 320 $\{+1\}$ symbols followed by 320 $\{-1\}$ symbols with the transmitted symbols timed from its local clock source.

Test mode 7 is for enabling measurement of the RS-FEC block error ratio of a link partner in a link between two PHYs, including RS-FEC encoder and decoder, transmit and receive analog front ends, and a cable connecting the PHYs. This mode reuses the 25GBASE-T1 normal (non-test) mode with zero data pattern. Instead of encoding data received from the MAC, continuous zero data pattern is encoded. On the receive side, after PCS FEC decoding processing, a zero data sequence is expected with no errors. Any block received with non-zero data bits is counted as an error and calculated in the RS-FEC block error ratio.

165.5.1.1 Test configurations

The following configurations, or their equivalents, as shown in Figure 165–25 and Figure 165–26, in stated respective tests, are defined for measuring the transmitter specifications for data communication only.

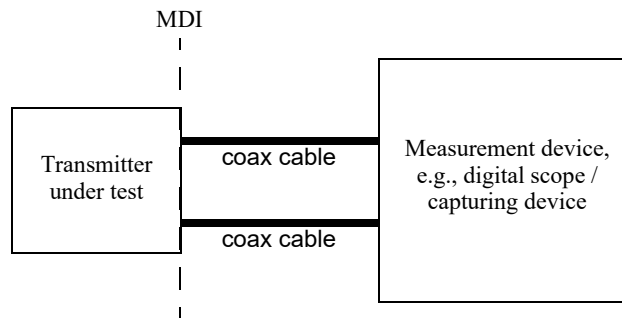


Figure 165–25—Transmitter test configuration 1 for transmitter droop, transmitter linearity, jitter and power spectral density measurement, and transmit power level measurements

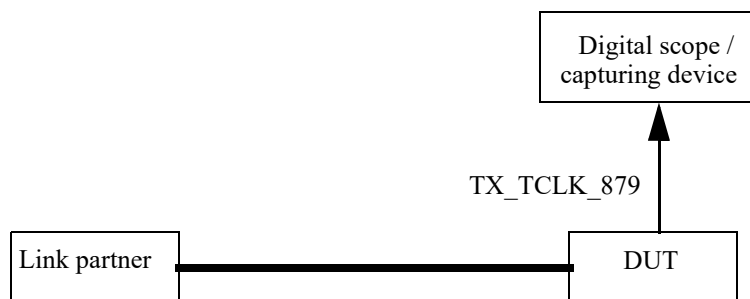


Figure 165–26—Transmitter test configuration 2 for MASTER and SLAVE clock jitter measurement

165.5.2 Test Points

For reference, test points for a Clause 165 PHY are defined in one direction as shown in Figure 165–27. A Clause 165 PHY uses bidirectional signal lines in full duplex operation. As a result, TP1 and TP4 are

physically identical, as are TP0 and TP5. They are shown here separately for reference in discussing parameters at the transmitter vs the receiver.

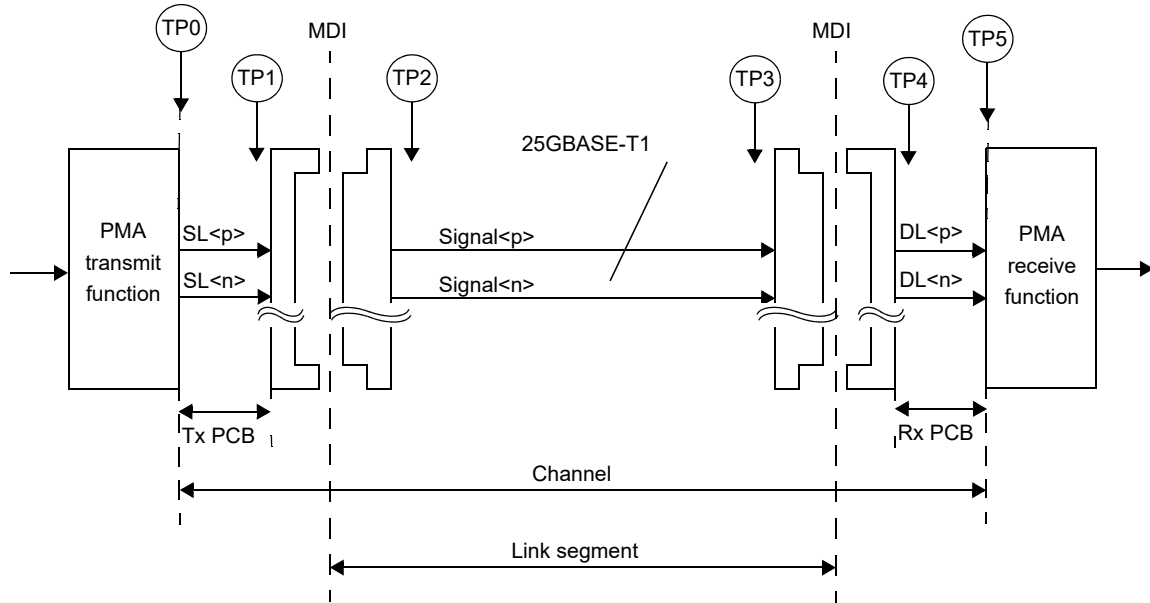


Figure 165-27—25GBASE-T1 link

Note that the source lane (SL) signals SL<p> and SL<n> are the positive and negative sides of the transmitter end's differential signal pair and the destination lane (DL) signals DL<p> and DL<n> are the positive and negative sides of the receiver end's differential signal pair.

Table 165-13 describes the defined test points illustrated in Figure 165-27.

Table 165-13—Test points

Test points	Description
TP0 to TP5	The channel including the transmitter and receiver differential controlled impedance PCB insertion loss and the link segment insertion loss.
TP1 to TP4	All link segment measurements are made between TP1 and TP4 as illustrated in Figure 165-27.
TP0 to TP2 TP3 to TP5	A mated connector pair has been included in both the transmitter and receiver specifications defined in 165.5.3 and 165.5.4. The recommended maximum insertion loss from TP2 to TP0 or from TP3 to TP5 including the test configuration is provided in 165A.2.1.
TP2	Unless specified otherwise, all transmitter measurements defined in 165.5.3 are made at TP2.
TP3	TP3 represents the link partner's TP2 test point.

165.5.3 Transmitter electrical specifications

The PMA provides the Transmit function specified in 165.4.2.2 in accordance with the electrical specifications of this clause. The transmitter shall be AC-coupled (i.e., it shall present a high DC common-mode impedance at the MDI). There can be various methods for AC-coupling in actual implementations.

Unless specified otherwise, all transmitter measurements and tests defined in 165.5.3 are made at TP2 utilizing a test system configuration that meets the specifications in 165.5.5 and a fourth-order Bessel-Thomson low-pass filter with 16 GHz at -3 dB bandwidth.

A mated connector pair has been included in the transmitter specifications defined in this subclause.

The recommended maximum insertion loss from TP0 to TP2 (or TP3 to TP5) including the test configuration is provided in Annex 165A. Annex 165A also includes parameters related to recommended differential controlled impedance printed circuit board parameters.

Where a load is not specified, the transmitter shall meet the requirements of this clause with a $100\ \Omega$ resistive differential load connected to each transmitter output. Transmitter electrical tests are specified with a load tolerance of $\pm 0.1\%$.

165.5.3.1 Maximum output droop

With the transmitter in test mode 6 and using the transmitter test configuration 1 shown in Figure 165–25, the magnitude of both the positive and negative droop shall be less than 15%, measured with respect to an initial value at 4 ns after the zero crossing and a final value at 16 ns after the zero crossing (12 ns period).

165.5.3.2 Transmitter linearity

With the transmitter in test mode 4 and using the transmitter test configuration 1 shown in Figure 165–25, the test defined in 120D.3.1.2 shall be performed. The ideal PAM4 level of 1/3 should be used for effective symbol levels of ES1 and ES2. The transmitter SNDR, as specified in 120D.3.1.6, shall exceed 38 dB.

165.5.3.3 Transmitter timing jitter

The transmitter timing jitter is measured by capturing the TX_TCLK_879 waveform in both MASTER and SLAVE configurations while in test mode 1 using the transmitter test configuration 2 shown in Figure 165–26. When in test mode 1, the link is up, and the two PHYs have established link (link_status is set to OK), the RMS value of the MASTER TX_TCLK_879 jitter relative to an unjittered reference shall be less than 0.4 ps, when measured over the bandwidth from 1 MHz to 100 MHz (integrated), and less than 0.8 ps when measured over the bandwidth from 10 kHz to 1 MHz (integrated). The peak-to-peak value of the MASTER TX_TCLK_879 jitter relative to an unjittered reference shall be less than 4 ps.

When in test mode 1, the link is up, and the two PHYs have established link (link_status is set to OK), the RMS value of the SLAVE TX_TCLK_879 jitter relative to an unjittered reference shall be less than 0.8 ps. The peak-to-peak value of the SLAVE TX_TCLK_879 jitter relative to an unjittered reference shall be less than 8 ps.

TX_TCLK_879 jitter shall be measured over an interval of $0.4\text{ ms} \pm 10\%$. The band-pass bandwidth of the capturing device shall be 200 MHz (this is equivalent to phase noise integration of the clock over a bandwidth of at least 100 MHz from the carrier frequency). The unjittered reference is a constant clock frequency extracted from each record of captured TX_TCLK_879. The unjittered reference is based on linear regression of frequency and phase that produces minimum time interval error.

165.5.3.3.1 Transmit MDI random jitter in MASTER mode

In addition to jitter measurement for transmit clock, MDI jitter is measured when in test mode 2 with the square wave pattern (see Table 165–14) and using test configuration 1 as shown in Figure 165–25. The RMS value of the MDI output jitter relative to an unjittered reference shall be less than 0.4 ps, when measured over the bandwidth from 1 MHz to 100 MHz (integrated), and less than 0.8 ps when measured over the bandwidth from 10 kHz to 1 MHz (integrated). The peak-to-peak value of the MDI output jitter

relative to an unjittered reference shall be less than 4 ps. Jitter shall be measured over an interval of $0.4 \text{ ms} \pm 10\%$. The band-pass bandwidth of the measurement device shall be larger than 200 MHz. The unjittered reference is a constant clock frequency extracted from each record of captured differential output on the MDI. The unjittered reference is based on linear regression of frequency and phase that produces minimum time interval error.

Table 165–14—Jitter test modes

Bit 1.2313.1	Bit 1.2313.0	Test pattern
0	0	Square wave: TX_TCLK_879
0	1	JP03A (as specified in 94.2.9.1)
1	0	JP03B (as specified in 94.2.9.2)
1	1	Reserved

165.5.3.3.2 Transmit MDI deterministic jitter in MASTER mode

Jitter measurements in this subclause are performed with the transmitter enabled in MASTER timing mode in test mode 2, with the patterns as defined by Table 165–14, and timed with a local clock.

To measure the peak-to-peak deterministic jitter (DJ_{pk-pk}) follow the steps as specified in 94.3.12.6.1, with the following modifications to step 5:

$$f_n = 2.5 \text{ MHz}, T = 27.2 \text{ ns}.$$

Using this method, DJ_{pk-pk} shall be less than 3.6 ps.

To measure peak-to-peak even-odd jitter (EOJ_{pk-pk}) follow the steps as specified in 94.3.12.6.2.

Using this method, EOJ_{pk-pk} shall be less than 1.6 ps.

165.5.3.4 Transmitter power spectral density (PSD) and power level

In test mode 5 (normal operation), the transmit power shall be in the range of -1 dBm to 2 dBm and the power spectral density of the transmitter, measured into a 100Ω load using test configuration 1 shown in Figure 165–25 shall be between the upper and lower masks specified in Equation (165–9) and Equation (165–10), respectively. The upper and lower masks for the 25 Gb/s data rate are shown in Figure 165–28.

$$UPSD(f) = \begin{cases} -90 - K & \text{dBm/Hz} & 0 < f \leq 1500 \\ -89 - K - \frac{f}{600 \times 2.5} & \text{dBm/Hz} & 1500 < f \leq 7500 \\ -82 - K - \frac{f}{250 \times 2.5} & \text{dBm/Hz} & 7500 < f \leq 13750 \end{cases} \quad (165-9)$$

$$LPSD(f) = \begin{cases} -96 - K & \text{dBm/Hz} & 5 < f \leq 1000 \\ -95 - K - \frac{f}{400 \times 2.5} & \text{dBm/Hz} & 1000 < f \leq 5000 \\ -90 - K - \frac{f}{200 \times 2.5} & \text{dBm/Hz} & 5000 < f \leq 7500 \end{cases} \quad (165-10)$$

where

f is the frequency in MHz

$$K = 10\log_{10}(2.5) \quad (165-11)$$

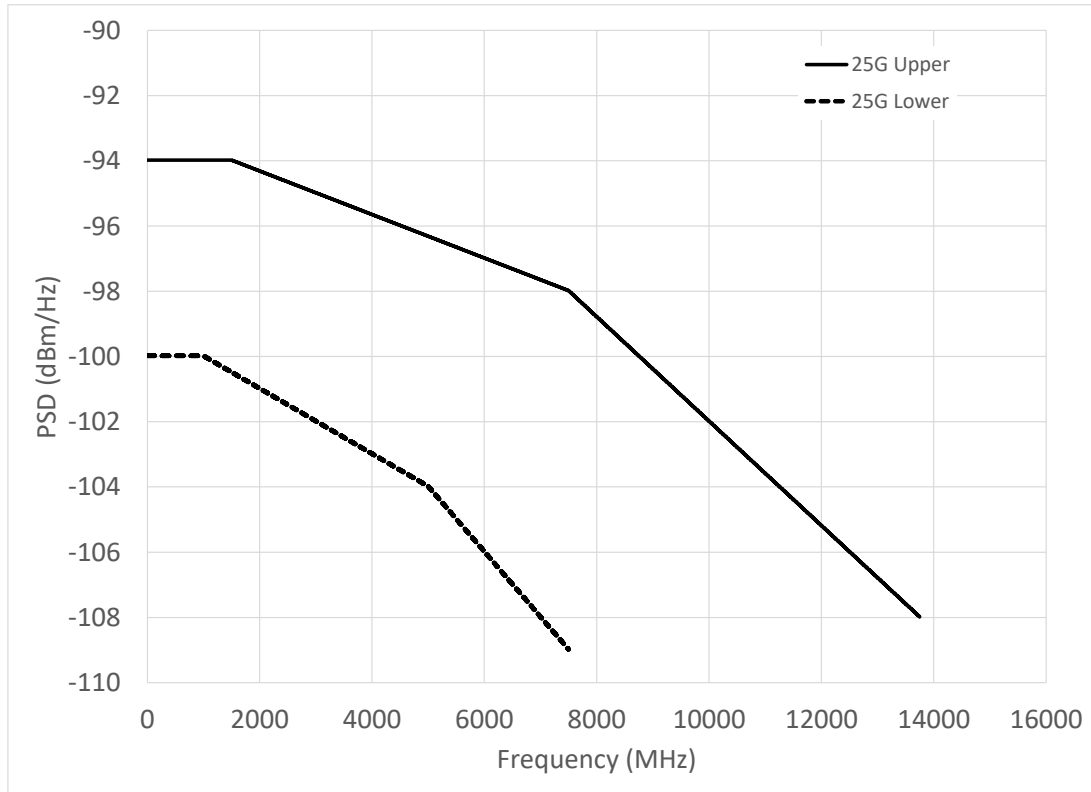


Figure 165-28—Transmitter power spectral density, upper and lower masks

165.5.3.5 Transmitter peak differential output

When measured with 100 Ω termination, the transmit differential signal at the MDI shall be less than 1.3 V peak-to-peak. This limit applies to all transmitted symbol sequences, including SEND_S, SEND_T, and SEND_N.

165.5.3.6 Transmitter clock frequency

The symbol transmission rate of the MASTER PHY shall be within the range 14.0625 GHz $\pm$ 50 ppm.

For a MASTER PHY, when the transmitter is in the LPI transmit mode, the transmitter clock short-term rate of frequency variation shall be less than 0.1 ppm/second. The short-term frequency variation limit shall also apply when switching to and from the LPI mode.

165.5.4 Receiver electrical specifications

The PMA provides the Receive function specified in 165.4.2.3 in accordance with the electrical specifications of this clause using cabling that is within the limits specified in 165.7.

165.5.4.1 Receiver differential input signals

Differential signals received at the MDI that were transmitted from a remote transmitter within the specifications of 165.5.3 and have passed through a link specified in 165.7 are expected to be decoded with a BER less than 10^{-12} after RS-FEC decoding, and sent to the 25GMII after link reset completion. The frame error ratio shall be less than 7.8×10^{-9} for 800 octet frames with minimum IPG or greater than 220-octet IPG.

165.5.4.2 Alien crosstalk noise rejection

This specification is provided to verify the receiver's tolerance to alien crosstalk noise. The test is performed with a noise source consisting of a signal generator with Gaussian distribution, bandwidth, and magnitude shown in Table 165–15. The receive DUT is connected to the noise source through a directional coupler, as shown in Figure 165–29, with a link segment as defined in 165.7. The BER is expected to be less than 10^{-12} , and to satisfy this specification the frame loss ratio is less than 10^{-9} for 125-octet packets measured at the MAC/PLS service interface.

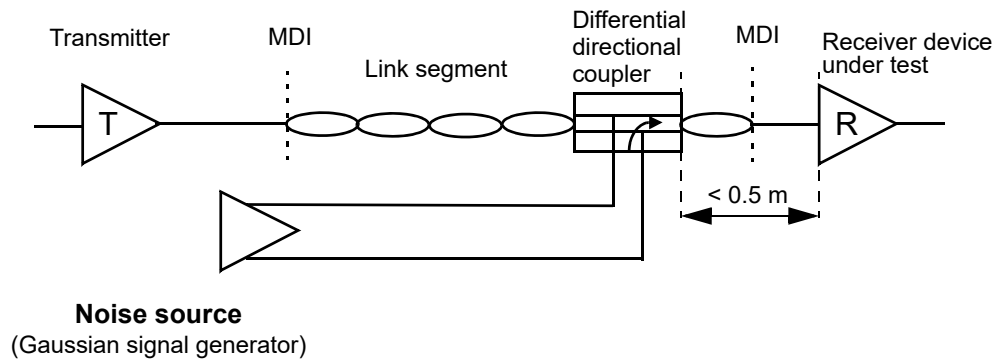


Figure 165–29—Alien crosstalk noise rejection test setup

Table 165–15—Alien crosstalk noise source

PHY type	Noise bandwidth (MHz)	Added noise at MDI (dBm/Hz)
25GBASE-T1	8750	–156

165.5.5 Test configurations

Specifications for a host test configuration (HTC) for tests at TP2, and for a link segment test configuration (LSTC) for measurements between TP1 and TP4 are defined. The test configurations are specified in a mated state in 165.5.5.3.

165.5.5.1 Host test configuration

The test configuration for measurements at TP2/TP3 is shown in Figure 165–30. Due to the symmetry of the transceiver, this test configuration provides access to TP2 and TP3.

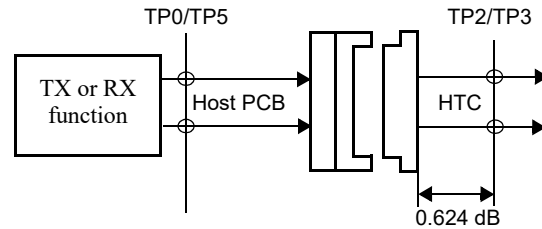
The HTC printed circuit board (PCB) insertion loss values determined using Equation (165–12) is used as the HTC reference insertion loss.

$$IL_{tref}(f) = \left(0.03048 \left(\frac{f}{1000} \right) + 0.17018 \left(\frac{f}{1000} \right)^{0.45} \right) \text{ dB} \quad (165-12)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

The reference insertion loss at TP2 or TP3 of the HTC is 0.624 dB at 7031.25 MHz using Equation (165-12). The effects of differences between the insertion loss of an actual test configuration and the reference insertion loss are to be accounted for in the measurements.



NOTE—dB losses shown are reference losses at 7031.25 MHz, actual configuration losses can be different, but are to be accounted for in measurements.

Figure 165-30—Host test configuration

165.5.5.2 Link segment test configuration

The link segment test configuration (LSTC) is used for measuring the link segment defined in 165.7 at TP1 and TP4. The relation of the LSTC and the TP1 and TP4 test points is illustrated in Figure 165-31. One LSTC is used at either end of the link segment under test.

The link segment test configuration PCB and test point insertion loss values determined using Equation (165-12) are used as the test configuration reference insertion loss. The effects of differences between the insertion loss of an actual test configuration and the reference insertion loss are to be accounted for in the measurements. The reference insertion loss of the link segment test configurations is 0.624 dB at 7031.25 MHz.

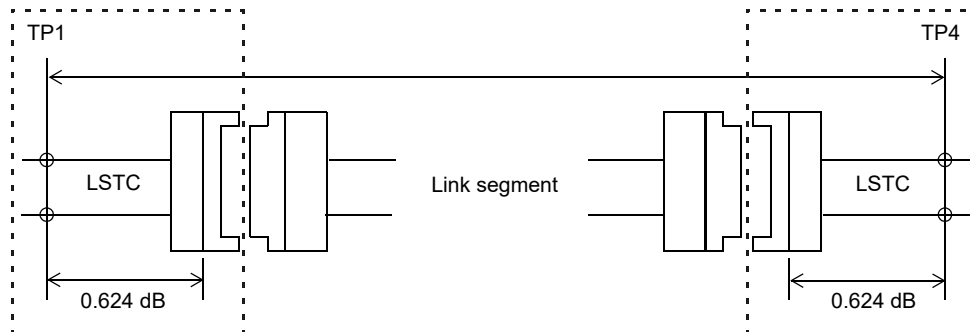


Figure 165-31—Link segment test configuration

165.5.5.3 Mated test configurations

The HTC and LSTC are specified in a mated state illustrated in Figure 165–32, to enable connections to measurement equipment.

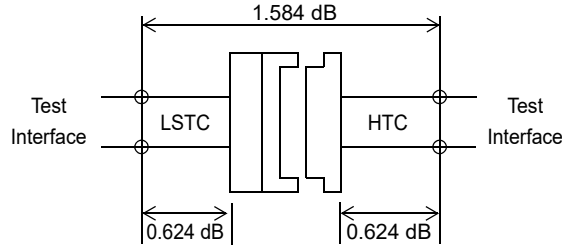


Figure 165–32—Mated test configurations

165.5.5.3.1 Insertion loss

The measured insertion loss of the mated test configurations shall meet the values determined using Equation (165–13) and Equation (165–14).

$$IL(f) \leq IL_{mtfmax}(f) = 1.05 \times \left(0.06109 \left(\frac{f}{1000} \right) + 0.3404 \left(\frac{f}{1000} \right)^{0.45} \right) + 0.2 \sqrt{\left(\frac{f}{2500} \right)} \quad (\text{dB}) \quad (165-13)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

$$IL(f) \leq IL_{mtfmin}(f) = 0.95 \times \left(0.06109 \left(\frac{f}{1000} \right) + 0.3404 \left(\frac{f}{1000} \right)^{0.45} \right) + 0.2 \sqrt{\left(\frac{f}{2500} \right)} \quad (\text{dB}) \quad (165-14)$$

where

$IL(f)$ is the measured mated test configuration insertion loss

f is the frequency in MHz; $1 \leq f \leq 9000$

The reference insertion loss of the mated test configuration is determined using Equation (165–15).

The reference insertion loss of the mated test configuration is 1.584 dB at 7031.25 MHz.

$$IL_{mtfref}(f) = 0.06109 \left(\frac{f}{1000} \right) + 0.3404 \left(\frac{f}{1000} \right)^{0.45} + 0.2 \sqrt{\left(\frac{f}{2500} \right)} \quad (\text{dB}) \quad (165-15)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

165.5.5.3.2 Return loss

The return loss of the mated test configurations shall meet the values determined using Equation (165–16).

$$RL_{mtf}(f) \geq \left\{ \begin{array}{ll} 25 & 1 \leq f < 1500 \\ 25 - 16.6 \log_{10} \left(\frac{f}{1500} \right) & 1500 \leq f < 3000 \\ 20 & 3000 \leq f < 9000 \end{array} \right\} \text{ (dB)} \quad (165-16)$$

where

f is the frequency in MHz

165.5.5.3.3 Mode Conversion

The mode conversion of the mated test configurations shall meet the values determined using Equation (165-17).

$$ConversionLoss_{mtf}(f) \geq 30 \text{ (dB)} \quad (165-17)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

165.5.5.3.4 Crosstalk

The crosstalk loss of the mated test configurations shall meet Equation (165-18).

$$XTF_{mtf}(f) \geq \left\{ \begin{array}{ll} 78 & 1 \leq f < 215 \\ 83 - 15 \log_{10} \left(\frac{f}{100} \right) & 215 \leq f < 9000 \end{array} \right\} \text{ (dB)} \quad (165-18)$$

where

f is the frequency in MHz

165.6 Management interface

25GBASE-T1 makes use of the management functions that may be provided by the optional MDIO (Clause 45), and the communication and self-configuration functions provided by the optional Auto-Negotiation (Clause 98).

165.6.1 Optional support for Auto-Negotiation

If Auto-Negotiation is supported and enabled the mechanism described in Clause 98 shall be used.

Auto-Negotiation is performed as part of the initial setup of the link, and allows the PHY at each end to advertise their capabilities and to automatically select the operating mode for communication on the link. Auto-Negotiation signaling is used for the following primary purposes for 25GBASE-T1:

- a) To negotiate that the PHY is capable of supporting 25GBASE-T1 transmission.
- b) To determine the MASTER-SLAVE relationship between the PHY at each end of the link.

165.7 Link segment characteristics

25GBASE-T1 is designed to operate over link segments comprising a shielded balanced pair of conductors that meet the requirements specified in this subclause. This link segment supports an effective data rate of 25 Gb/s in each direction simultaneously.

165.7.1 Link transmission parameters

The transmission characteristics for the 25GBASE-T1 link segment are specified to support operation over automotive temperature and electromagnetic conditions.

165.7.1.1 Insertion loss

The insertion loss of each 25GBASE-T1 link segment shall meet the values determined using Equation (165–19).

$$\text{Insertion loss}(f) \leq 0.00135(f) + 0.3564(f)^{0.45} + 0.495\left(\frac{f}{7500}\right)^6 \quad (\text{dB}) \quad (165-19)$$

where

f is the frequency in MHz; $10 \leq f \leq 9000$

The 25GBASE-T1 link segment insertion loss limit is illustrated in Figure 165–33.

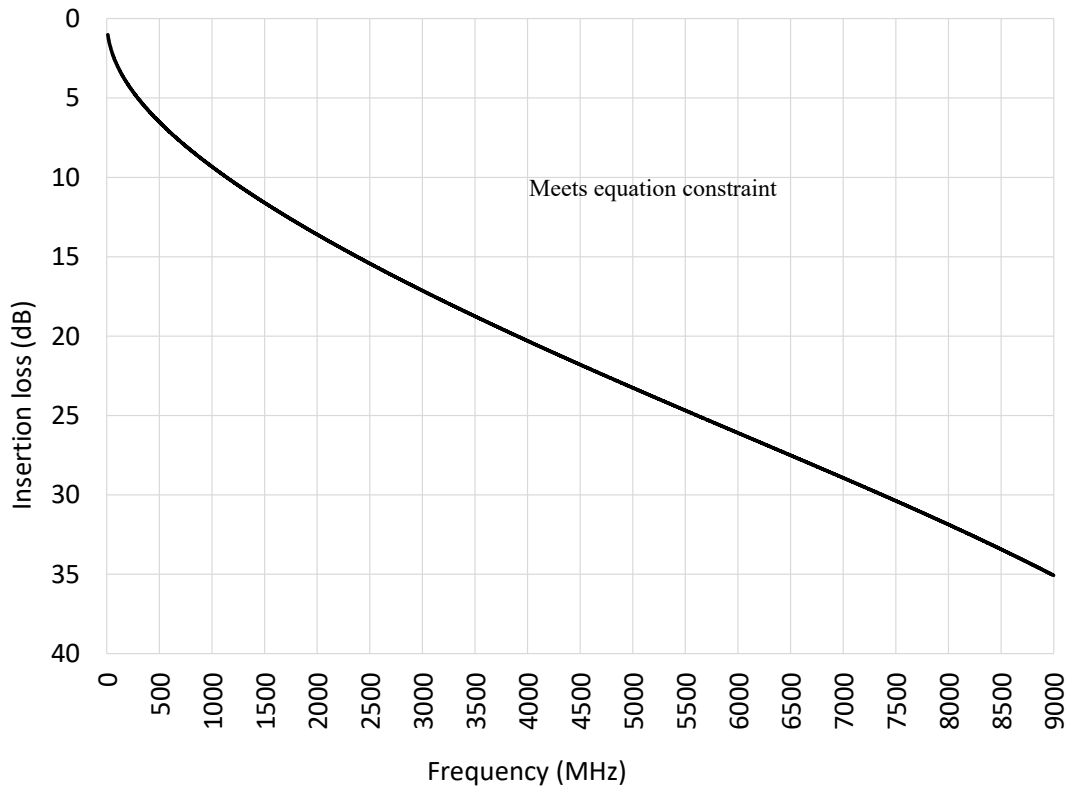


Figure 165–33—Insertion loss calculated limit in Equation (165–19)

165.7.1.2 Differential characteristic impedance

The nominal differential characteristic impedance of the link segment is 100 Ω.

165.7.1.3 Return loss

165.7.1.3.1 25GBASE-T1 link segment return loss

In order to limit the noise at the receiver due to impedance mismatches each 25GBASE-T1 link segment shall meet the values determined by using Equation (165–20) at all frequencies from 10 MHz to 9000 MHz. The reference impedance for the return loss specification is 100 Ω.

$$25G_Return_Loss \geq \left\{ \begin{array}{ll} 20 & 10 \leq f < 480 \\ 20 - 10 \log_{10} \left(\frac{f}{480} \right) & 480 \leq f < 7600 \\ 8 & 7600 \leq f < 9000 \end{array} \right\} \text{ (dB)} \quad (165-20)$$

where

f is the frequency in MHz; $10 \leq f \leq 9000$

The 25GBASE-T1 return loss is illustrated in Figure 165–34.

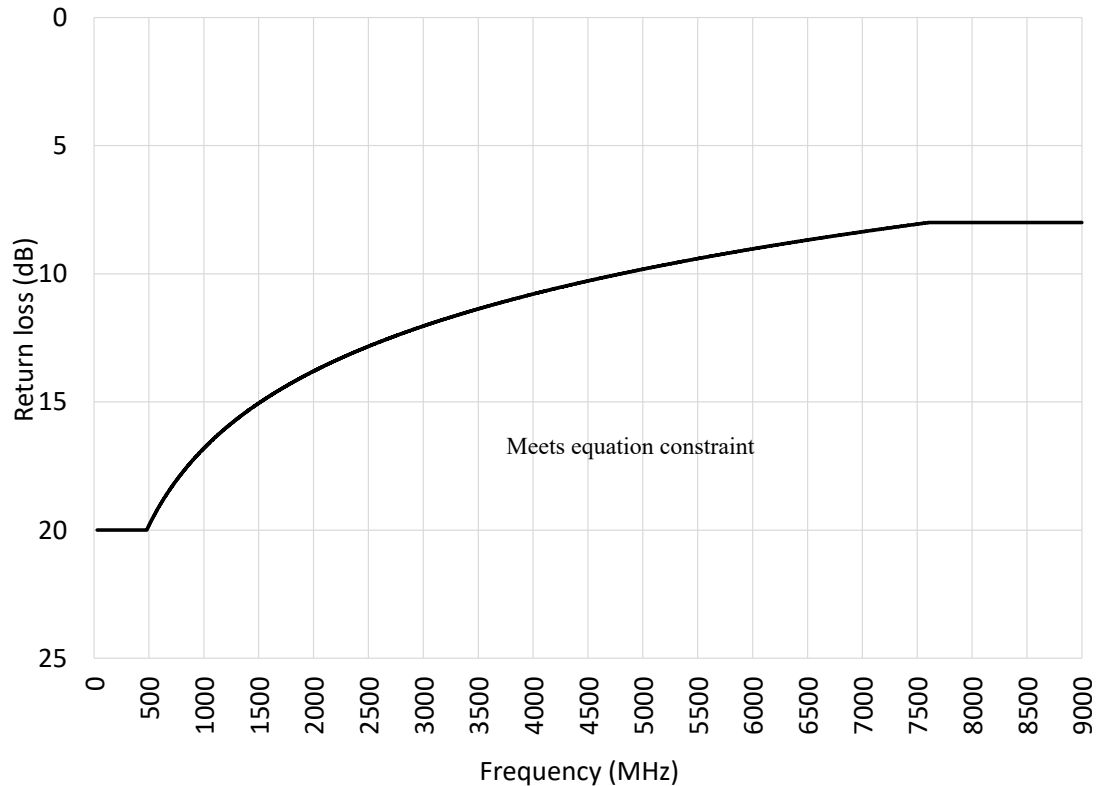


Figure 165–34—Return loss calculated limit in Equation (165–20)

165.7.1.3.2 Characteristics of the echo from the link segment

The following subclauses define additional metrics and requirements to limit the characteristics of the echo from the link segment. There are two separate, but similar requirements, called the “residual echo metric” (REM) and the “echo tail metric” (ETM) which are fundamentally time domain metrics. The specified algorithms derive the time-domain responses from complex-valued frequency domain measurements for improved sensitivity. In both metrics, the resulting echo impulse response is divided into segments of length N_{seg} , and the energy in each segment is computed. Following that, the two metrics differ, both in their assumptions and specification.

In the first metric, REM, a number ($N_{discard}$) of the highest-energy segments of the impulse response are discarded, and the total echo energy from the remaining segments is computed and compared to the requirement.

The second metric, ETM, is designed to characterize the behavior of the lower level reflections excluding the cabling connectors and major discontinuities. In ETM, the round-trip delay of the link segment is estimated first, and any echo beyond a single round-trip reflection is discarded from the tail of the impulse response. Then, an iterative calculation is performed, discarding the initial segments from the initial m_s segments to a value m_e segments. Additionally, for each iteration, the remaining total echo energy is computed, as $ETM(m)$, after discarding the remaining $N_{discard_etm}$ highest-energy segments from the tail of the impulse response excluding the m initial segments. These values of $ETM(m)$ are then compared to a limit line for values of m from m_s to m_e . It should be noted that the algorithms, particularly the ETM algorithm, were developed with the physical properties of automotive cabling in mind and are potentially unsuitable beyond the intended application.

165.7.1.3.3 Residual echo metric (REM)

This subclause defines a metric to limit noise from echo outside of major discontinuities in a link segment. The echo outside these major discontinuities is referred to as the residual echo. This metric is determined using the procedure in 165.7.1.3.4 using the parameters in Table 165–16.

Table 165–16—REM parameters

Parameter	Value	Description
Δf	2.5 MHz	The sample frequency spacing for the frequency domain transfer function measurements
N	4096	Number of sampling points to use for the time domain representation of the echo impulse response
N_{seg}	4	Number of samples in each segment
$N_{discard}$	16	Number of largest segments to discard

165.7.1.3.4 Calculating residual echo metric

Step 1. The frequency domain transfer function for the differential mode channel echo, S_{11} , is measured at the link segment side of the MDI with the far end terminated in 100 Ω resistance. For example, if the cable is terminated in a plug, the measurement is on the cabling between the (de-embedded) plug and the far end termination. This measurement is performed for both ends of the link segment and provides the magnitude and phase of the transfer function, measured with frequency spacing Δf . The measured signal can be represented as a complex sequence E_k as given by Equation (165–21).

$$E_k = \begin{cases} S_{11}(k\Delta f) \\ S_{22}(k\Delta f) \end{cases} \quad (165-21)$$

Step 2. The frequency domain transfer function is converted to time domain impulse response with sampling interval, T , according to the following method:

Step 2a. The phase of E_k is adjusted to make the values for $k = 0$ and $k = K_N$ real. The adjustment is done by dropping any imaginary component at DC and applying linear phase adjustment to E_k , corresponding to fractional delay of the time domain signal, and is given by Equation (165-22) and Equation (165-23).

$$\begin{aligned} H_k &= E_k e^{-jk\theta}, 0 < k \leq K_N \\ H_0 &= \text{real}(E_0) \end{aligned} \quad (165-22)$$

where

$$\begin{aligned} \theta &= \frac{\text{angle}(E_{K_N})}{K_N} \\ K_N &= \frac{N}{2} \end{aligned} \quad (165-23)$$

Step 2b. The impulse response of the signal is computed by applying Hermitian symmetric extension of the signal for k from $K_N + 1$ to $N - 1$, according to Equation (165-24):

$$H_k = \text{conj}(H_{2K_N - k}) \quad (165-24)$$

where $k \in \{K_N + 1, \dots, 2K_N - 1\}$

and then computing the inverse Fourier transform according to Equation (165-25):

$$h_n = \frac{1}{K_N} \sum_{k=0}^{2K_N-1} H_k e^{j\frac{2\pi kn}{2K_N}} \quad (165-25)$$

Step 3. The first $N/2$ samples of the echo impulse response, h_n , are split into segments with N_{seg} samples in each segment. The sum of the squares for each segment is computed by adding the squared impulse response in each segment according to Equation (165-26).

$$P_r = \sum_{k=rN_{seg}}^{(r+1)N_{seg}-1} h_k^2 \quad (165-26)$$

Step 4. The $N_{discard}$ largest P_r values are excluded from the calculations by setting their value to zero in the residual echo value according to Equation (165-27).

$$RE_r = \begin{cases} 0 & \text{if } P_r \text{ is one of } N_{discard} \text{ largest } P_r \text{ values} \\ P_r & \text{for all other } r \end{cases} \quad (165-27)$$

Step 5. The residual echo metric, REM, is calculated as the sum of all the residual echo values from Step 4 according to Equation (165–28):

$$REM = 10\log_{10}\left(\sum_r RE_r\right) \text{ (dB)} \quad (165-28)$$

165.7.1.3.5 Limit on residual echo metric

The REM value of each end of the link segment, defined by the calculation described in 165.7.1.3.4, shall comply with Equation (165–29):

$$REM(N_{discard}) \leq \min(REM_{max} - IL(f_c) - REM_{offset}) \text{ (dB)} \quad (165-29)$$

where

$$\begin{aligned} f_c & \text{ is 4 GHz} \\ REM_{max} & \text{ is } -30 \text{ dB} \\ REM_{offset} & \text{ is 20 dB} \end{aligned}$$

165.7.1.3.6 Echo tail metric (ETM)

This subclause defines a metric to limit the distribution in time of echo outside major discontinuities in a link segment. This is referred to as the echo tail, and the echo tail metric is determined using the procedure in 165.7.1.3.7 using the parameters in Table 165–17.

Table 165–17—ETM parameters

Parameter	Value	Description
Δf	2.5 MHz	The sample frequency spacing for the frequency domain transfer function measurements
N	4096	Number of sampling points to use for the time domain representation of the echo impulse response
N_{seg}	4	Number of samples in each segment
$N_{discard_etm}$	6	Number of largest segments to discard for ETM calculations

165.7.1.3.7 Calculating echo tail metric

Step 1. Perform Step 1 & Step 2 as specified in 165.7.1.3.4 to compute the time domain impulse response for the echo, h_n , from frequency domain measurements of the differential mode channel echo, S_{11} , which is measured at the link segment side of the MDI with the far end terminated in 100 Ω resistance.

Step 2. Determine the time span of the echo response to the far-end termination from the frequency domain measurements of the insertion loss, which are represented as complex sequences $IL_{k,i}$ according to Equation (165–30):

$$\begin{aligned} IL_{k,1} &= S_{21}(k\Delta f) \\ IL_{k,2} &= S_{12}(k\Delta f) \end{aligned} \quad (165-30)$$

Step 2a. Identify the unwrapped phase of the frequency response according to Equation (165–31):

$$\theta_{k,i} = \text{unwrap}(\text{angle}(IL_{k,i})) \quad (165-31)$$

Step 2b. Estimate the propagation delay by calculating the slope from a linear fit to the phase according to Equation (165–32):

$$d_i = \frac{N}{2\pi N_{seg}} \times \frac{M \sum_{k=k_s}^{k_s+M-1} (k \times \theta_{k,i}) - \left(\sum_{k=k_s}^{k_s+M-1} k \right) \times \left(\sum_{k=k_s}^{k_s+M-1} \theta_{k,i} \right)}{M \sum_{k=k_s}^{k_s+M-1} k^2 - \left(\sum_{k=k_s}^{k_s+M-1} k \right)^2} \quad (165-32)$$

With $k_s = 40$, and $M = 1600$, the linear fit is calculated over the frequency range of 100 MHz to 4.1 GHz. The propagation delay, d_i , is expressed in terms of the number of segments.

The span of the echo response is the round-trip delay or twice the propagation delay as given by Equation (165–33):

$$L_e = 2 \times \text{floor}(\min(d_1, d_2)) \quad (165-33)$$

Step 3. Repeat Step 3a through Step 3d iteratively for each integer value of m from $m = m_s = 13$ to $m = m_e = 154$ to compute $\text{ETM}(m)$.

Step 3a. Define the partial echo response at point m according to Equation (165–34):

$$g_n^m = \begin{cases} 0 & n < m \times N_{seg} \\ h_n & m \times N_{seg} \leq n < L_e \times N_{seg} \\ 0 & L_e \times N_{seg} \leq n \end{cases} \quad (165-34)$$

where h_n is the time domain impulse response of the echo computed in Step 1.

Step 3b. The first $N/2$ samples of the partial echo impulse response, g_n^m , are split into segments with N_{seg} samples in each segment. The sum of the squares for each segment is computed by adding the squared impulse response in each segment according to Equation (165–35).

$$P_r^m = \sum_{k=rN_{seg}}^{(r+1)N_{seg}-1} (g_k^m)^2 \quad (165-35)$$

Step 3c. The $N_{discard_etm}$ largest P_r^m values are excluded from the calculations by setting their value to zero in the echo tail value according to Equation (165–36).

$$ET_r(m) = \begin{cases} 0 & \text{if } P_r^m \text{ is one of } N_{discard_etm} \text{ largest } P_r^m \text{ values} \\ P_r^m & \text{for all other } r \end{cases} \quad (165-36)$$

Step 3d. The echo tail metric for lag m , $ETM(m)$, is calculated as the sum of the echo tail values from Step 3c according to Equation (165–37):

$$ETM(m) = 10\log_{10}\left(\sum_r ET_r(m)\right) \text{ (dB)} \quad (165-37)$$

165.7.1.3.8 Limit on echo tail metric

The ETM value of each end of the link segment, defined by the calculation described in 165.7.1.3.7, shall comply with Equation (165–38):

$$ETM(m) \leq REM_{Limit} - 16 \times \frac{m - m_s}{m_e - m_s} \quad m_s \leq m < m_e \quad \text{(dB)} \quad (165-38)$$

where REM_{Limit} is the limit of REM as defined in the right-hand side of Equation (165–29), $m_s = 13$, and $m_e = 154$.

165.7.1.4 Coupling attenuation

In order to limit the noise at the receiver as well as emissions, when tested using the IEC 62153-4-7 triaxial tube in tube method as specified in [Annex 149A](#), where shielded balanced pair cabling is used, the 25GBASE-T1 link segment shall meet the coupling attenuation values determined by using Equation (165–39).

$$\text{Coupling attenuation}(f) \geq \left\{ \begin{array}{ll} 70 & 30 \leq f < 750 \text{ MHz} \\ 50 - 20\log_{10}\left(\frac{f}{7500}\right) & 750 \leq f \leq 9000 \text{ MHz} \end{array} \right\} \text{ (dB)} \quad (165-39)$$

where

f is the frequency in MHz; $30 \leq f \leq 9000$

The coupling attenuation is illustrated in Figure 165–35.

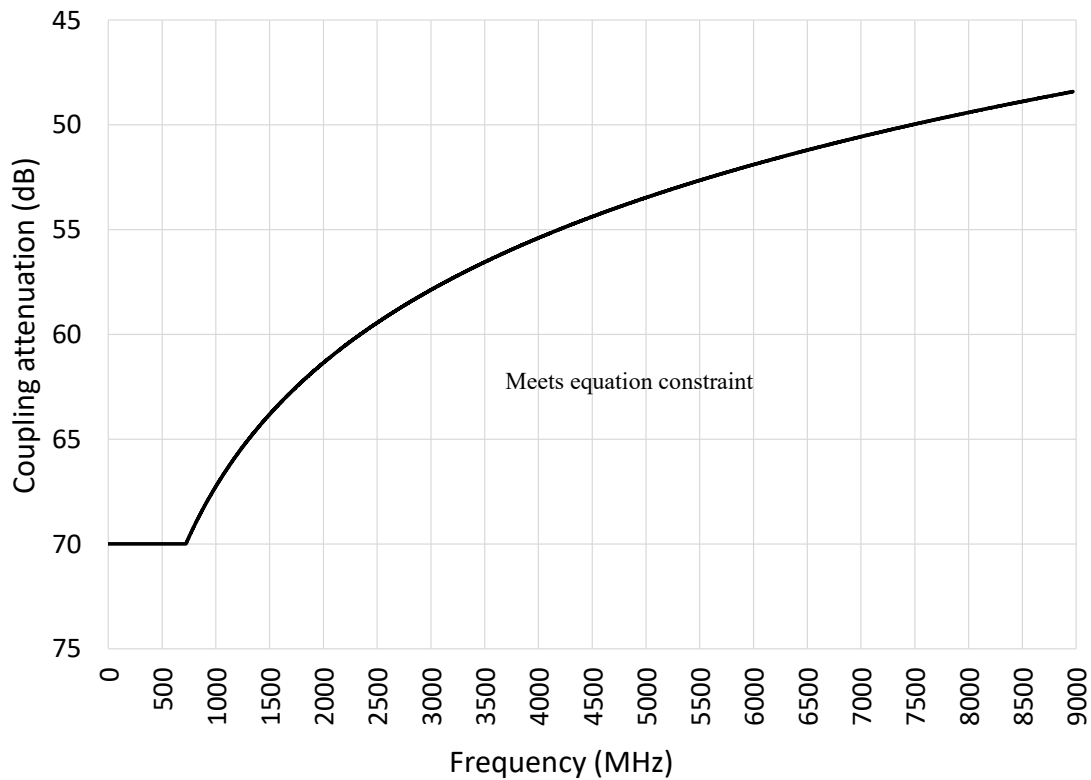


Figure 165-35—Coupling attenuation calculated limit in Equation (165-39)

165.7.1.5 Screening attenuation

Where shielded balanced pair cabling is used, the minimum screening attenuation for a link segment is 45 dB for all frequencies between 30 MHz and 4000 MHz. Screening attenuation is tested as specified in IEC 62153-4-7 using triaxial tube-in-tube method. Additional screening attenuation test methodologies are defined in [Annex 149A](#). For frequencies greater than 4000 MHz, the coupling attenuation specification in 165.7.1.4 is sufficient.

165.7.1.6 Maximum link delay

The propagation delay of a link segment shall not exceed 60 ns at all frequencies between 2 MHz and 9000 MHz. The delay specification represents an acceptable margin over the delay of 11 m of automotive cabling expected to be used.

165.7.2 Coupling parameters between link segments

Noise coupled between the disturbed link segment and the disturbing link segment is referred to as *alien crosstalk noise*. Power sum alien near-end crosstalk (PSANEXT) loss and power sum alien attenuation to crosstalk ratio far-end (PSAACRF) are specified to limit the total alien NEXT and alien FEXT coupled between link segments. The test methodologies are specified in [Annex 97B](#).

For implementations with multiple 25GBASE-T1 ports on the same MDI connector assembly, coupling between ports on the MDI connector is not considered to be part of the alien crosstalk specifications. For further information, see 165A.5.

165.7.2.1 Power sum alien near-end crosstalk (PSANEXT)

The differential pair-to-pair near-end crosstalk (NEXT) loss between the disturbed link segment and the disturbing link segment is specified to meet the bit error ratio objective by limiting the alien crosstalk at the near end of a link segment. Multiple disturber alien NEXT loss is specified as the power sum of the individual alien NEXT disturbers. The PSANEXT loss is derived using Equation (97–25) over the frequency range defined for Equation (165–40).

The power sum ANEXT loss between a disturbed link segment and the disturbing link segment shall meet the values determined using Equation (165–40).

$$\text{PSANEXTloss}(f) \geq \min\left(75, 80 - 15 \log_{10} \frac{f}{100}\right) \text{ dB} \quad (165-40)$$

where

f is the frequency in MHz; $10 \leq f \leq 9000$

PSANEXT is illustrated in Figure 165–36.

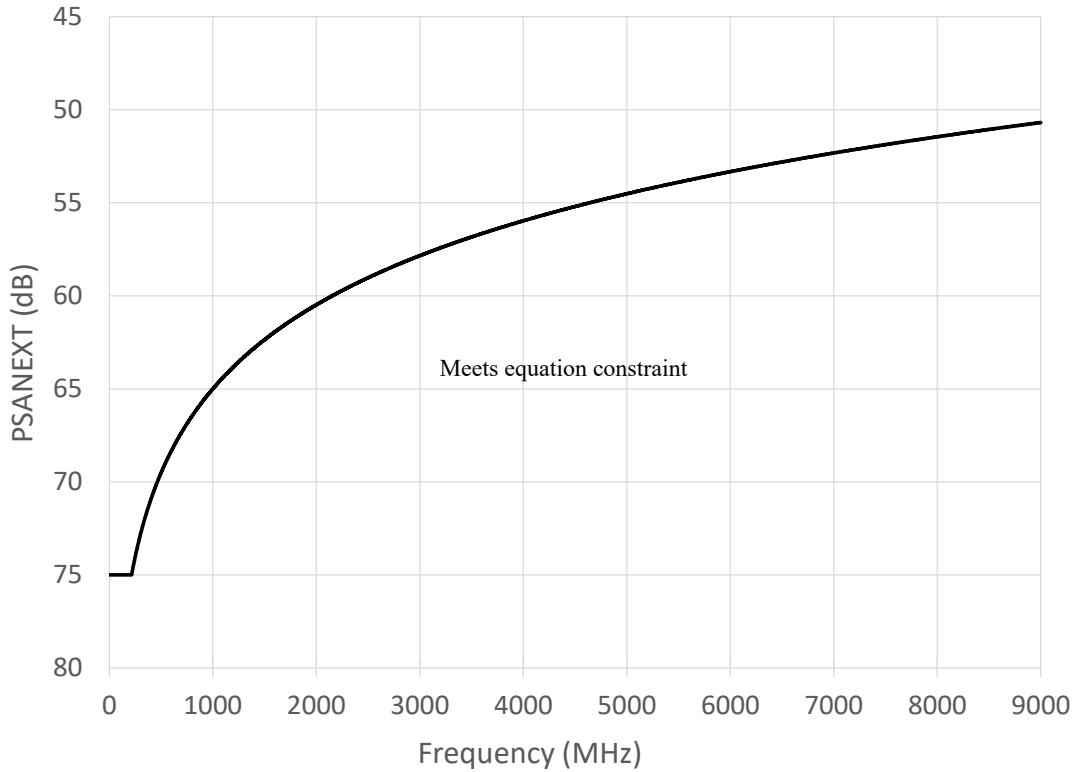


Figure 165–36—PSANEXT calculated limit in Equation (165–40)

165.7.2.2 Power sum alien attenuation to crosstalk ratio far-end (PSAACRF)

In order to limit the alien crosstalk at the far-end of a link segment, the differential pair-to-pair alien far-end crosstalk (FEXT) loss between the disturbed link segment and the disturbing link segment is specified to meet the bit error ratio objective. Multiple disturber attenuation to crosstalk ratio far-end ACRF is specified as the power sum of the individual alien ACRF disturbers to limit the total alien FEXT coupled into a link

segment. The PSAACRF is derived using Equation (97–27) over the frequency range defined for Equation (165–41).

The power sum AACRF between a disturbed link segment and the disturbing link segment shall meet the values determined using Equation (165–41).

$$\text{PSAACRF}(f) \geq \min\left(75, 86 - 20\log_{10}\frac{f}{100}\right) \text{ dB} \quad (165-41)$$

where

f is the frequency in MHz; $10 \leq f \leq 9000$

PSAACRF is illustrated in Figure 165–37.

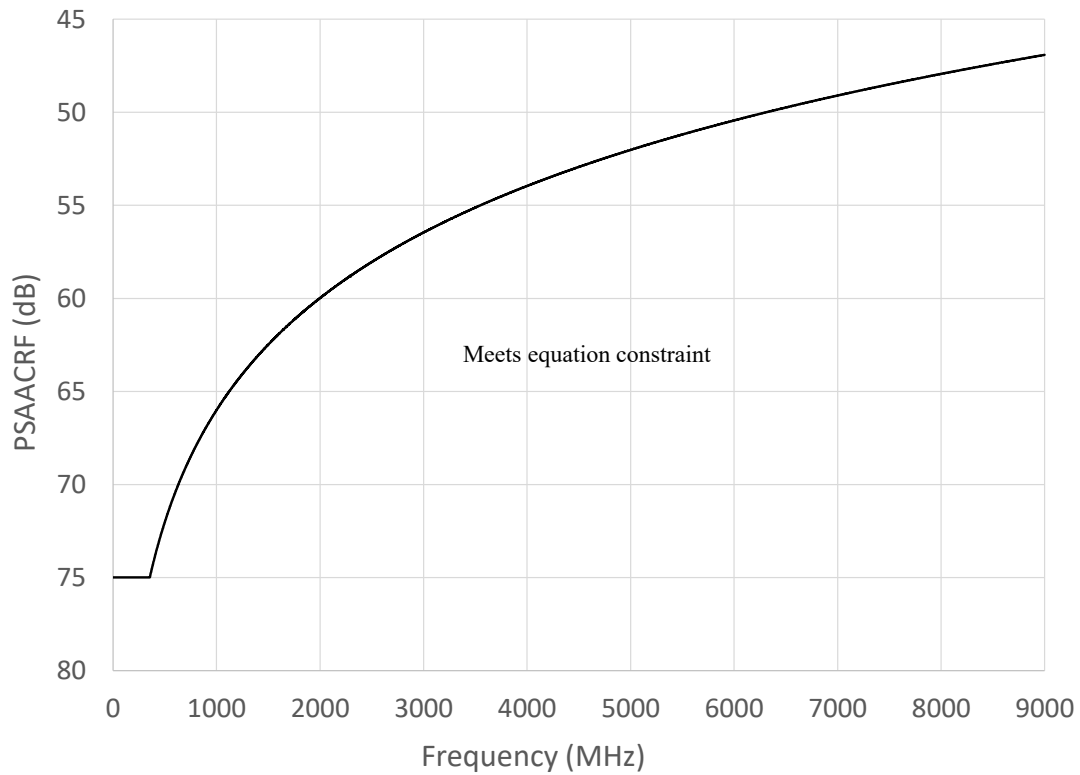


Figure 165–37—PSAACRF calculated limit in Equation (165–41)

165.8 MDI specification

165.8.1 MDI connectors

Where shielded balanced pair cabling is used, the mechanical interface to the shielded balanced cabling is a 2-pin connector with a shield. Further specification of the mechanical interface is beyond the scope of this standard.

165.8.2 MDI electrical specification

The electrical requirements specified in 165.5.3 and 165.5.4 shall be met when the PHY is connected to the MDI connector mated with a specified connector to a shielded balanced pair of conductors.

165.8.2.1 MDI return loss

The differential impedance at the MDI (see Figure 165–38) for each transmit/receive channel shall be such that any reflection (due to differential signals incident upon the MDI with a test port having a differential impedance of 100 Ω) is attenuated relative to the incident signal per Equation (165–42).

$$MDI_Return_Loss(f) \leq \left\{ \begin{array}{ll} 20 - 20\left(\log_{10} \frac{50}{f}\right) & 10 \leq f < 50 \\ 20 & 50 \leq f < 480 \\ 20 - 10\log_{10}(f/480) & 480 \leq f < 4800 \\ 10 - 18\log_{10}(f/4800) & 4800 \leq f \leq 9000 \end{array} \right\} \text{ (dB)} \quad (165-42)$$

where

f is the frequency in MHz

For 25GBASE-T1, the maximum applicable frequency for the MDI return loss is 9000 MHz.

The MDI return loss is illustrated in Figure 165–38.

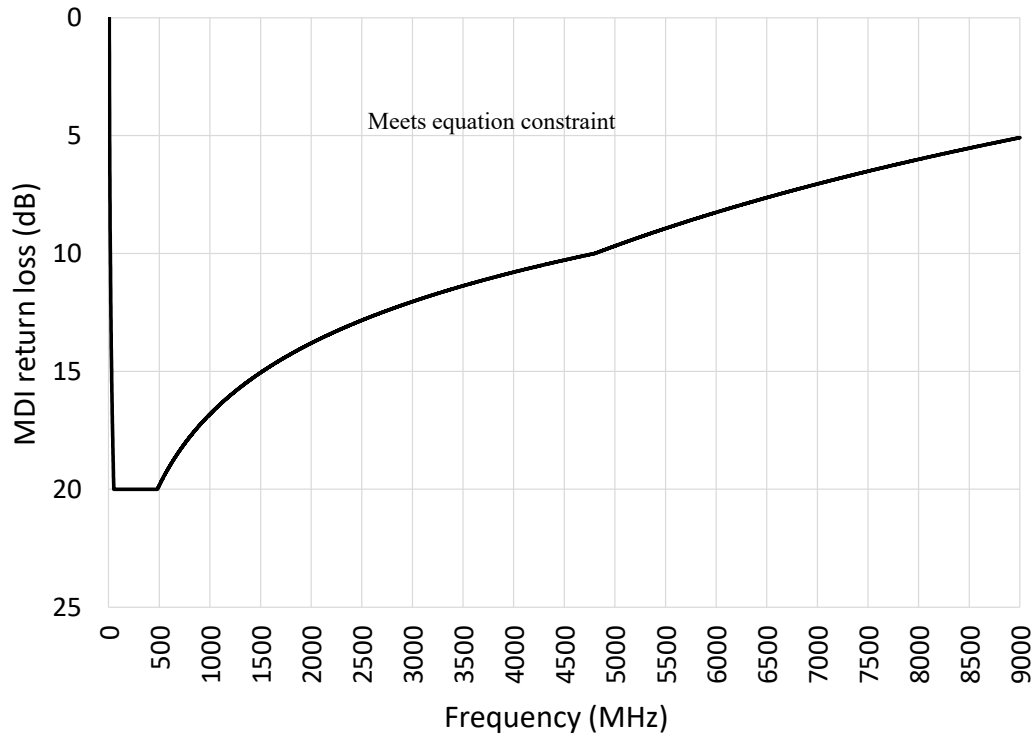


Figure 165–38—MDI return loss calculated limit in Equation (165–42)

165.8.3 MDI fault tolerance

See 96.8.3.

165.9 Environmental specifications

165.9.1 General safety

All equipment subject to this clause is expected to conform to all applicable local, state, national, and application-specific standards.

165.9.2 Network safety

All cabling and equipment subject to this clause is expected to be mechanically and electrically secure in a professional manner.

165.9.2.1 Environmental safety

All equipment subject to this clause, when used in the automotive environment, is expected to conform to the potential environmental stresses with respect to their mounting location, as defined in the following specifications:

- a) General loads: ISO 16750-1
- b) Electrical loads: ISO 16750-2, ISO 7637-2:2008, and ISO 8820-1
- c) Mechanical loads: ISO 16750-3, ASTM D4728, and ISO 12103-1
- d) Climatic loads: ISO 16750-4, IEC 60068-2-1, IEC 60068-2-27, IEC 60068-2-30, IEC 60068-2-38, IEC 60068-2-52, IEC 60068-2-64, and IEC 60068-2-78
- e) Chemical loads: ISO 16750-5 and ISO 20653

Automotive environmental conditions are generally more severe than those found in many commercial and industrial environments. The target automotive, industrial, or commercial environment(s) require careful analysis prior to implementation.

165.9.2.2 Electromagnetic compatibility

A system integrating a PHY intended for automotive applications is expected to comply with all applicable local and national codes. When used in an automotive environment, the PHY is expected to meet the following motor vehicle EMC requirements:

- a) Radiated/conducted emissions: CISPR 25, IEC 61967-1, IEC 61967-4, and IEC 61000-4-21
- b) Radiated/conducted immunity: ISO 11452, IEC 62132-1, IEC 62132-4, and IEC 61000-4-21
- c) Electrostatic discharge: ISO 10605, IEC 61000-4-2, and IEC 61000-4-3
- d) Electrical disturbances: IEC 62215-3, ISO 7637-2, and ISO 7637-3

Exact test setup and test limit values are application-specific.

165.10 Delay constraints

In full duplex mode, predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) also demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices.

The sum of the transmit and receive data delays for an implementation of the PHY shall not exceed the limits shown in Table 165–18. Transmit data delay is defined from the input of a given unit of data at the 25GMII to the presentation of the same unit of data by the PHY to the MDI. Receive data delay is defined from the input of a given unit of data at the MDI to the presentation of the same unit of data by the PHY to the 25GMII.

NOTE—The physical medium interconnecting two PHYs introduces additional delay in a link.

Table 165–18—Delay limits

Mode	Interleave	Bit times	pause_quanta	Delay (ns)
25GBASE-T1	1	25 600	50	1024
25GBASE-T1	2	36 864	72	1474.56
25GBASE-T1	4	58 880	115	2355.2
25GBASE-T1	8	102 400	200	4096

165.11 Protocol implementation conformance statement (PICS) proforma for Clause 165, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 25GBASE-T1⁷

165.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 165, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 25GBASE-T1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

165.11.2 Identification

165.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

165.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cy-2023, Clause 165, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 25GBASE-T1.
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3cy-2023.)	

Date of Statement	
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⁷*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

165.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Auto-Negotiation.	98		O	Yes [] No []
*EEE	Support of EEE capability.	165.1.3.3		O	Yes [] No []
*INS	Installation/cabling.	165.7	Items marked with INS include installation practices and cabling specifications applicable when the link segment is balanced pair cabling, and not applicable to backplane link segments.	O	Yes [] No []
*OAM	Support of 25GBASE-T1 PCS level operations, administration, and maintenance.	165.3.9		O	Yes [] No []
*MDIO	Support of MDIO capability.	45.1	Register and Interface support.	O	Yes [] No []

165.11.4 PICS proforma tables for Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and baseband medium, type 25GBASE-T1

165.11.4.1 General

Item	Feature	Subclause	Value/Comment	Status	Support
G1	If Auto-Negotiation is implemented, it meets the requirements of Clause 98.	98		AN:M	Yes [] N/A []
G2	Capable of operating as MASTER or SLAVE, per runtime configuration.	165.1.3		M	Yes []
G3	If Auto- Negotiation is not used, a MASTER-SLAVE relationship is established by management or hardware configuration of the PHYs.	165.1.3		!AN:M	Yes [] N/A []

165.11.4.2 Physical Coding Sublayer (PCS)

165.11.4.2.1 PCS Transmit

Item	Feature	Subclause	Value/Comment	Status	Support
PCT1	PCS Reset.	165.3.2.1	Described in 165.3.2.1.	M	Yes []
PCT2	Management interface restore to operation.	165.3.2.1	Within 10 ms from the setting of bit 1.2309.15.	M	Yes []
PCT3	PCS Transmit bit ordering.	165.3.2.2	Figure 165–6.	M	Yes []
PCT4	If a PMA_TXMODE.indication message has the value SEND_Z, PCS Transmit passes a vector of zeros at each symbol period to the PMA via the PMA_UNITDATA.request primitive.	165.3.2.2		M	Yes []
PCT5	If a PMA_TXMODE.indication message has the value SEND_T, PCS Transmit generates a sequence (T_n) defined in 165.3.5.1 to the PMA via the PMA_UNITDATA.request primitive.	165.3.2.2		M	Yes []
PCT6	If a PMA_TXMODE.indication message has the value SEND_N, the PCS is in the normal mode of operation and the PCS Transmit function uses a 65B coding technique specified in 149.3.2.2.4 to represent data or control.	165.3.2.2		M	Yes []
PCT7	All 25GMII control code values that do not appear in Table 149–2 are not transmitted and are treated as an error if received.	165.3.2.2.5		M	Yes []
PCT8	RS-FEC encoding.	165.3.2.2.14	The RS-FEC encoding generates the 90 10-bit parity symbols (900 bits total) using the 8460-bit vector, consisting of tx_group130x65B and the 10-bit OAM_field as the message symbols.	M	Yes []
PCT9	Interleave set to depth setting.	165.3.2.2.15	Interleaver depth set to value requested by link partner during Infofield exchange.	M	Yes []
PCT10	Round-robin interleaving scheme.	165.3.2.2.15	Figure 165–8.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
PCT11	The PCS Transmit aggregates <i>L</i> RS-FEC input frames into an interleaved RS-FEC input superframe.	165.3.2.2.15		M	Yes []
PCT12	EEE transmit state diagram.	165.3.2.2.22	Figure 165–14.	EEE:M	Yes [] N/A []
PCT13	Sleep signal.	165.3.2.2.22	Transmitted when an LPI control character in the last 64B/65B block of an RS-FEC frame is detected by the PCS Transmit function.	EEE:M	Yes [] N/A []
PCT14	After the sleep signal is transmitted, LPI control characters are input to the PCS scrambler continuously until the PCS Transmit function exits the LPI transmit mode.	165.3.2.2.22		EEE:M	Yes [] N/A []
PCT15	While the PMA asserts SEND_N, the lpi_tx_mode variable controls the transmit signal through the PMA_UNITDATA.request primitive.	165.3.2.2.22	Described in 165.3.2.2.22.	M	Yes []

165.11.4.2.2 PCS Receive

Item	Feature	Subclause	Value/Comment	Status	Support
PCR1	PCS 64B/65B Receive state diagram.	165.3.2.3	Figure 149–18 and Figure 149–19.	M	Yes []
PCR2	PCS receive bit ordering.	165.3.2.3	Figure 165–7.	M	Yes []
PCR3	Frame and block synchronization.	165.3.2.3.1	Described in 165.3.2.3.1.	M	Yes []
PCR4	MASTER PHY receiver descrambler generator polynomial.	165.3.2.3.2	Equation (149–6).	M	Yes []
PCR5	SLAVE PHY receiver descrambler generator polynomial.	165.3.2.3.2	Equation (149–5).	M	Yes []
PCR6	RS-FEC parity bits.	165.3.2.3.3	Defined in 165.3.2.2.14.	M	Yes []

165.11.4.2.3 Test-pattern generators

Item	Feature	Subclause	Value/Comment	Status	Support
TPG1	Test-pattern generator operation	165.3.3	As specified in 149.3.3.	M	Yes []

165.11.4.2.4 Side-stream scrambler

Item	Feature	Subclause	Value/Comment	Status	Support
SSS1	Side-stream scrambler operation	165.3.4	As specified in 149.3.4.	M	Yes []

165.11.4.2.5 LPI signaling

Item	Feature	Subclause	Value/Comment	Status	Support
LPI1	An EEE-capable PHY synchronizes refresh intervals during LPI mode.	165.3.6.1		EEE:M	Yes [] N/A []
LPI2	Alert.	165.3.6.1	Start at the fifth frame after any eighth PHY frame boundary starting at the beginning of the frame following a refresh PHY frame.	EEE:M	Yes [] N/A []
LPI3	tx_refresh_active.	165.3.6.1	Derived from the transmitted PHY frames as shown in Table 165–5.	EEE:M	Yes [] N/A []
LPI4	tx_alert_start_next.	165.3.6.1	Derived from the transmitted PHY frames as shown in Table 165–6.	EEE:M	Yes [] N/A []
LPI5	During the quiet period.	165.3.6.2	PHY stops transmitting.	EEE:M	Yes [] N/A []

165.11.4.2.6 Functions and state diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
FSD1	lpi_tx_sleep_timer.	165.3.7.2.3	Period equal to 16 RS-FEC frame periods.	EEE:M	Yes [] N/A []
FSD2	lpi_tx_alert_timer.	165.3.7.2.3	Period equal to four RS-FEC frame periods.	EEE:M	Yes [] N/A []
FSD3	lpi_tx_wake_timer.	165.3.7.2.3	Period equal to lpi_wake_time RS-FEC frame periods.	EEE:M	Yes [] N/A []
FSD4	lpi_rx_wake_timer.	165.3.7.2.3	Period equal to lpi_wake_time RS-FEC frame periods.	EEE:M	Yes [] N/A []
FSD5	DECODE function.	165.3.7.2.4	Decode the block based on code specified in 165.3.2.2.2.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
FSD6	ENCODE function.	165.3.7.2.4	Encode the block as specified in 165.3.2.2.2.	M	Yes []
FSD7	RFER monitor state diagram functions.	165.3.7.3	Implemented.	M	Yes []
FSD8	PCS Transmit state diagram functions.	165.3.7.3	Implemented.	M	Yes []
FSD9	PCS Receive state diagram functions.	165.3.7.3	Implemented.	M	Yes []
FSD10	RFER function during LPI receive operation.	165.3.7.3	Not performed from the time that the PCS 64B/65B Receiver enters the state RX_L, until the state RX_W is exited.	EEE:M	Yes [] N/A []
FSD11	RX_FRAME value.	165.3.7.2.6	RX_FRAME is FALSE during the first 8 WAKE frames.	EEE:M	Yes [] N/A []

165.11.4.2.7 PCS loopback

Item	Feature	Subclause	Value/Comment	Status	Support
PCSL1	Placed in loopback mode.	149.3.8.3	When MDIO register 3.0.14 is set to a one.	MDIO:M	Yes [] N/A []
PCSL2	When in PCS loopback mode.	149.3.8.3	Accept data on the transmit path from the 25GMII and return it on the receive path to the 25GMII.	M	Yes []
PCSL3	When in PCS loopback mode.	149.3.8.3	Transmit a continuous stream of 65B RS-FEC encoded PAM4 symbols to the PMA sublayer.	M	Yes []
PCSL4	When in PCS loopback mode.	149.3.8.3	Ignore all data presented to it by the PMA sublayer.	M	Yes []

165.11.4.2.8 OAM

Item	Feature	Subclause	Value/Comment	Status	Support
OAM1	State diagram behavior.	165.3.9.4	Conforms to Figure 149–24 and Figure 149–25 .	OAM:M	Yes [] N/A []
OAM2	When the PCS frame is operating in interleaved mode of 2x, 4x, or 8x the first symbol (OAM<0>) is inserted in the first RS frame in the superframe so that the full OAM frame can be packed into eight superframes in the 2x interleaved mode, into four superframes in the 4x interleaved mode, and into two superframes in the 8x interleaved mode.	165.3.9.2.1		OAM:M	Yes [] N/A []
OAM3	Partially transmitted OAM frame.	165.3.9.2.1	Described in 165.3.9.2.1.	OAM:M	Yes [] N/A []
OAM4	When OAM is not implemented.	165.3.9.2.1	10-bit OAM field is set to all 0's.	OAM:M	Yes [] N/A []
OAM5	OAM frame data.	149.3.9.2.2	Reserved fields is set to 0.	OAM:M	Yes [] N/A []
OAM6	OAM frame data symbol encoding.	149.3.9.2.13	Using Reed-Solomon code (16,14).	OAM:M	Yes [] N/A []
OAM7	Parity calculation.	149.3.9.2.13	Same result as the shift register implementation shown in Figure 149–23 .	OAM:M	Yes [] N/A []
OAM8	OAM Reed-Solomon shift register.	149.3.9.2.13	Initialized to zero.	OAM:M	Yes [] N/A []
OAM9	BASE-T1 OAM frame acceptance criteria.	149.3.9.2.14	Described in 149.3.9.2.14 .	OAM:M	Yes [] N/A []
OAM10	If a PHY is already in LPI mode and receives SNR<1:0> set to 01 by its link partner.	149.3.9.2.15	Immediately exit LPI.	OAM:M	Yes [] N/A []
OAM11	mr_tx_received.	165.3.9.4.3	Clear on read.	OAM:M	Yes [] N/A []

165.11.4.3 Physical Medium Attachment (PMA)

165.11.4.3.1 PMA Reset function

Item	Feature	Subclause	Value/Comment	Status	Support
PR1	PMA Reset.	165.4.2.1	Described in 165.4.2.1.	M	Yes []

165.11.4.3.2 PMA Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
PMAT1	When the PHY Control state diagram is not in the DISABLE_TRANSMITTER state, PMA Transmit.	165.4.2.2	Continuously transmit pulses modulated by the symbols given by tx_symb onto the MDI.	M	Yes []
PMAT2	During Link Synchronization, when sync_link_control = DISABLE and Auto-Negotiation is either not enabled or is not implemented, the sync_tx_symb output by the PHY Link Synchronization function is used in place of tx_symb as the data source for PMA Transmit.	165.4.2.2		AN:M	Yes [] N/A []
PMAT3	When lpi_tx_mode = ALERT, the PN sequence defined in 165.4.2.6 is used in place of tx_symb as the data source for PMA Transmit.	165.4.2.2		EEE:M	Yes [] N/A []
PMAT4	Signals generated by PMA Transmit.	165.4.2.2	Comply with 165.5.3.	M	Yes []
PMAT5	TX_TCLK source when the PMA_CONFIG.indication parameter config is MASTER.	165.4.2.2	Sourced from a local clock source while meeting the transmit jitter requirements of 165.5.3.3.	M	Yes []
PMAT6	MASTER-SLAVE relationship.	165.4.2.2	Include loop timing.	M	Yes []
PMAT7	TX_TCLK source when the PMA_CONFIG.indication parameter config is SLAVE.	165.4.2.2	Sourced from the recovered clock of 165.4.2.8 while meeting the transmit jitter requirements of 165.5.3.3.	M	Yes []
PMAT8	PMA transmit fault function.	165.4.2.2	Mapped to the transmit fault bit as specified in 45.2.1.7.4.	MDIO:M	Yes [] N/A []
PMAT9	Average launch power of the transmitter when the PMA_transmit_disable variable is set to TRUE.	165.4.2.2.1	Less than –53 dBm.	M	Yes []

165.11.4.3.3 PMA Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
PMAR1	Quality of received symbols.	165.4.2.3	RFER of less than 2×10^{-10} after RS-FEC decoding, over a channel meeting the requirements of 165.7.	M	Yes []
PMAR2	If the MDIO Interface is implemented, then this function contributes to the receive fault bit specified in 45.2.1.7.5 and 45.2.1.243.7.	165.4.2.3		MDIO:M	Yes [] N/A []

165.11.4.3.4 PHY Control function

Item	Feature	Subclause	Value/Comment	Status	Support
PCF1	PHY Control state diagram.	165.4.2.4	Figure 165–22.	M	Yes []
PCF2	12-octet Infocfield.	165.4.2.4	Include the fields in 165.4.2.4.2 through 165.4.2.4.8.	M	Yes []
PCF3	Infocfield is be transmitted at least 256 times with each change to octets 7 to 10.	165.4.2.4		M	Yes []
PCF4	Infocfield reserved bits.	165.4.2.4.1	Set to zero on transmit and ignored when received by the link partner.	M	Yes []
PCF5	Start of frame delimiter hexadecimal value.	165.4.2.4.2	0xBBA700.	M	Yes []
PCF6	PMA_state<7:6>.	165.4.2.4.4	Communicate the state of the transmitting transceiver to the link partner.	M	Yes []
PCF7	Any Message Field value not listed in Table 165–7 or Table 165–8 is not transmitted.	165.4.2.4.4		M	Yes []
PCF8	Any Message Field value not listed in Table 165–7 or Table 165–8 is ignored at the receiver.	165.4.2.4.4		M	Yes []
PCF9	The Message Field setting for the first transmitted PMA frame is the first row of Table 165–7 for the MASTER and the first or second row of Table 165–8 for the SLAVE.	165.4.2.4.4		M	Yes []
PCF10	For a given Message Field setting, the next Message Field setting is the same Message Field setting or the Message Field setting corresponding to a row below the current setting.	165.4.2.4.4		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
PCF11	The optional EEE capability is enabled only if both PHYs set the capability bit $EEE_{En} = 1$.	165.4.2.4.5		EEE:M	Yes [] N/A []
PCF12	The optional BASE-T1 OAM capability is enabled only if both PHYs set the capability bit $OAM_{En} = 1$.	165.4.2.4.5		OAM:M	Yes [] N/A []
PCF13	The capability bit values are considered as valid only when the loc_rcvr_status bit is 1.	165.4.2.4.5		M	Yes []
PCF14	Remaining PHY capability bits are reserved and set to 0.	165.4.2.4.5		M	Yes []
PCF15	DataSwPFC24.	165.4.2.4.6	Set to an integer multiple of 32.	M	Yes []
PCF16	DataSwPFC24 on switch from PAM2 to PAM4.	165.4.2.4.6	Minimum of 4081 and maximum of 4785.	M	Yes []
PCF17	CRC16 polynomial.	165.4.2.4.8	$(x + 1)(x^{15} + x + 1)$ of the previous 7 octets.	M	Yes []
PCF18	CRC16.	165.4.2.4.8	Figure 165–19.	M	Yes []
PCF19	CRC16 delay elements.	165.4.2.4.8	All initialized to zero.	M	Yes []
PCF20	Startup sequence.	165.4.2.4.10	Comply with Figure 165–22.	M	Yes []
PCF21	During startup, prior to entering the COUNTDOWN state, the SLAVE aligns its transmit 65B RS-FEC frame to within +0/–4 partial PHY frames of the MASTER as seen at the SLAVE MDI.	165.4.2.4.10		M	Yes []
PCF22	SLAVE Infocfield partial frame count.	165.4.2.4.10	Match the MASTER Infocfield partial frame count for the aligned frame.	M	Yes []
PCF23	MASTER startup timing.	165.4.2.4.10	Comply with values in Table 165–10.	M	Yes []
PCF24	SLAVE startup timing.	165.4.2.4.10	Comply with values in Table 165–11.	M	Yes []

165.11.4.3.5 Link Monitor function

Item	Feature	Subclause	Value/Comment	Status	Support
LMK1	Link Monitor function.	165.4.2.5	Comply with Figure 165–23.	M	Yes []

165.11.4.3.6 PHY Link Synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
PLS1	Link Synchronization PHY PMA training when Auto-Negotiation is disabled or not implemented.	165.4.2.6	Described in 165.4.2.4.	!AN:M	Yes [] N/A []
PLS2	Shift register sequence $S_n[]$ rate.	165.4.2.6	703.125 MHz.	M	Yes []
PLS3	MASTER PN sequence generator.	165.4.2.6	Equation (165–5).	M	Yes []
PLS4	SLAVE PN sequence generator.	165.4.2.6	Equation (165–6).	M	Yes []
PLS5	PN sequence generator shift registers when entering TRANSMIT_DISABLE state.	165.4.2.6	Reset to $S_n[7:0] = 0000\ 0001$.	M	Yes []
PLS6	25GBASE-T1 mapping of $S_n[0]$ to T_n .	165.4.2.6	If $S_n[0] = 0$ then $T_n = +1\ +1\ \dots\ +1$ (repeated 20 times). If $S_n[0] = 1$ then $T_n = -1\ -1\ \dots\ -1$ (repeated 20 times).	M	Yes []
PLS7	The synchronization state diagram is used to synchronize 25GBASE-T1 PHY prior to the link training.	165.4.2.6		M	Yes []
PLS8	If Clause 98 Auto-Negotiation function is enabled, then the Auto-Negotiation function is used as the mechanism for PHY synchronization and the synchronization state diagram in Figure 165–21 remains in the SYNC_DISABLE state.	165.4.2.6		AN:M	Yes [] N/A []
PLS9	send_s_sigdet.	165.4.2.6.1	This variable is set FALSE no later than 1 μ s after the signal goes quiet on the MDI.	M	Yes []
PLS10	send_s_timer.	165.4.2.6.2	Expires 1.25 μ s $\pm$ 0.025 μ s after being started.	M	Yes []
PLS11	sigdet_wait_timer.	165.4.2.6.2	Expires 5 μ s $\pm$ 0.025 μ s after being started.	M	Yes []
PLS12	silent_wait_timer.	165.4.2.6.2	Expires 1.25 μ s $\pm$ 0.025 μ s after being started.	M	Yes []

165.11.4.3.7 Refresh monitor function

Item	Feature	Subclause	Value/Comment	Status	Support
RMF1	Refresh monitor function.	165.4.2.7	Comply with Figure 165–24.	EEE:M	Yes [] N/A []

165.11.4.3.8 Clock Recovery function

Item	Feature	Subclause	Value/Comment	Status	Support
CRF1	Clock Recovery function.	165.4.2.8	Provide a clock suitable for signal sampling so that the RS FER indicated in 165.4.2.3 is achieved.	M	Yes []

165.11.4.3.9 MDI

Item	Feature	Subclause	Value/Comment	Status	Support
MDIS1	Symbol response $s(t)$.	165.4.3.1	Comply with the electrical specifications given in 165.5.3.	M	Yes []

165.11.4.3.10 PMA State variables

Item	Feature	Subclause	Value/Comment	Status	Support
PMAS1	minwait_timer.	165.4.4.2	Expires $975\ \mu\text{s} \pm 50\ \mu\text{s}$ after being started.	M	Yes []
PMAS2	lpi_refresh_rx_timer.	165.4.4.2	Period equal to 50 complete quiet-refresh signal periods, equivalent to 1.59744 ms.	EEE:M	Yes [] N/A []

165.11.4.4 PMA electrical specifications

165.11.4.4.1 Test modes

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	Test modes.	165.5.1	Provided to allow for testing of the transmitter jitter, transmitter distortion, transmitter PSD, transmitter droop, and BER testing.	M	Yes []
TM2	Test mode enabling.	165.5.1	Enabled by setting a control register, 1.2313.15:13, or equivalent functionality.	MDIO:M	Yes [] N/A []
TM3	The test modes only change the data symbols provided to the transmitter circuitry and do not alter the electrical and jitter characteristics of the transmitter and receiver from those of normal operation.	165.5.1		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TM4	Test Mode 1.	165.5.1	Described in 165.5.1.	M	Yes []
TM5	Test Mode 2.	165.5.1	Described in 165.5.1.	M	Yes []
TM6	Test Mode 3.	165.5.1	Described in 165.5.1.	M	Yes []
TM7	Test Mode 4.	165.5.1	Described in 165.5.1.	M	Yes []
TM8	Test Mode 5.	165.5.1	Described in 165.5.1.	M	Yes []
TM9	Test Mode 6.	165.5.1	Described in 165.5.1.	M	Yes []
TM10	Test Mode 7.	165.5.1	Described in 165.5.1.	M	Yes []

165.11.4.4.2 Transmitter electrical specifications

Item	Feature	Subclause	Value/Comment	Status	Support
TES1	Coupling.	165.5.3	Operate with AC coupling to the MDI.	M	Yes []
TES2	Resistive differential load.	165.5.3	Meet electrical requirements of this clause with a 100 Ω resistive differential load connected to transmitter output if load is not specified.	M	Yes []
TES3	Maximum output droop.	165.5.3.1	Positive and negative droop is less than 15%.	M	Yes []
TES4	Transmitter linearity.	165.5.3.2	Defined in 120D.3.1.2.	M	Yes []
TES5	Transmitter SNDR for 25GBASE-T1.	165.5.3.2	Exceed 38 dB.	M	Yes []
TES6	RMS value of the MASTER TX_TCLK_879 jitter from 1 MHz to 100 MHz.	165.5.3.3	Less than 0.4 ps.	M	Yes []
TES7	Peak-to-peak value of the MASTER TX_TCLK_879 jitter.	165.5.3.3	Less than 4 ps.	M	Yes []
TES8	RMS value of the SLAVE TX_TCLK_879 jitter.	165.5.3.3	Less than 0.8 ps.	M	Yes []
TES9	Peak-to-peak value of the SLAVE TX_TCLK_879 jitter.	165.5.3.3	Less than 8 ps.	M	Yes []
TES10	TX_TCLK_879 jitter measurement interval.	165.5.3.3	Measured over an interval of 0.4 ms $\pm$ 10%.	M	Yes []
TES11	TX_TCLK_879 jitter capture device bandwidth.	165.5.3.3	Larger than 200 MHz.	M	Yes []
TES12	RMS value of the MDI output jitter from 1 MHz to 100 MHz.	165.5.3.3.1	Less than 0.4 ps.	M	Yes []
TES13	Peak-to-peak value of the MDI output jitter.	165.5.3.3.1	Less than 4 ps.	M	Yes []
TES14	MDI jitter measurement interval.	165.5.3.3.1	Measured over an interval of 0.4 ms $\pm$ 10%.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TES15	MDI jitter capture device bandwidth.	165.5.3.3.1	Larger than 200 MHz.	M	Yes []
TES16	DJpk-pk jitter.	165.5.3.3.2	Less than 3.6 ps.	M	Yes []
TES17	EOJpk-pk jitter.	165.5.3.3.2	Less than 1.6 ps.	M	Yes []
TES18	Transmit power level.	165.5.3.4	In the range of –1 dBm to 2 dBm.	M	Yes []
TES19	PSD of the transmitter.	165.5.3.4	See Equation (165–9) and Equation (165–10).	M	Yes []
TES20	Transmitter peak differential output.	165.5.3.5	Less than 1.3 V peak-to-peak.	M	Yes []
TES21	Transmitter clock frequency.	165.5.3.6	Within the range 14.0625 GHz $\pm$ 50 ppm.	M	Yes []
TES22	Clock short-term rate of frequency variation.	165.5.3.6	Less than 0.1 ppm/second when in LPI mode.	EEE:M	Yes [] N/A []
TES23	Clock short-term rate of frequency variation.	165.5.3.6	Less than 0.1 ppm/second when switching to and from LPI mode.	EEE:M	Yes [] N/A []

165.11.4.4.3 Receiver electrical specifications

Item	Feature	Subclause	Value/Comment	Status	Support
RES1	Receiver differential input signals.	165.5.4.1	The frame error ratio is less than 7.8×10^{-9} for 800 octet frames with minimum IPG or greater than 220-octet IPG.	M	Yes []

165.11.4.4.4 Mated test configurations

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Insertion loss.	165.5.5.3.1	See Equation (165–13) and Equation (165–14).	M	Yes []
TF2	Return loss.	165.5.5.3.2	See Equation (165–16).	M	Yes []
TF3	Mode conversion.	165.5.5.3.3	See Equation (165–17).	M	Yes []
TF4	Crosstalk.	165.5.5.3.4	See Equation (165–18).	M	Yes []

165.11.4.5 Link segment characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
LSC1	Insertion loss.	165.7.1.1	See Equation (165–19).	M	Yes []
LSC2	Return loss.	165.7.1.3.1	See Equation (165–20).	M	Yes []
LSC3	Residual echo metric.	165.7.1.3.5	See Equation (165–29).	M	Yes []
LSC4	Echo tail metric.	165.7.1.3.8	See Equation (165–38).	M	Yes []
LSC5	Coupling attenuation.	165.7.1.4	See Equation (165–39).	INS:M	Yes [] N/A []
LSC6	Maximum link delay.	165.7.1.6	Not to exceed 60 ns at all frequencies between 2 MHz and 9000 MHz.	M	Yes []
LSC7	PSANEXT.	165.7.2.1	See Equation (165–40).	M	Yes []
LSC8	PSAACRF.	165.7.2.2	See Equation (165–41).	M	Yes []

165.11.4.6 MDI specifications

Item	Feature	Subclause	Value/Comment	Status	Support
MDI1	The electrical requirements specified in 165.5.3 and 165.5.4 are met when the PHY is connected to the MDI connector mated with a specified connector to a shielded balanced pair of conductors.	165.8.2		M	Yes []
MDI2	Return loss.	165.8.2.1	See Equation (165–42).	M	Yes []

165.11.4.7 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
DC1	Sum of the transmit and receive data delays.	165.10	Not to exceed the limits in Table 165–18.	M	Yes []

Annex 98B

(normative)

IEEE 802.3 Selector Base Page definition

98B.3 Technology Ability Field bit assignments

Change the row for bits “A6 through A8” and insert a new row for bit “A6” above it in Table 98B–1 as follows (unchanged rows not shown):

Table 98B–1—Technology Ability Field bit assignments

bit	Selector description
...	
A6	25GBASE-T1 ability
A 6 through A8	Reserved
...	

98B.4 Priority Resolution

Insert the following new entry in the dashed list in 98B.4 before the entry for 10GBASE-T1 as follows:

— 25GBASE-T1

Annex 149B

(informative)

OAM status

149B.1 Purpose

Change the first sentence in 149B.1 as follows:

This annex describes a suggested assignment of the OAM status bits for use with the Clause 149 and
Clause 165 MultiGBASE-T1-PHYs.

Insert new Annex 165A after Annex 164C (inserted by IEEE Std 802.3cs-2022) as follows:

Annex 165A

(informative)

Tx function to Rx function channel characteristics

165A.1 Overview

Annex 165A provides information on insertion loss and return loss parameters of the channel defined between Tx function and Rx function illustrated in Figure 165A–1. Table 165–13 describes the test points illustrated in Figure 165A–1. See Figure 165A–2 for TP2 and TP3 test points.

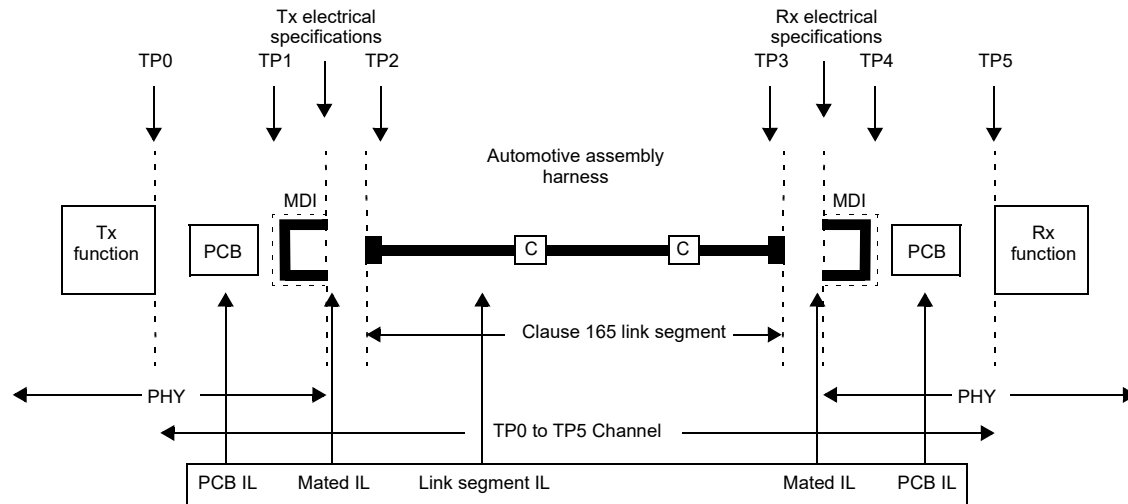


Figure 165A–1—Channel Tx function to Rx function

165A.2 TP2 to TP0 and TP3 to TP5 reference insertion loss

The recommended maximum insertion loss from TP2 to TP0 and TP3 to TP5 including the test configuration is shown in Figure 165A–2. Figure 165A–2 includes the host PCB, a mated connector pair, and the host test configuration (HTC) reference insertion losses. All losses shown are at 7031.25 MHz.

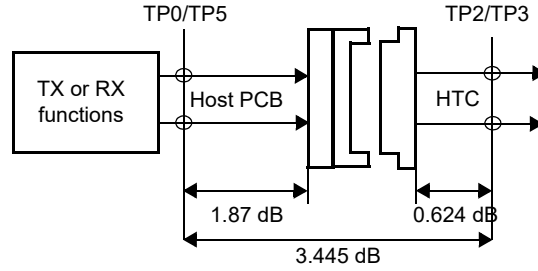


Figure 165A-2—TP2 to TP0 and TP3 to TP5 reference insertion loss

165A.2.1 Host PCB trace loss

The recommended maximum insertion loss from TP0/TP5 to the host-side PCB connection of the MDI is determined using Equation (165A-1) and illustrated in Figure 165A-1. This maximum recommended loss is based on a 76.2 mm trace length, and is 1.871 dB at 7031.25 MHz.

$$IL_{PCB}(f) \leq IL_{PCBmax}(f) = 0.09144 \left(\frac{f}{1000} \right) + 0.51054 \left(\frac{f}{1000} \right)^{0.45} \text{ dB} \quad (165A-1)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

The recommended minimum insertion loss from TP0/TP5 to the host-side PCB connection of the MDI is determined using Equation (165A-2) and illustrated in Figure 165A-1. This minimum recommended loss is based on a 25.4 mm trace length, and is 0.624 dB at 7031.25 MHz.

$$IL_{PCB}(f) \geq IL_{PCBmin}(f) = 0.03048 \left(\frac{f}{1000} \right) + 0.17018 \left(\frac{f}{1000} \right)^{0.45} \text{ dB} \quad (165A-2)$$

where

f is the frequency in MHz; $1 \leq f \leq 9000$

165A.3 TP0 to TP5 channel insertion loss

The channel insertion loss from TP0 to TP5 is determined using Equation (165A-3), Equation (165A-4), and Equation (165A-5).

$$IL_{Channel}(f) \leq 2 \times IL_{PCBmax}(f) + 2 \times IL_{MDIConnector}(f) + IL_{LinkSegment}(f) \text{ dB} \quad (165A-3)$$

$$IL_{MDIConnector}(f) \leq 0.1 \sqrt{\frac{f}{2500}} \text{ dB} \quad (165A-4)$$

$$IL_{LinkSegment}(f) \leq 0.00135(f) + 0.3564(f)^{0.45} + 0.495 \left(\frac{f}{7500} \right)^6 \text{ dB} \quad (165A-5)$$

where

f is the frequency in MHz for Equation (165A–3), Equation (165A–4), and Equation (165A–5); $10 \leq f \leq 9000$

The 25GBASE-T1 PHY utilizes 4-level pulse amplitude modulation (PAM4) transmitted at 14.0625 GBd. Table 165A–1 shows the recommended maximum insertion losses for the channel and channel components at the Nyquist frequency for this PHY.

Table 165A–1—TP0 to TP5 channel insertion loss

PHY	Rate GBd	Nyquist frequency (MHz)	IL_{PCBmax} (dB)	$IL_{LinkSegment}$ (dB)	$IL_{MDIConnector}$ (dB)	$IL_{channel}$ (dB)
25GBASE-T1	14.0625	7031.25	1.871	29.02	0.168	33.098

165A.4 Example models for Tx/Rx function to MDI return loss

The channel topology between Tx function and Rx function consists of the Tx function to MDI, the link segment, and the MDI to Rx function. The channel return loss can be modeled as a concatenation of the Tx function to MDI return loss, the link segment return loss, and the MDI to Rx function return loss.

Example implementations for the Tx function and Rx function to MDI topologies are as shown in Figure 149C–3 and Figure 149C–4. For Clause 165 PHYs, the informative values for components are given in Table 165A–2.

Table 165A–2—Analysis parameters and values

Element	Unit	Minimum	Nominal	Maximum
R_T	Ω	45	50	55
Z_O	Ω	45	50	55
C_T	pF	—	0.1	—
L_P	μH	—	6.8	—
C_P	pF	—	0.24	—
C_{ESD}	pF	—	0.4	—
C_{AC}	nF	—	22	—

165A.5 Coupling between ports on multiport designs

When multiple ports of 25GBASE-T1 are implemented on the same board, care should be taken to avoid coupling between ports. The coupling between adjacent ports on a multiport MDI connector or between adjacent balanced pairs is recommended to be approximately the same level, but no greater, than that specified for power sum alien near-end crosstalk in Equation (165–40).

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